Prima BT

Mass Spectrometers
User Guide
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Issue 2

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Revision History

Revision Level	Date	Comments
Issue 1	21/10/13	Initial Version
Issue 2	19/11/13	Serial I/O pin outs; section C General re-wording

Contents

Chapter 1
Introduction

This manual is provided for users, operators, and owners of Thermo Scientific Mass Spectrometers.

This manual covers the Thermo Scientific Prima BT instrumentation and options.

Information in this guide is organized in chapters and covers site preparation, installation, commissioning, operation, and maintenance. Technical descriptions of the analyser and various inlet configurations are provided as individual appendices.

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Chapter 2
Safety Information

Overview

This chapter contains general safety and operating information applicable to analytical systems that must be understood by all persons installing, using, or maintaining the system. This information is designed to aid personnel in the safe installation, operation, and service of the analyser and sample systems. It is not designed to replace or limit appropriate safety measures applicable to work performed by personnel. Any additional safety and operating measures that are required must be determined by and followed by personnel performing work on the system.

It is the responsibility of the user / operator / integrator (hereon referred to as user) to perform risk assessments of the site location, installation and operation of the instrumentation. This manual highlights hazards associated with the instrument and provides details of precautions to be taken to ensure safe operation. The user must account for the information provided and put in place procedures to ensure for the safety of personnel and plant.

To ensure personal safety, system integrity, and optimum performance, users should thoroughly understand the content of this manual before installing, using, or performing maintenance on this product.

Failure to follow appropriate safety procedures or inappropriate use of the equipment described in this manual can lead to equipment damage or injury.

Any person working with or on the equipment described in this manual is required to evaluate all functions and operations for potential safety hazards before commencing work. Appropriate precautions must be taken as necessary to prevent potential damage to equipment or injury to personnel.

Only qualified technical personnel should access the analyser enclosure.

If in doubt as to the instructions and information provided or if you have additional questions regarding the product, a service or support request, contact your nearest Thermo Fisher Scientific representative for process mass spectrometers.

Safety Conventions and Symbols

The following conventions and symbols are used throughout this manual to alert users to potential hazards or important information. **Failure to heed the warnings and cautions in this manual can lead to injury or equipment damage.**

Warning / Danger

Warning / Danger statements notify users of procedures, practices, conditions, etc. which may result in serious injury or death if not carefully observed or followed. The triangular icon displayed with a warning varies depending on the type of hazard (only two are displayed here).

Prohibited

These statements explicitly PROHIBIT an action. Ignoring an instruction related to a hazard highlighted with this symbol may cause an unsafe condition that could result in an accident.

Caution

Cautions notify users of operating procedures, practices, conditions, etc. that may result in equipment damage or cause injury if not carefully observed or followed.

Caution

Statements accompanied by this caution symbol **explicitly require** an action be taken. Ignoring an instruction related to a hazard or instruction highlighted with this symbol may cause an unsafe condition which could result in an accident.

Note

Notes emphasize important or essential information or a statement of company policy regarding an operating procedure, practice, condition, etc.

The information in this manual is designed to aid personnel to correctly and safely install, operate, and / or maintain the system described; however, personnel are still responsible for considering all actions and procedures for potential hazards or conditions that may not have been anticipated in the written procedures. **If a procedure cannot be performed safely, it must not be performed until appropriate actions can be taken to ensure the safety of the equipment and personnel.**

The procedures in this manual are not designed to replace or supersede required or common sense safety practices. All safety warnings listed in any documentation applicable to equipment and parts used in or with the system described in this manual must be read and understood prior to working on or with any part of the system.

Failure to correctly perform the instructions and procedures in this manual or other documents pertaining to this system can result in equipment malfunction, equipment damage, and / or injury to personnel.

Warning! Deviation from the instructions may result in equipment malfunction, equipment damage, and / or injury to personnel.

System Safety

Read this manual before operating the instrument! In particular, review the following cautionary statements.

Caution! Only qualified personnel trained in the operation and maintenance of the Prima BT should access the inner enclosure.

Caution! Never disregard any cautionary or warning instructions. Read and observe all instructions on printed labels fixed to the instrument.

Caution! Failure to operate the instrument as specified could impair the protection features provided and described in this guide.

Hazards exist within the instrument enclosure, and the remainder of this chapter details any necessary precautions to be taken. Read the following safety notes carefully before attempting to operate or work on the instrument.

Connecting the Instrument to the Electrical Supply

Caution!

- Check that the mains supply satisfies the requirements detailed in Chapter 3: Site Requirements.
- Connection must be made by qualified personnel only. Observe the instructions provided in Chapter 3: Site Requirements, and ensure that the connection is consistent with local safety regulations.

Repair & Maintenance

When performing repair or maintenance on the instrument:

Warning! Before attempting to carry out repairs or maintenance, isolate the mains electrical supply.

Caution! Maintenance should be carried out by qualified personnel only.

Changing the Rotary Pump Oil

Warning! Allow the pump oil to cool before draining.

Switching on the Rotary Pump

Caution! Check that the oil level is between the two markers on the oil level window before switching on the rotary pump.

High Temperatures

Warning! The following assemblies may attain a surface temperature as listed:

- Rapid Multi-Stream Sampler (RMS) 80°C
- Single Point Inlet system 120°C.
- Single Point Inlet System (High Temperature Version): 230°C.
- RMS Sample Probe: 120°C. (Internal)
- Inlet Probe Assembly: 120°C. (Internal)
- Rotary pump operating temperature: 75°C.

Rotary Pump Exhaust Outlet

Warning! Ensure that the rotary pump exhaust is unobstructed. Failure to do so **will** cause dangerous internal pressures in the pump. This **will** result in blown seals and bursting of the oil box and could result in physical injury.

Inlet System Exhaust

Warning! A leak-tight exhaust must be fitted to the inlet system when sampling gas streams which are:

- Explosive

- Toxic

- Asphyxiant

Gas Mixtures in the Inlet System

Warning! All gases flowing into multi-stream inlets (e.g. RMS) mix in the common exhaust. Consideration should be given to the creation of potentially explosive or toxic mixtures in the valve and exhaust as a result of incompatible sample streams. See "Exhaust Gas Connections" in Chapter 3.

Protective Clothing

Warning! It is recommended that the following personal protective equipment (PPE) is used when performing the following tasks:

- Changing rotary pump oil: Safety glasses, overalls and rubber gloves.
- Changing or removing the glass leak: Safety glasses.
- Connecting sample lines: Safety glasses.
- Changing a filament: Rubber gloves.

Hazardous Materials

Warning! The substances listed below are potentially hazardous.

- Thoria filaments: Thoria is naturally radioactive and should not be ingested. When handling wear rubber gloves. Old filament assemblies can, however, be treated in general waste.
- Rotary pump oil: Must not be ingested. It is a possible irritant. Keep away from skin, especially open wounds. Waste oil is to be disposed of in an environmentally responsible manner.
- Gas lines: It is the user's responsibility to check the integrity of gas lines taking samples to the instrument. Special care must be taken when dealing with toxic gases.

Chapter 3
Site Requirements

Note: Further information on many of the items covered in this chapter can be found elsewhere in this guide or on the installation drawing for the particular instrument configuration.

Physical Installation

The instrument comprises a single bench top enclosure with external (floor standing) mechanical rotary vacuum pump.

Caution! The analyser weighs approximately **100kg**
Any bench or support frame intended for use with this instrument must be rated accordingly.
The instrument should be lifted from the shipping pallet and positioned, using a fork lift fitted with 1” deep forks.

Note:

- The basic enclosure dimensions are;
  width 80 cm
  height 50 cm
  depth 56 cm

- All gas line connections, sample and calibration, are located on the left hand face as viewed from the front.
- Access for service is from the front once cosmetic and inner covers have been removed. Placing the instrument in any other orientation may impede access.
  Leave at least 5cm behind the instrument so as not to obstruct the ventilation grille.
- Consult the instrument general assembly drawing for more detailed information on space and access requirements.

Computer and Communications

A computer specification can be requested from Thermo Fisher Scientific if required. Most modern PCs are suitable, but they should ideally incorporate at least one serial port. This functionality can be provided through a USB-to-serial converter.

Contact the factory regarding compatibility with different versions of Microsoft® Windows® operating system.

If a permanent PC connection is planned, the most common configuration, physical space, and a mains power supply are required. The PC can be local to the instrument or remote depending on requirements and the area classification of the instrument location.

The PC is connected to the instrument via a serial link connected to the HOST port. User connection panel is on the left hand side of the instrument. Other serial connections, such as link to DCS, are also made here.

Environment

General

Note: The instrument should be located indoors in a temperature regulated environment. Contact with water sources, chemicals, etc. should also be avoided.

Temperature

The operating temperature range should be between 12°C and 30°C. Sudden step changes in temperature (>5°C) should be avoided.
The instrument is air cooled. The fan/filter intake on the top of the instrument must be unobstructed and clean of particulate build-up to ensure adequate cooling within this temperature range.

Humidity

Relative humidity should be 90% maximum, non-condensing.

Vibration

The vibration frequency and amplitude should be within the limits shown in the following table.

Table 3–1. Vibration Limits

Frequency	Maximum Amplitude
≤ 20 Hz	5 mm
≤ 60 Hz	1 mm
≤ 200 Hz	0.05 mm
> 200 Hz	0.01 mm

Where necessary, the instrument should be isolated from sources of vibration by use of suitable mountings. Transmission of vibration along sample lined and conduit should also be considered.

Particulate Contamination

Note:

- Where possible, the instrument should be positioned in an environment free of particulate contamination.
- This instrument is air cooled and attention should be given to the condition of the fan filter which should be inspected and cleaned on a regular basis to ensure effective cooling is maintained. Frequency of inspection will depend on the level of dust in the immediate environment.

Altitude

Altitude must be less than 2000 m.

Inlet Gas Connections

Calibration and sample gas lines to single or multi-point inlet options are connected with Swagelok compression fittings (1/4” or 6 mm, as defined in the order). All fittings are supplied with PTFE plugs fitted (ferrules packed separately) to provide a seal during transit. These can be used as seals for unused ports, but it is recommended that for long term use, particularly at elevated temperatures, Swagelok plugs are substituted.

Calibration Gases

A calibration gas panel (six way manifold) is used in conjunction with some inlet types to control calibration gas flow. A line pressure of 1 bar(g) (adjustable for fine tuning) will give the required flow rate, typically 150 cm^3/min

For some inlets (e.g. 16 way RMS and single point solenoid) flow is monitored internally by the system. If the inlet configuration selected does not have this feature external flow monitoring will be required.

Calibration gases lines connect to the panel via 1/4" or 6 mm compression fittings, as defined in the order.

Calibration gases required for a given application are specified with the quotation or application document.
Contact the factory if this document is not available.

Note: Depending on complexity and constituents, some calibration gas mixtures may have long purchase lead times.

Calibration gases should be of the highest quality with a certificate of analysis from the manufacturer giving a concentration tolerance for each component. Helium gas (for instrument background measurement) should be at least 99.995% pure.

Gas consumption will depend on specified use and gas type. Each line selected during calibration will be required to flow for between 15 seconds and 5 minutes, depending on factors such as gas type, line length, line integrity, etc.

In addition to calibration cycles, some gases may also be designated for instrument test or check functions. The frequency and duration of these additional duties need to be accounted for in establishing likely consumption rates and to schedule the ordering of replacement cylinders.

Note: The accuracy of analytical data produced by the instrument is directly related to the accuracy, quality, and stability of the calibration gases. In addition to the accuracy of certification, attention should also be given to factors such as shelf life and storage conditions (temperatures and bottle orientation, for example) that could result in effects such as condensation, stratification, or degradation, all of which could influence the quality of calibration. Seek advice from your gas supplier.

Double stage regulators with 0-2 bar(g) (adjustable) output stages and 40 bar (maximum) outlet pressure relief should be used with the gas cylinders (pressure relief is not required if the cylinder pressure is less than 40 bar). It is the user's responsibility to provide calibration gas cylinders, regulators, and pipe work from each cylinder to the calibration panel and safe venting of pressure relief valves.

Caution! Regulators and line materials must be compatible with the gases used. The use of good quality regulators ensures reliable calibration gas flow, especially for unattended operations.

Sample Gases

It is the user's responsibility to run and install sample lines to the instrument's inlet assembly. The line material must be compatible with the sample gas stream composition and operating temperatures.

Caution!

- Each sample stream should be reviewed with regard to sample conditioning / filtering.
- As a minimum, each line should be filtered to prevent particulate contamination.
- Even where dust levels in the sample gas are very low, it is recommended that a $2\ \mu m$ particulate filter with a capacity appropriate to the dust loading of the sample gas be fitted.
- If significant quantities of dust particles $< 2\ \mu m$ are present, a smaller pore size filter should be used.
- Precautions must be taken to prevent liquid in any form (liquid condensate, foam, aerosols, etc.) from entering the inlet system.
  Suitable traps and / or heated lines, avoiding cold spots, should be used.

Sample flow must be regulated.
Typical flow rate is:

- 0.5 l/min

Flows in the range 0.1 to 10.0 l/min can exist in certain configurations / installations. Consult the factory or the manual for the specific inlet type fitted.

Exhaust Gas Connections

There are two gas outlets from the instrument that should exhaust outside the immediate work area either to atmosphere or a suitable vent / extraction system.

- Rotary pump exhaust: 1/2” hose connection.
- Inlet system exhaust:
  Connection specific to the inlet type. 0.5" or 12mm compression for the 16 way RMS.
  1/4” or 6 mm for all other inlet types.

Note: The following should be taken into account:

- Exhaust temperature can reach 120°C depending on inlet type.
  An exhaust line compatible with the gas composition and temperatures must be used.
- To avoid condensate returning to the inlet assembly, it may be necessary to heat the line or fit a suitable liquid trap. Fitting an exhaust line, angled downwards over its full length, will aid in drainage of any condensate and in many instances may be sufficient.
- Users are responsible for assessing the composition and physical characteristics of sample streams, the risks of condensation, and taking appropriate measures to protect the inlet system.

Caution!

- An inert gas feed can be added to the inlet exhaust stream to dilute corrosive, reactive, condensable, or hazardous (e.g. explosive) gas mixtures.
- **In some instances explosive gas mixtures may form by mixing different process gases in the common exhaust of the inlet system.**
- In other cases, mixed streams can react together to form condensable or polymeric materials.
- The addition of an inert gas stream can reduce or avoid the need for the use of special materials, heat tracing of the exhaust line, etc.
- Inert gas flow requirements will be application specific.
- **Assessing the suitability of an inert gas addition to the exhaust stream, its provision, and monitoring are the responsibility of the user**.

Caution! Interconnection of the Rotary and Inlet exhaust lines should be avoided if at all possible. The rotary pump exhaust contains oil mist that could result in serious contamination if allowed to enter the RMS inlet and subsequently the mass spectrometer. If interconnection is unavoidable, it should be positioned as far downstream as possible and a guaranteed positive gas flow provided to the RMS exhaust to reduce the chances of back flow (see above statements on inert gas flow).

Refer to the chapter relating to the specific inlet system for further details on the exhaust requirements.

It is the user's responsibility to supply all the necessary pipe work to connect the two exhaust outlets to a suitable location.

Power

This section provides details on the power requirements for the instrument.

This equipment must be earthed

Voltage and Frequency

This equipment is rated at 1.5kVA

A single-phase power supply with earth is required;
Either
220/240 V 50/60Hz
or
110-115/120 V 50/60Hz

Caution!
The voltage range of the equipment is preconfigured.
Refer to the ratings label on the right rear of the instrument and ensure the correct voltage is available prior to connecting the supply chord and switching on.

Requirements

For systems with an external RV8 rotary pump and an RMS inlet, the normal average power consumptions is 1.5 kVA. A typical start-up current profile (from cold) is shown in Figure 3–1.

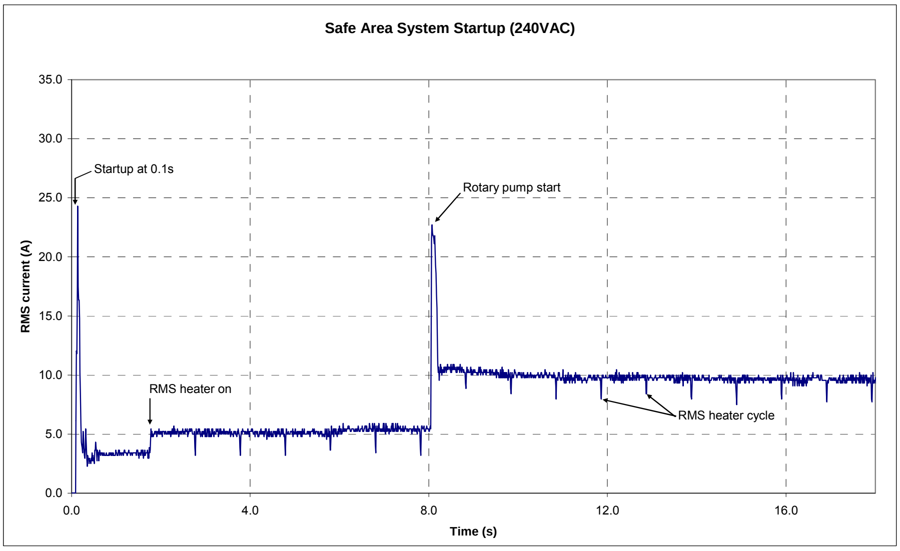


Figure 3–1. Typical startup current profile for safe area system (240 Vac)

Over-current Protection

The incoming power supply must include an over-current protection device rated at a maximum of 20 A continuous operation, but should be capable of withstanding the appropriate startup period as shown above without triggering. This normally requires the use of a slow-blow fuse or a circuit breaker specified for motor starting.

Power Quality

The system power should be free from supply ‘drop-out’ over 1/2 cycle or more. If the local power supply is unreliable or noisy, then power conditioning (e.g. uninterruptible power supply, constant voltage transformer) of appropriate capacity should be installed.

Power Connection

Refer to Chapter 5: Installation & Interconnection for details.

Other Services

For all services, the connection type (fractional or metric) will default to that supplied for the inlet system.

Turbo Vent IN

When the vacuum system shuts down, it automatically vents so that the internal pressure rises up to ambient. Atmospheric air contains water vapour which, if present in the vacuum chamber, will extend the time to pump down when the vacuum system is restarted.

A Turbo vent facility is provided to allow the system to be vented with a clean dry gas (most commonly nitrogen), which significantly reduces the time required for the subsequent pump down.

Connection is 1/4” / 6 mm compression. Pressure should be in the range 0.5 to 2 bar(g), minimal flow requirement. An onboard filter / regulator reduces the delivered pressure to 0.2 bar(g). Nitrogen supply can be from a cylinder or a plant nitrogen line.

Options

Pump Purge Gas

Some, or all, of the following pump purge gas options may have been ordered, depending upon the instrument application. These are normally only necessary for applications involving condensable, corrosive, or toxic sample gases. The purge gas is normally nitrogen, though in some cases instrument air may be acceptable. All connections are 1/4” or 6 mm compression.

Rotary Oil Box Purge

A rotary oil box purge may be required to ensure a positive flow of purge gas through the rotary pump oil box to dilute and sweep the gases out into the rotary pump exhaust line. The purge gas flow should be controllable up to 1 l/min (by means of an external control valve and flow meter).

Rotary Pump Ballast

A rotary pump ballast purge primarily prevents formation of liquids from any potentially condensable vapours being pumped, but it also provides a general purge for the pump oil. The purge gas flow should be controllable up to 5 l/min (by means of an external control valve and flow meter).

Turbo Pump Bearing Purge

A turbo pump bearing purge is used to protect the turbo pump bearing when corrosive gases are being pumped. The flow requirement is very low and is internally regulated. Nitrogen is the preferred purge gas in this case.

External I/O

Various options are available to handle I/O that is not available within the main instrument enclosure. Typically, this is for larger quantities of discrete I/O (analogue and / or digital) but may also be used for protocol converters. Such units are generally stand-alone, requiring a mains power supply. The unit would be located remotely from the main instrument connected via a serial link (most commonly RS422).

Separate documentation will be provided for this type of unit.

Chapter 4
Handling & Storage

Receiving the Instrument

Note: Thermo Scientific instruments are shipped by carriers who specialize in the handling of precision equipment. In the unlikely event that equipment is inadvertently damaged in transit, follow the instructions below to protect yourself and your company from any possible loss or liability.

The Instrument will be delivered in a crate and a fork lift or similar will be required to unload and move it.

On receipt, inspect for any obvious damage or evidence of rough handling. In particular pay attention to the shock indicator and tilt labels fitted.

If external damage is apparent:

- Do not refuse shipment. Instead, make a note of any damage on the receiving documents, and leave the instrument in its original packing.
- Request inspection from the carrier within 15 days of delivery (3 days for international), and contact Thermo Fisher Scientific to report the damage.
- Move the shipment to a protected location, preferably the installation site.
- Leave the boxes as complete as possible, and do not unpack or open the boxes without a representative from Thermo Fisher present or express instruction to do so by a Thermo Fisher employee. **Doing so may void your warranty or order.**

If there is no visible damage to the packaging the instrument should be unpacked if installation is imminent (see next section "Unpacking the Instrument"). If it is likely that installation will not take place within 8 weeks of receipt, it is recommended that the instrument be left in its original packaging and stored in a protected environment (see "Storing the Instrument").

Unpacking the Instrument

The instrument should be unpacked following the directions marked on the packaging itself. Should there be any visible signs of damage to the instrument once removed from its shipping packaging, contact Thermo Fisher with details.

It is recommended that any peripherals packed within the crate, or shipped separately, are left sealed in their own packaging until ready for use.

Storing the Instrument

Whether for an extended period or for a period prior to installation, the instrument should be stored in the following manner:

- In a secure location.
- In original packaging.
- Dry (protected from standing and falling water, etc.).
- Between 5°C and 40°C (40°F and 105°F).
- 90% maximum humidity, non-condensing.
- All packages together.

Chapter 5
Installation & Interconnection

Introduction

This section covers the work to be carried out before startup of the instrument. Before continuing, review "Handling & Storage" (Chapter 4).

All aspects addressed in Chapter 3: Site Requirements should be in place or provided for before continuing.

Positioning the Instrument

The site should meet the environmental specifications detailed in Chapter 3: Site Requirements.

The instrument is intended to be positioned on a flat surface at a nominal height of 70cm – 90cm so as to allow service access from the front and to the inlet assembly on the left-hand side.

A separate mechanical rotary pump is required for the vacuum system and should normally be floor standing.

Caution!
The instrument should be positioned to give un-obstructive access to the mains input connector and isolation switch on the left hand side.
A gap of at least 5cm should be maintained between the rear of the instrument and any vertical surfaces, such as a wall or bench panelling, so as not to obstruct the rear air vent and vacuum line connections to the Rotary pump.
If the Rotary pump is positioned directly below the instrument the distance between the underside of the bench and the surface the pump stands on needs to be a minimum of 50cm so as not to impede the vacuum lines between the pump and the instrument.
Similarly a minimum 50cm clearance should be left above the instrument to facilitate removal of covers, access for service and to ensure adequate fresh air flow to the cooling fan intakes.

Rotary Pump Connection

The pumping lines for the external rotary pump are on the rear panel directly below the air vent.
In the photograph below these are shown orientated to run the vacuum lines to the right hand side of the instrument when viewed **from the front.**

On the instrument the Inlet vacuum line is connected to the right hand fitting and the Turbomolecular pump is the left hand fitting as viewed **from the rear.**

The orientation of these ports can be adjusted so the vacuum lines can be conveniently routed depending on where the rotary pump has been positioned.
Lightly loosen the outer 17mm nut, re-orientate the fitting to the required position and secure by re-tightening the nut.

At the rotary pump assembly the inlet line is connected to the vertical through port and the Turbomolecular line to the horizontal side arm. Tube fittings are compression seals and do NOT require excessive tightening. Finger tight and an additional quarter turn should be sufficient to form a seal.

OVERTIGHTENING WILL RESULT IN DAMAGE TO THE SEALS AND POTENTIALLY CAUSE LEAKS.

Cable Connections

Power Connection

Consult Chapter 3: Site Requirements for details of the power supply requirements.
A 16A power cord assembly is provided for use exclusively with the instrument. In the event that it becomes necessary to renew this item, the replacement must be of at least the same specification. The power supply and site outlet must be compatible with the supply current and comply with relevant local electrical regulations.

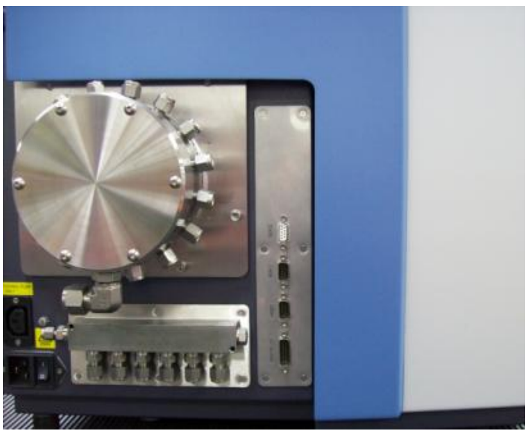
A shrouded IEC outlet socket rated at 10A, 250V AC, supplies the external rotary pump. It must not be used for any other purpose. The power cord (supplied) is rated for this purpose. Any replacement cord must be of at least the same specification.

Signal Connection

All user connection are made to sockets in a panel to the LH side of the instrument..

Cabling should be kept as short as possible from cable entry to connection point.

There are 3 9 way D type connectors designated HOST, SER-1 and SER-0 and a single 15 way D for digital I/O



Gas Connections

Sample gas, calibration gas, and exhaust gas connections should be made as specified in Chapter 3: Site Requirements.

Pipe runs should not restrict service access to the instrument.

All gas lines should be leak checked after installation.

Where heat tracing of lines is required, this should be fully tested.

Other Services

Certain options may require additional services (e.g. compressed air, nitrogen). These are specified in Chapter 3: Site Requirements.

As stated above, care should be taken with tubing / pipe runs to avoid restricting service access.

Computer System

A separate mains power feed is required for the PC.

The Prima BT system is connected to a Windows-based computer running the Thermo Scientific GasWorks software. The HOST serial port of the instrument connects to a serial port on the computer. If necessary, a USB-to-serial adapter can be provided.

Chapter 6
Commissioning

General

Initial startup of the instrument will normally be carried out by a trained engineer and is typically included as part of the purchase contract. The full startup and configuration procedure is detailed and can be complex for some applications. It is therefore recommended that initial startup should not be attempted by the user, unless they have significant experience with the instrument on the specific application.

When commissioning is requested, a confirmation of site readiness will be required. This will normally be in the form of the Site Requirements Checklist, which is to be returned to the office organizing the work. The user must complete this form to confirm that the installation site is ready before the engineer will be allowed to travel to site.

Procedure

The procedure followed by the commissioning engineer will include the following steps.

Note: User and or site personnel may be required for some of these steps. The schedule, content, and availability of user and or site personnel should be agreed before work commences.

1. Verification of Site

   A brief inspection will be carried out to ensure that the site has been prepared and all site services are available in accordance with the site requirements detailed in Chapter 3.

1. Verification of Hardware

   An inspection of the hardware will be carried out to verify that no visible transit damage has occurred. The delivered hardware will also be checked with the user to verify that it is consistent with the order.

2. Startup

   The instrument will be powered up. A number of tests will be carried out to verify that the instrument is working correctly. These tests are the same for all instruments, regardless of application.

3. Application Configuration

   The software database will be configured for the application and the specific installation. This includes:

   a. Setting up the analysis method for the gases in the process mixture(s).

   b. Configuring the sample streams.

   c. Setting up the calibration method(s).

4. Application Verification

   The above configuration will be functionally tested by running a full calibration, followed by analysis of process gas mixtures. Performance testing will be carried out by analyzing a calibration gas (typically the “mixed gas”, which will generally be similar to the process gas in composition) over a period of time. Other standard gas mixtures may also be used if they fall within the concentration limits defined in the performance specification. In some cases, achieving a particular performance during testing on a specific mixture or mixtures will be part of an agreed acceptance criterion.

5. Communications and I/O (where applicable)

   Communications will be configured and tested. Wherever possible, initial testing will be to a PC running a test program for the protocol in question (e.g. the installation engineer’s laptop PC or the GasWorks PC).

   Any discrete I/O will also be configured and tested as above.

6. Training

   Approximately two days of the commissioning will be allocated to operator training. The schedule, content, and availability of personnel should be agreed before work begins.

   More detailed training courses are available and should be considered when some experience has been gained with the instrument. For further details and a quotation, contact the local Thermo Fisher Scientific office or representative.

7. Acceptance

   At the end of a successful startup, unless the contract indicates otherwise, a formal acceptance of the system will be required. A standard Thermo Fisher Scientific acceptance document will be used.

Chapter 7
Startup & Shutdown

Startup Procedure

These procedures assume that the system has been installed and commissioned as earlier in this manual.

The startup procedure for the instrument is as follows:

1. Ensure that the rotary pump is filled with oil to a point just below the upper marker on the oil level window. Ensure that the pump outlet is not obstructed.

2. Apply power to the instrument and switch the Mains ON at the IEC connector:

3. At this point, both the rotary and the turbo pumps will operate.

4. Switch on the PC. Double-click on the GasWorks icon. GasWorks may have been configured to automatically connect to the instrument CPU (this will have been set by checking **connect automatically on startup** under **GasWorks Session** > **Preferences** > **Instrument**).

   If it has not been configured to connect automatically, click on the **Connect** icon ( ) to connect to the instrument CPU (Figure 7–1).

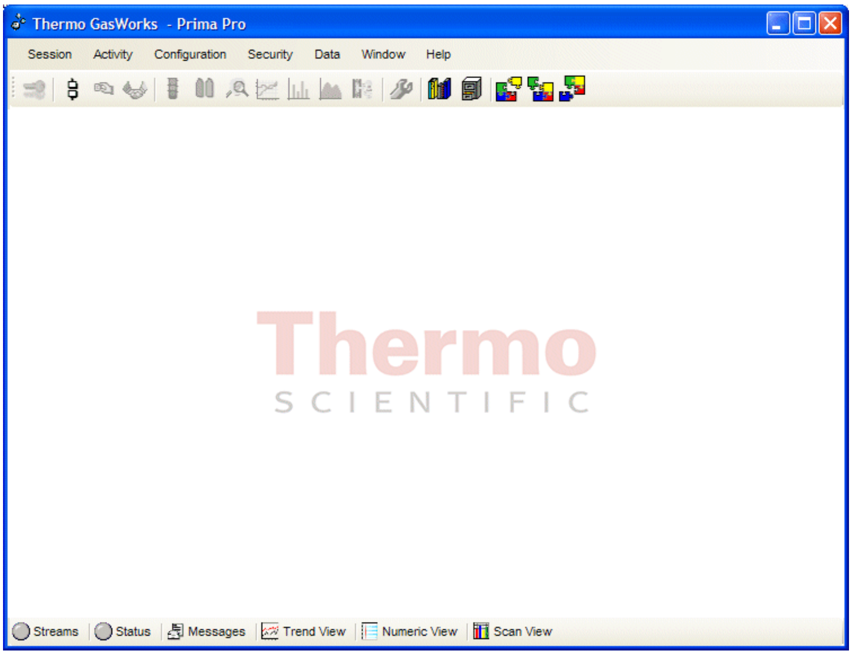


Figure 7–1. Connect to GasWorks

The instrument status light in the bottom toolbar will indicate when the instrument vacuum is at the correct level and filament emission is achieved. The instrument status light is an acknowledgeable alarm indicator. It should be GREEN, indicating a “go” condition, where all the measured instrument parameters are within their configured alarm thresholds. If any measured instrument parameter is outside its configured alarm threshold, then the instrument status light will be shown in RED, indicating a “no-go” condition. If the instrument status light is shown UNCOLOURED (GREY), either a complete cycle of instrument parameter measurements has not been completed or the host PC has not connected to the instrument CPU.

If the pressure is not below the vacuum gauge trip level then the emission will not switch on and the ion energy will remain at zero.

Note that a single mouse click on the Instrument Status toolbar button will open the instrument status window (Figure 7–2), which shows the alarm status of each of the instrument parameters.

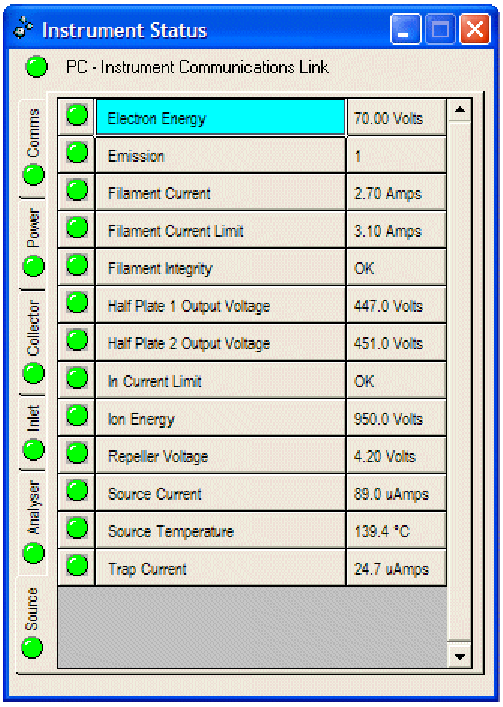


Figure 7–2. Instrument Status window

A click on a particular parameter status light will show the current value, alarm limits, and alarm acknowledgement button (Figure 7–3). If all parameters are within limits but the status light is red, clicking on the **Acknowledge All** button will change the instrument status light to green.

Wait until all parameters are within limits (the status light is green), before starting scheduled analysis.

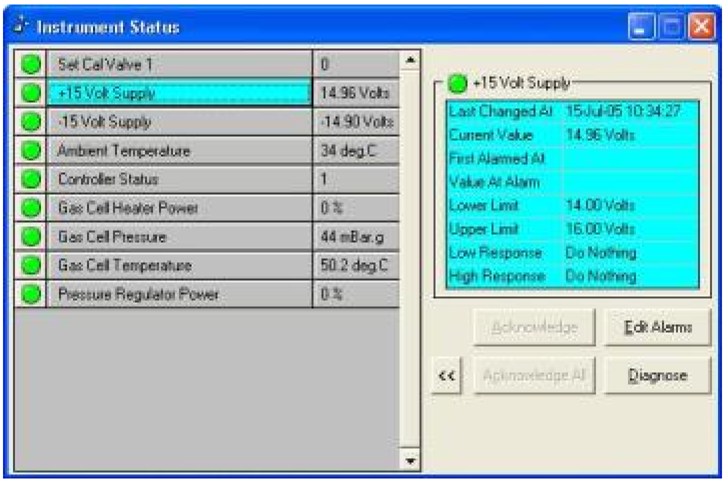


Figure 7–3. Parameter status window

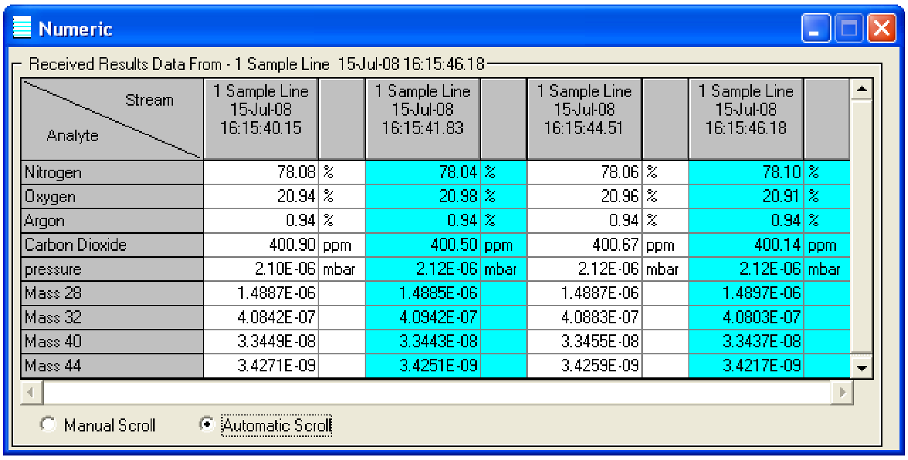
Vacuum Gauge Trip Level

Before the filament (and ion energy) can be switched on, the pressure in the vacuum chamber must be less than the vacuum gauge high pressure trip level (normally set to 3.2 x 10^{-5} mbar) and also higher than the vacuum gauge low pressure trip level (1 x 10^{-7} mbar).

On initial switching it is quite normal for the filament to outgas slightly as it heats up. This will result in an increase in the measured pressure of the vacuum system. The trip logic therefore has a limited amount of hysteresis to accommodate the pressure rise. To prevent the trip circuit from immediately switching the filament off again, the "off trip" is set higher than the "on trip" so a pressure reading above the set point is normal.

Starting Scheduled Analysis

Click on the **Traffic Lights** button ( ) to start a scheduled analysis. Each enabled stream will be selected according to its sequence, the concentrations measured, and output via data communications to an I/O unit or other host computer. If the numeric display window is selected, the concentrations will be displayed on the host PC monitor (Figure 7–4). Similarly, if a trend display has been configured and is selected, concentration trends with time can be viewed (Figure 7–5).



Stream / Analyte	1 Sample Line 15-Jul-08 16:15:40.15		1 Sample Line 15-Jul-08 16:15:41.83		1 Sample Line 15-Jul-08 16:15:44.51		1 Sample Line 15-Jul-08 16:15:46.18	
Nitrogen	78.08	%	78.04	%	78.06	%	78.10	%
Oxygen	20.94	%	20.98	%	20.96	%	20.91	%
Argon	0.94	%	0.94	%	0.94	%	0.94	%
Carbon Dioxide	400.90	ppm	400.50	ppm	400.67	ppm	400.14	ppm
pressure	2.10E-06	mbar	2.12E-06	mbar	2.12E-06	mbar	2.12E-06	mbar
Mass 28	1.4887E-06		1.4885E-06		1.4887E-06		1.4897E-06	
Mass 32	4.0842E-07		4.0942E-07		4.0883E-07		4.0803E-07	
Mass 40	3.3449E-08		3.3443E-08		3.3455E-08		3.3437E-08	
Mass 44	3.4271E-09		3.4251E-09		3.4259E-09		3.4217E-09	

Figure 7–4. Numeric trend window

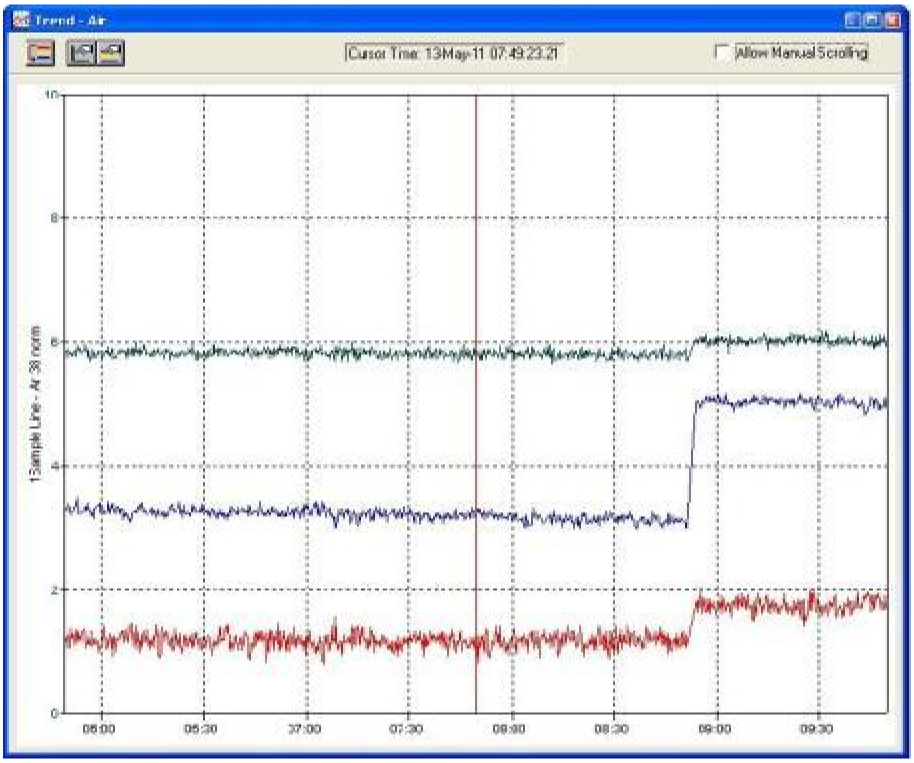


Figure 7–5. Trend display

Checking Instrument Tuning and Mass Alignment

For a detailed description of instrument tuning parameters and mass alignment please refer to Chapter 8: Status & Tuning Parameters.

Note: Control Centre should not be opened whilst Schedule is running. Before running Control Centre, ensure Schedule is stopped.

A single click on the **Control Centre** button () will open the Control Centre window (Figure 7–6), which has the following tabs:

- Emission
- Peak shape
- Sensitivity
- Detectors
- Mass alignment
- Heaters

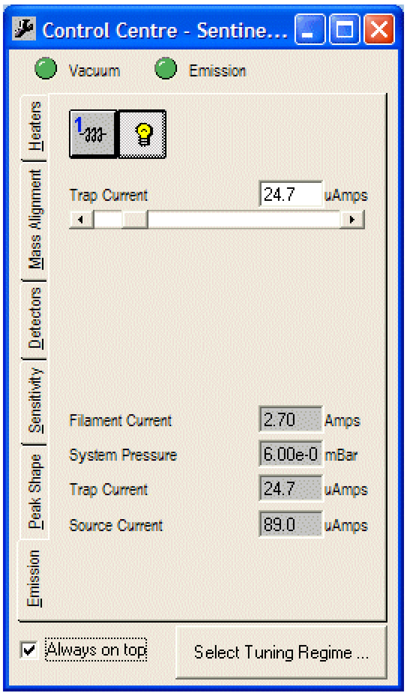


Figure 7–6. Control Centre window

Each tab groups a set of adjustable instrument parameters and associated system readings.

Each adjustable parameter is displayed with a value and slider bar.

Each parameter can be adjusted in one of three ways:

- By typing in the required setting and pressing Enter.
- By clicking on either the left or right arrow to change to a lower or higher setting respectively.
- By dragging the position indicator along the adjustment bar.

Several tuning regimes can be configured in which different “sets” of tuning settings can be grouped. In practice there is a “default” tuning regime for all masses above a certain cut-off in the range of 5 to 19 and a “low resolution “tuning regime for all masses below this cut-off.

Emission

The first tab is by default Emission. This tab allows the following:

- Switching the filament on and off.
- Changing the selection of either filament 1 or filament 2.
- Adjusting the trap current.

Peak Shape

The Peak Shape tab displays the ion intensity over a defined mass window (Figure 7–7).

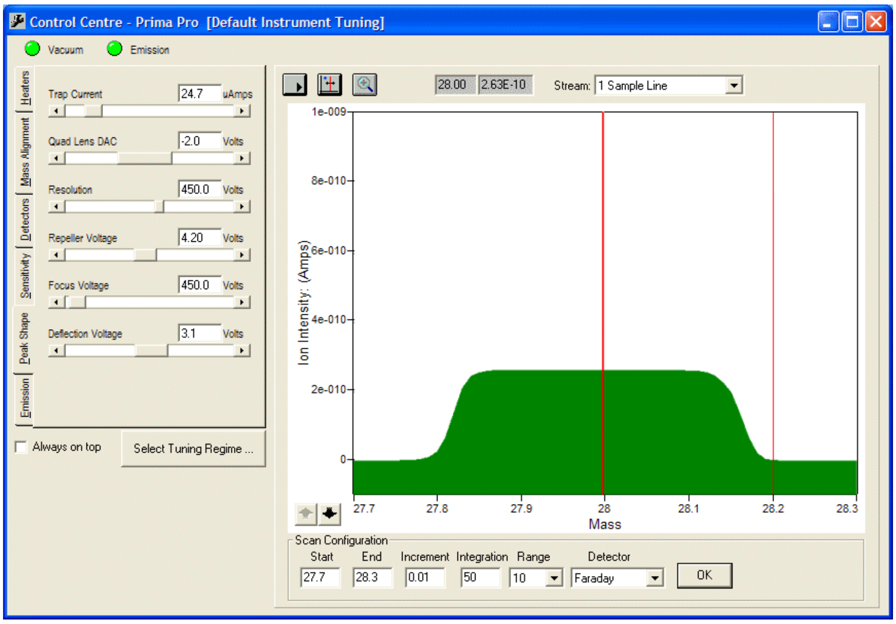


Figure 7–7. Peak Shape tab

Once a suitable peak has been located, the peak shape (height and width, extent of flat top, and steepness of peak sides) may be optimized by adjusting the following parameters:

- Trap Current
- Quadrupole Lens
- Resolution
- Repeller Voltage
- Focus Voltage
- Deflection Voltage

Sensitivity

The effect on ion intensity at a selected mass (default mass 28.0) of changing the repeller, focus, deflection and resolution voltages can be viewed on the Sensitivity tab (Figure 7–8). Each parameter can be selected in turn from the Function drop down list, and a graph of peak intensity versus voltage produced. A cursor can be selected, set at the optimum position on the graph, and the associated tuning parameter setting stored using the **Set DAC** button.

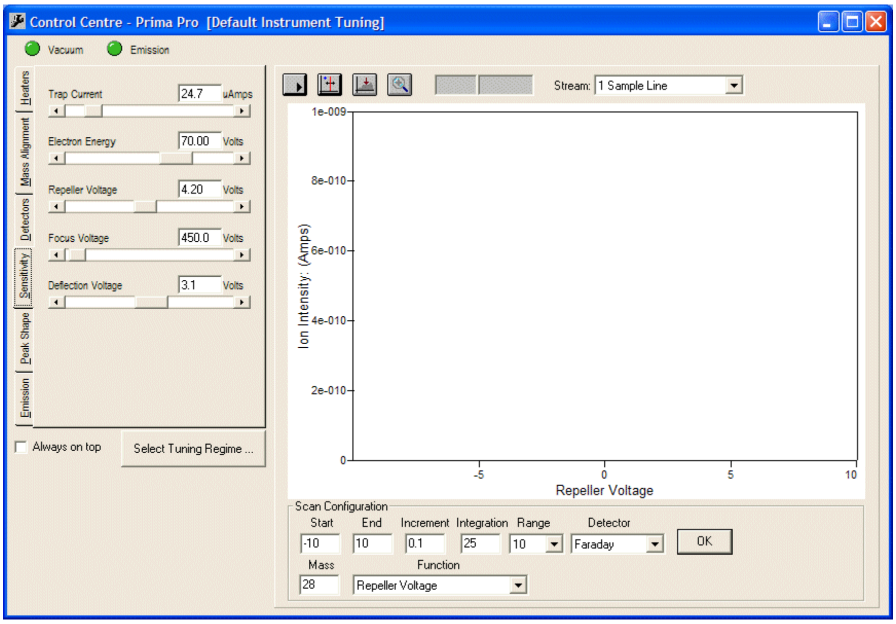


Figure 7–8. Sensitivity tab

Detectors

The Detectors tab (Figure 7–8) is only available for dual detector systems with both Faraday and Secondary Emission Multiplier (SEM) detectors.

As both detectors are off-axis, a deflection voltage is applied to divert the ion beam into the selected detector. Deflector voltage settings for both detectors are optimized on this tab.

The SEM voltage can be adjusted (i.e. calibrated) to set a specified gain (typically 103 or 104).

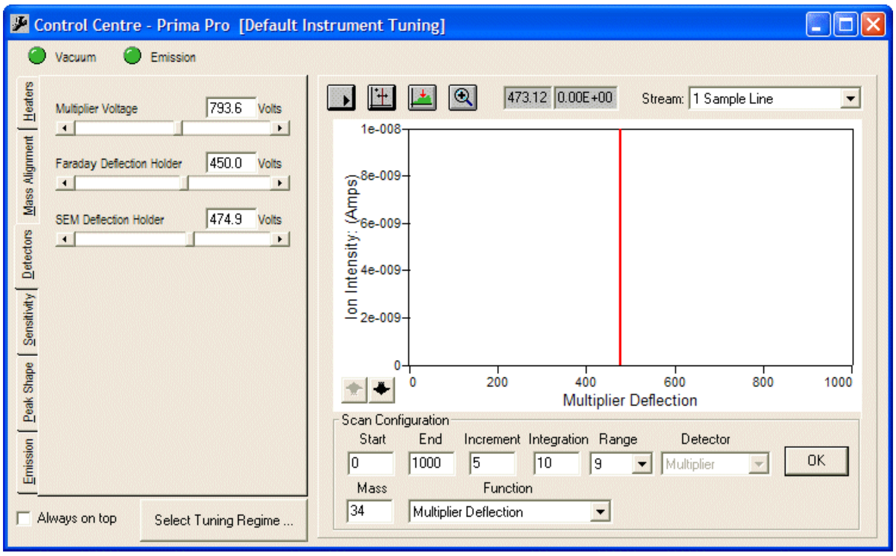


Figure 7–9. Detectors tab

Mass Alignment

The Mass Alignment tab (Figure 7–10) is used to display peaks over a defined mass range in a similar way to that used on the Peak Shape tab. In this instance, a cursor is used to assign mass numbers to identified peaks. Typically, two or three masses need to be 'locked' over the mass range of interest.

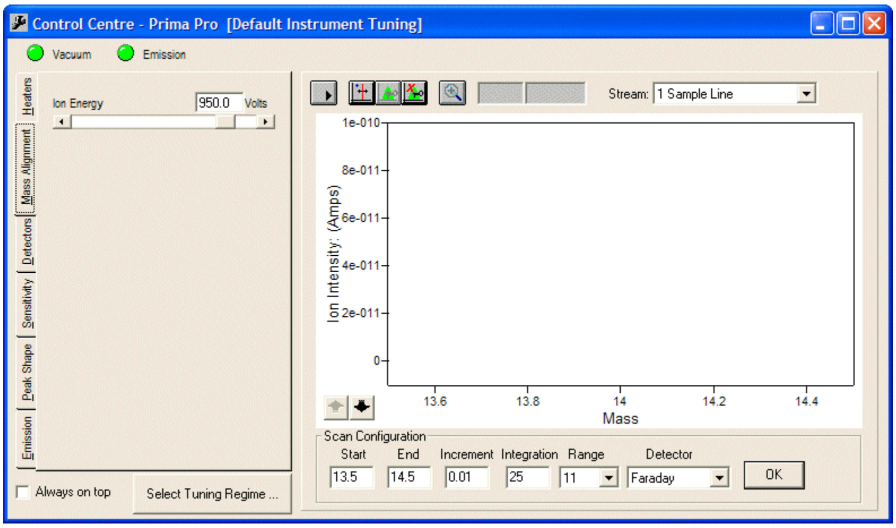


Figure 7–10. Mass Alignment tab

The Prima BT mass position is related to magnetic field by the following equation

$$\text{Mass} = k_m \times \text{field}^2 + c_m$$

where k_m is a constant of proportionality and c_m is an offset, with both of these constants having a slight mass dependency. Values of km and cm are determined for adjacent pairs of locked masses; at masses beyond the locked mass range, the values of km and cm are taken from the closest locked mass pair. If an incorrect mass assignment is made the mass table can be deleted and the process repeated.

Since the Prima BT instrument is a magnetic sector mass spectrometer, which produces a focused ion beam at the detector, the peak shape obtained is 'flat-topped', i.e. uniform response is observed over a finite mass width, e.g. 0.3 amu at mass 28. Provided the measurement taken at the mass of interest is on the peak's flat top, high precision analysis will be observed. If masses are aligned within the central 2/3 of the flat top region, this is normally sufficient to guard against any drift in the mass scale.

Heaters

The Heaters tab (Figure 7–11) allows for adjustment and setting of the source temperature, inlet probe heater power, RMS body temperature, and RMS sample tube temperature.

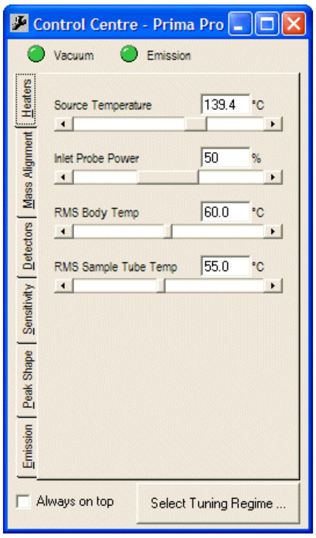


Figure 7–11. Heaters tab

Checking Instrument Status Parameters

Open the Instrument Status window (see Figure 7–2). The tabs and the readings displayed under each tab are listed below.

Analyser tab

- Cabinet temperature: Within the range 10°C to 35°C.
- Electronics temperature: Within the range 10°C to 45°C.
- System pressure: Typically 5 x 10^{-6} mbar, but application dependent. Refer to installation records for configured system pressure.

Source tab

- Ion energy: Approximately 1000 V (for 150 amu mass range) or 800 V (for 200 amu mass range).
- Electron energy: Typically set to 70 V.
- Filament current: Depends on filament type:
  - 2.4-2.9 A for a Thoria coated iridium filament
  - 1.5-2.5 A for a W/ThO coiled filament
  - 3.0- 3.8 A for a Re filament

Note If filament is close to zero, check that Trap Current is set to at least 20 µA and EE is at least 50 V.

- Trap Current: Within the range 10 µA to 100 µA.
- Source Current: Within 2 to 15 times the value of Trap Current.
- Repeller Voltage: Normally within the range of 0 V to 10 V.

Power Tab

- Magnet Current: With 1000 V ion energy for any mass, the magnet current is given approximately by 2.3 x $(\text{mass}/28)^{0.5}$.

Shutdown

Introduction

The shutdown procedure will vary slightly depending on application and on what the system is doing at the time of shutdown.

In general, it is better to leave the instrument running rather than repeatedly shutting down for periods of inactivity. This is true even if the instrument will be unused for several weeks. However, shutdowns will be required for routine maintenance, etc.

The emergency shutdown procedure (later in this chapter) should also be followed in the event of a power failure.

Shutdown Procedure

The following steps should be carried out in sequence.

1. Shut off all sample gas flows.

   When the system shuts down, all the heaters will go off and gas handling parts of the system will begin to cool. Where there is a risk of residual sample gases condensing, it is recommended that the inlet system be purged with a suitable inert gas such as nitrogen before power is removed.

2. Switch OFF the filament on the Emission tab of the Control Centre software.

3. Disconnect GasWorks software with the  icon.

4. Switch OFF the Mains Electrical ISOLATOR (see “Startup” earlier in this chapter) to de-energize the system electronics. Isolate power externally to fully de-energize the system.

Following is the emergency shutdown procedure.

Emergency Shutdown

1. Switch off power at the Mains ON/OFF switch. (see "Startup" earlier in this chapter).

2. The system heaters will now be de-energized and gas handling parts of the system will be cooling down.

   All sample gas flows should be switched off as soon as possible. Where there is a risk that residual sample gases will begin to condense, the inlet system should be purged with a suitable inert gas such as nitrogen as soon as possible.

Chapter 8
Status & Tuning Parameters

Instrument Status

Instrument status and tuning parameter values may be checked by clicking on the Instrument Status button in the GasWorks software. Parameters are grouped under separate tabs according to the following headings from bottom to top:

- Source
- Analyser
- Inlet
- Collector
- Power
- Comms

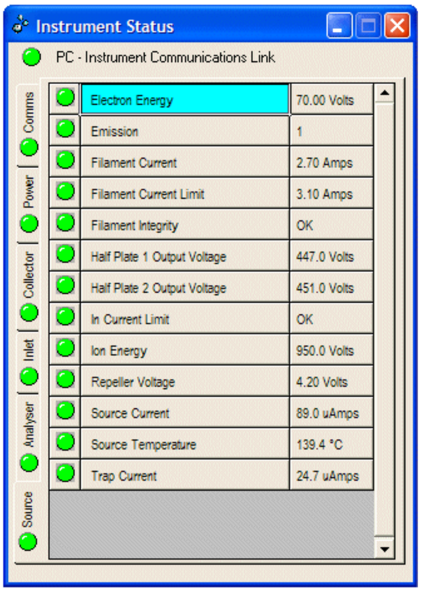


Figure 8–1. Instrument Status window

Source Tab

Electron Energy

This supply provides the energy to accelerate electrons (produced by the filament) through to the ion source. Electrons subsequently interact with molecules of the sample gases to create ions, a process referred to as ionization.

The electron energy can be adjusted through on the Sensitivity tab of Control Centre; however it is typically set between 40 and 70 V.

The main effect of varying the Electron Energy parameter is to vary the types and ratios of ions, the “fragmentation pattern” produced by electron impact of the sample molecules.

In addition to dislodging one or more electrons from the sample molecule (to produce singly or multiply charged positive ions), the electron impact can also break molecular bonds to generate smaller fragments. Lower electron energy tends to give a lower abundance of fragment and multiply charged ions, which in certain applications can help reduce the extent of component overlap and therefore analytical complexity.

Emission

The Emission status is a “composite” status channel that shows when all related parameters are within limits.

Filament Current

When the emission is switched on, the current through the filament will be approximately 2.4 to 2.8 A when fitted with Thoria filaments. When fitted with non standard Tungsten filaments, the current will be closer to 3.7 A.
If the Filament Current value is close to zero, access Control Centre, and check that the Trap Current setting (from Emission tab) is at least 10 µA and the electron energy setting is at least 40 V.

Filament Current Limit

Filament Current Limit is provided to protect the filament from being overdriven and is preset for the filament type in the ASU.

The current limits for common filaments are shown in the table below.

Table 8–1. Current limits for common filaments

Filament	Current Limit (Amps)
Thoriated iridium	3.1
Tungsten / ThO alloy	3.0

If the limit is set too low, then the filament may take some time to regulate when it is first switched on, or may fail to emit at all.

If the value is too high, then there is a danger that the filament might blow, burn out, due to excess current.

When switched on, filaments will run in limit for a short period until the surface attains temperature and the required level of electron emission is reached. Ideally, current limits are set about 0.4 A higher than the normal running current.

Filament Integrity

Filament integrity will indicate **OK** when the filament circuit is continuous and **Failed** when open circuit, normally indicating a filament failure. If maintenance has just been performed on the instrument check that all cable harnesses have been reconnected before assuming a filament failure.

Filament selection between filaments 1 and 2 is made in Control Centre from the Emission tab.

If the first running filament fails, consideration should be given to scheduling a routine source change. This is particularly important where analytical data is critical and an unscheduled shutdown caused by failure of a second filament could have additional consequences.

Half Plate 1 and 2 Output Voltages

Two 1/2 plates positioned after the source block are used to focus and steer the ion beam. The potential difference between the two plates represents the deflection voltage and can be varied in the range -50 V to +50 V by adjusting the Deflection Voltage setting on the Peak Shape tab in Control Centre.

The mean voltage for the 1/2 plates represents the focus potential and can be varied in the range 0 V to 800 V by adjusting the Focus Voltage on the Peak Shape tab in Control Centre.

In Current Limit

In Current Limit indicates that the filament current is in limit and that emission control is not yet established. This is normal for a few seconds while the filament heats up after switching on.

OK indicates that emission control has been attained and the filament is NOT in current limit.

Ion Energy

This supply provides the energy to accelerate the ionized sample molecules out of the source and into the magnetic field where they will separate according to their respective mass / charge ratio. The Ion Energy value is nominally 1000 V for 150 amu mass range instruments and is set in on the Mass Alignment tab of Control Centre.

A lower voltage setting will achieve a higher mass range as the mass range is inversely proportional to ion energy. See Appendix A: Technical Description: for more detail.

The mass range is therefore about 200 amu at an ion energy setting of 750 V.

Note All the tuning parameters need to be re-optimized and mass scale recalibrated if the ion energy setting is changed.

Repeller Voltage

The ion repeller is an electrode within the source block that ensures ions move towards the ionization chamber exit slit. The voltage applied to this plate affects sensitivity and may influence the peak shape, linearity, and stability. The voltage applied to the ion repeller is usually in the range 2 to 10 V, although it can be adjusted over the range -10 V to +20 V.

Source Current

The Source Current represents the electrons, emitted by the filament, that fail to traverse the source and are therefore not collected at the trap electrode. The proportion of electrons that traverse the source and reach the trap electrode normally result in a source current that is 2 to 5 times greater than that of the trap current setting.

A source current ten times or greater than that of the trap suggests that the filament may either be poorly aligned, have become distorted, or the ion source has become contaminated. All of which could contribute to a reduction in the number of electrons reaching the trap electrode.

Stability and linearity tend to be worse at higher source current / trap current ratios.

Source Temperature

The ion source is heated by a cartridge heater in the ion source block, and temperature is monitored by a platinum resistance thermometer (PRT).
Source temperature is normally set within the range of 140°C to 200°C on the Heaters tab of Control Centre but can be application dependant. Limits will be set accordingly.

Trap Current

The trap electrode is used to regulate and control the electron emission through a control loop to the filament current drive. Increasing this parameter will increase sensitivity, but care must be taken to ensure that the biggest peaks can still be measured on the lowest amplifier gain range.

As a guide, mass 28 should not be much set greater than 7×10^{-10} A with air in the inlet. This is usually achieved with trap currents in the range 20 µA to 100 µA.

Analyser Tab

ASU Controller Status

A value of **1** in the ASU (analyser supplies unit) Controller Status field indicates the ASU to be operating within normal limits.

Other status conditions are as follows.

Table 8–1. ASU Controller Status conditions

Status	Value
No Communications	-1
Uninitialised	0
OK	1
Flashover Detected	3
Power Supply Low	4

Enclosure Temperature

The temperature in the enclosure is monitored by a sensor located to the right of the fan tray assembly and should read less than 30°C. This appears as cabinet temperature on the status page.

Electronics Temperature

The temperature in the electronics is monitored by a sensor on the ASU assembly and should read less than 50°C.

Magnet Controller Status

A value of **1** in the Magnet Controller Status field indicates this unit is operating within normal limits. Other status conditions are as follows.

Table 8–2. Magnet Controller Status conditions

Status	Value	
No communications	-1	
Magnet out of control	0	Until first initialisation after power up.
Magnet in control	1	Ready for use.
Magnet out of range	2	Last request was out of range.
Magnet not available	3	Unknown.
Magnet zero required	4	Requires initialisation.
Magnet field busy	5	Settling last requested field.
Magnet frequency busy	6	Integrating VFC.

Status	Value	
Magnet zeroing busy	7	Zeroing the sense coil.
Magnet temp busy	8	Measuring the magnet temperature.
Magnet failure	9	Controller or circuit failure.
Magnet pulse busy	10	Busy driving the magnet.
Magnet field init busy	11	Initialising the system (about 60 s)

Mass Filter

The approximate magnetic field setting, in Tesla, is indicated by the Mass Filter parameter. The value for Mass 28 with 1000 V ion energy is typically 0.4 Tesla.

The relationship between mass and magnetic field and ion energy is given by:

$$\text{Mass} \; \alpha \; (\text{magnetic field})^2 \,/\text{ion energy}$$

A four times increase in magnetic field strength therefore doubles the mass detected.

Pressure Set Point

At pressures above the value set in the Pressure Set Point parameter, power is removed from the filament supply, source heater, SEM supply, and source high voltage supplies to protect the analyser from damage which may result if run at high pressure.

System Pressure

The System Pressure value indicates the pressure as reported by a Penning vacuum gauge head that is mounted on the top right of the vacuum chamber and is a measure of the high vacuum produced by the turbo molecular pump.

The pressure, as measured by the Penning gauge, is a combination of the amount of sample gas entering the vacuum chamber through the capillary / leak assembly and gas that ‘evolves’ from the internal surfaces of the vacuum system. This latter effect is known as out-gassing.

When the inlet is blocked, the vacuum falls to its background level, which is usually around 10^{-7} mbar. When the inlet is open, the operating vacuum should be in the range 1 - 10 x 10^{-6} mbar. This varies slightly from instrument to instrument and on the gas being sampled.

The upper limit of pressure that can be displayed by the Penning gauge is about 1 x 10^{-2} mbar. A stable reading at this level indicates that the pressure may be too high to be measured by the gauge and that there is either a significant system leak or a pumping fault.

Check that the turbo molecular pump is up to speed (indicated as Turbo Revs on the Instrument Status window). Check that the rotary pump is functioning correctly.

Note that it is detrimental to the gauge to leave it running for an extended period (e.g. more than a few hours) while measuring pressures greater than 10^{-3} mbar. Excessive ion burns on the gauge will occur, resulting in reduced gauge accuracy. This can only be recovered by cleaning the gauge electrodes.

Ions are produced as a result of a discharge formed in the high voltage field between the anode and cathode of the gauge head. As the pressure in the system falls, the number of ions in the discharge reduces and this can be directly associated with the pressure of the chamber.

The analyser is protected from damage due to excess pressure by two trip points based on the Penning gauge reading. The Penning gauge gives a voltage output related to the pressure reading according to:

$$p = 10^{U-10.5}$$

Alternatively, the pressure / signal relationship can be expressed as

$$U = 10.5 + \log_{10} P$$

The high pressure trip (see "Pressure Set Point" earlier in this chapter) prevents operation of the emission circuit, ion energy, multiplier, and source heater as described above.

A low pressure trip protects the same devices and is a level indicative of a potential gauge failure.

The threshold values for these two trips are equivalent to:

High Pressure Trip = 3×10^{-5} mbar (approximately)
Low Pressure Trip = 1×10^{-7} mbar (approximately)

Hysteresis is built into the high pressure trip to allow for the pressure rise that occurs when the filament begins to heat and outgas.

System Pressure Trip

System Pressure Trip will indicate **OK** if the system pressure is below the Pressure Set Point and **Fail** if the system pressure is above the Pressure Set Point.

Turbo Motor Current

The Turbo Motor Current is the current drawn by the turbo molecular pump motor.

Turbo Operating Hours

The Turbo Operating Hours field indicates the running hours for the turbo molecular pump. See the pump manufacturer's recommendations on service intervals.

Turbo Speed

Turbo Speed is the speed as a percentage of full speed of rotation. 100% is full speed.

Turbo Speed Interlock

OK is indicated in the Turbo Speed Interlock field when the turbo molecular is at speed and under normal operation.

Vacuum

OK is indicated in the Vacuum field when the system has reached normal operating levels.

Inlet

Pressure Raw

Pressure Raw is the raw output from the inlet differential pressure transducer and flow sensor as a DAC reading. This is fitted to RMS and solenoid assemblies only.

RMS Ambient Temperature

The RMS Ambient Temperature is the temperature recorded on the inlet controller electronics. It is typically approximately 45°C.

RMS Body Temp

The RMS Body Temperature as monitored by thermocouple in the RMS block is normally set in the range of 50 to 120°C depending on the application.

RMS Controller Status

The RMS Controller Status indicates a value of **1** when the unit is operating within normal limits.

RMS Sample Flow Sensor

The RMS Sample Flow Sensor is the calibrated output of the inlet flow sensor (differential pressure transducer). It is displayed in cc per minute.

RMS Flow Zero

The RMS Flow Zero is the zero flow output from the analogue flow sensor.

RMS Position

The RMS Position is the RMS rotary arm position.

Sample Tube Current

The RMS Sample Tube Current is the RMS sample tube heater current.

Sample Tube Temperature

The RMS Sample Tube Temperature is the RMS sample tube heater temperature set point and control.

Collector Tab

Faraday Deflector

The Faraday Deflector parameter is the voltage applied to the deflector electrode that directs the ion beam into the Faraday collector. The Faraday detector is mounted off-axis on both Faraday only and
Faraday / SEM dual detector systems.

Multiplier Deflector

The Multiple Deflector parameter is the voltage applied to the deflector that directs the ion beam into the SEM.

This is an optional micro channel plate (MCP) used to increase the dynamic range of the instrument to permit measurement of species at low (ppm and lower) concentrations.

Multiplier Voltage

The Multiplier Voltage controls the gain of the SEM. Prima BT analysers use an MCP. Varying the voltage output can vary the gain of this detector from x10 to approximately x10000. Typically, the gain is set to 2000 for a standard single MCP SEM and 10000 for a dual MCP SEM.

Negative and Positive Quad Lens Voltages

In conventional sector mass spectrometers compensation for errors in analyser geometry, introduced by magnet fringing field effects, beam aberrations and localized charge effects due to contamination, is achieved, with difficulty, by adjusting the magnet position. The Prima BT analyser overcomes this by fixing the position of the magnet and providing a quadrupole lens (quad lens) to compensate for and adjust the position of the refocused beam image. A complicated impractical series of magnet adjustments is therefore replaced with a single voltage adjustment of the quad lens.

The quad lens DAC voltage can be varied from -25 V to +25 V but, in practice, a variation of a few volts positive or negative is usually sufficient to achieve a symmetrical, flat-topped peak that provides good, long-term precision from the analyser.

Resolution

The Resolution Deflector electrode provides Z plane deflection of the ion beam through one of two available resolving slits. The electrode is part of the detector assembly and is positioned between the quad lens and the detector.

The use of two resolving slits permits the selection of a very wide resolving slit when low mass peaks are being measured. This feature results in the production of broad, flat-topped peaks all the way down to mass 1.

In conventional mass spectrometers, the resolving slit size is chosen so as to resolve adjacent peaks at the high mass end of the scale (peaks get closer together with increasing mass). The disadvantage to having high resolution is that low mass hydrogen and helium peaks become very narrow and therefore prone to drift when these peaks are to be measured.

The high resolution slit is on-axis and for peaks above the switching point, normally greater than mass 5, the voltage for the resolution deflector will be close to zero.

Below the switch point, the voltage will be around 250 V, for systems with 1000 V ion energy, deflecting the beam through the broader low resolution slit.

Power

Amplifier +24V

The Amplifier +24V parameter is the 24 V PDU output for the amplifier.

Analyser Supplies +24V

The Analyser Supplies +24V parameter is the 24 V PDU output analyser electronics (ASU).

Instrument CPU 24V

The Instrument CPU 24V parameter is the 24 V PDU output for the CPU card.

Fan 24V

The Fan 24V parameter is the 24 V power PDU output for the air enclosure fan assembly.

Inlet Probe Current

The Inlet Probe Current parameter is the current through the inlet probe cartridge heater.

Magnet Supply Current

Current passing through the electromagnet coils can be used to give an approximate indication of the current mass number. The approximate current to mass relationship is 4 [I^2] for systems with 1000 V ion energy.

Mass 100 therefore requires roughly 5 A through the coils. This relationship is very approximate. If the mass table has not yet been generated (or has been corrupted), this relationship can be used to check whether or not the magnet controls circuitry is operating.

Magnet Sync AC

The Magnet Sync AC parameter is an AC power synchronization signal to the magnet driver card, which improves precision and reduces mains frequency induced noise.

Power Controller Status

The Power Controller Status parameter indicates a value of **1** when unit is operating within normal limits.

Turbo 24V

The Turbo 24V parameter is the 24 V PDU output for the turbo molecular pump drive electronics.

Vent Valve 24V

The Vent Valve 24V parameter is the 24 V PDU power supply for the turbo molecular pump vent valve.

Communications

Host Comms Quiet

The Host Comms Quiet parameter checks for a break in communications between the host PC and the Prima BT analyser.

User I/O 0 Comms Quiet

If configured (within GasWorks Hardware Configuration), the User I/O 0 Comms Quiet status is checking for a break in communications between the user I/O card port 0 and a user device.

User I/O 1 Comms Quiet

If configured (within GasWorks Hardware Configuration), the User I/O 1 Comms Quiet status is checking for a break in communications between the user I/O card I/O 1 port and a user device..

VGiNet

The VGiNet status checks for breaks in the internal VGiNet communications.

Checking Mass Alignment

Use this procedure to check that peaks are being produced and that they are correctly identified and aligned with the mass scale.

e.g. Mass 28 for N_2 in Air

On the GasWorks toolbar click the  icon to open Control Centre.

If the system has not been tuned previously, use the Peak Shape and Sensitivity tabs to do so. Type in the required value for each parameter and press **Enter** to set the parameters as follows:

- Trap current = 20
- Quad lens = 0
- Resolution DAC = 0
- Repeller Voltage = 5
- Focus Voltage = 450
- Deflection Voltage: = 0

Select **Mass Alignment.** Set the Start and End masses (typically beginning with start mass 0, end mass 50), mass Increment (e.g. 0.03), Integration (e.g. 10), measurement Range (e.g. 10), and inlet Stream (e.g. air or N_2) parameters. Click on the **Start Scan** button.

Check that peaks are produced and that they are correctly aligned with the mass scale markers.

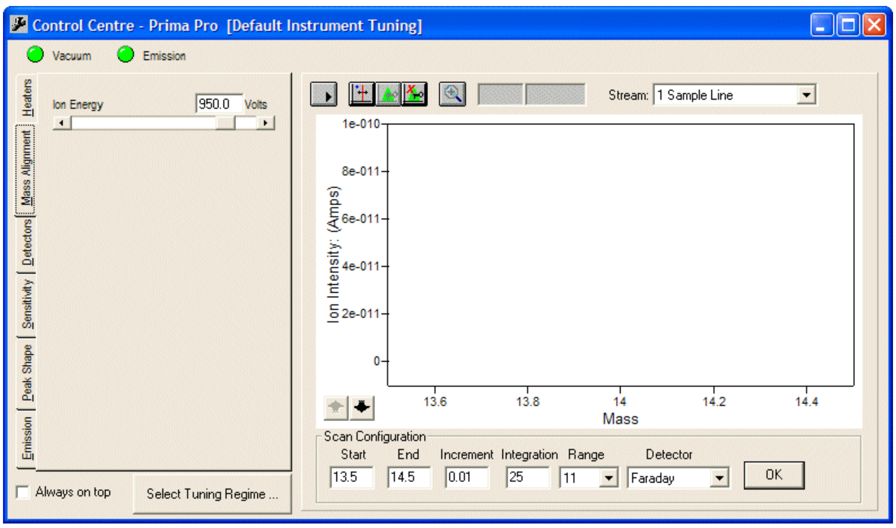


Figure 8–2. Mass Alignment tab

To display minor peaks, increase the gain range by clicking on the down arrow button next to the intensity axis. For an air sample, the principle peak will be at mass 28 due to 78.08% of nitrogen (N_2 = 2 x 14, where 14 is the atomic weight of nitrogen), and the second biggest peak will be at mass 32 (oxygen at 20.95% O_2 = 2 x 16).

There should also be a peak at mass 14; this is produced by both atomic nitrogen $(N)^+$ and doubly-charged nitrogen ions $(N_2)^{++}$. The mass spectrometer separates the sample ions according to their mass / charge ratios and some molecular fragmentation occurs in the ion source together with ionization of the gas molecules.

Further peaks of interest include 16, 17, and 18 (due to O^+ from oxygen and water, OH^+ and H_2O^+ from water), 40 (0.94% Argon), and 44 (350 ppm CO_2 12 + 2 x 16).

If alignment is poor, the simplest approach is to delete the locked peak table by stopping the scan and clicking on the **Delete Locked Peak Table** button: .

To create a new mass table, add a cursor to the display using the following button: .

Position the cursor on mass 14 peak and lock this as mass 14.00. Click the **Peak Lock** button: .

Enter **14.00** in the Peak Locking dialogue box (Figure 8–3).

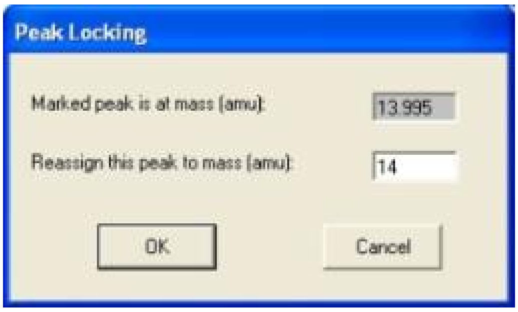


Figure 8–3. Peak Locking dialogue box

The value shown in the marked peak window is the Cursor position. Reposition the scan to mass 44 by changing the scan start and stop values and lock this peak to mass 44.00 as before.

The zoom-in facility can greatly facilitate accurate mass setting.

If a gas (or gases) other than air are used, knowledge of its composition will be needed to correctly identify which peaks are which masses.

The approach is as above and should use two or three masses which span the required analysis range.

For example, in an ethylene cracker application it is necessary to align the peaks only at masses 2 (H_2) and 78 (C_6H_6) to give accurate mass alignment throughout the 2 to 100 amu mass range.

Chapter 9
Fault Diagnosis

Introduction

This chapter provides information on basic fault diagnosis that should only be undertaken by an experienced technician trained in the maintenance of high voltage electronic equipment. The information provided is not exhaustive, and normal diagnostic procedures should be followed where specific details are not given.

Alarm Conditions

The status of the instrument is indicated in GasWorks software by the status window display. Any out of limit condition for any parameter is indicated by an alarm condition (normally indicated in red). Typical values, limits, and causes of alarm conditions are indicated in the table below.

Table 9–1. Diagnosing alarm conditions

Device	Typical Reading	Set Point Low	Set Point High	Typical Cause
Electron Energy	70.10 volts	30.00 volts	95.00 volts	ASU failure
Emission	1	0.5	1.5	Power parameters in alarm
Filament Current	2.60 amps	1.50 amps	3.50 amps	Still pumping down or filament failure
Filament Current Limit	3.06 amps			
Filament Integrity	(OK)			
Half Plate 1 Output Voltage	448.0 volts	300.0 volts	950.0 volts	ASU failure
Half Plate 2 Output Voltage	431.0 volts	300.0 volts	950.0 volts	ASU failure
In Current Limit	(OK)		(OK)	Ion source fault
Ion Energy	998.0 volts	750.0 volts	1050.0 volts	ASU failure
Repeller Voltage	6.96 volts	-10.00 volts	10.00 volts	ASU failure
Source Current	85.0 µamps		1000.0 µamps	Faulty filament
Source Temperature	140.1°C	138.0°C	142.0°C	Still warming up or source heater failure
Trap Current	29.9 µamps	10.0 µamps	100.0 µamps	Filament failure

Device	Typical Reading	Set Point Low	Set Point High	Typical Cause
ASU Controller Status	1	0.5	1.5	ASU failure
Electronics Temperature	28.20°C	10.00°C	55.00°C	Ambient temperature or cabinet filter blocked
Magnet Controller Status	1	0.5	7	Magnet busy or failure
Mass Filter	0.45690 Tesla			
Pressure Set Point	3.44e-05 mbar			
System Pressure	3.13e-06 mbar	2.00e-07 mbar	2.50e-05 mbar	Still pumping or leak or pump failure
System Pressure Trip	(OK)		(OK)	Still pumping or leak or pump failure
Turbo Disable	(Off)		(Off)	Turbo switched to “stop” on PDC
Turbo Motor Current	0.62 amps	0.01 amps	4.00 amps	Turbo adjustment or failure
Turbo Operating Hours	7264 hours			
Turbo Speed	100%	95%		Still pumping or leak or pump failure
Turbo Speed Interlock	(OK)			Still pumping or leak or pump failure
Vacuum	(OK)	(OK)		Still pumping or leak or pump failure
RMS Body Temp	59°C	30°C	100°C	Still heating up or heater failure
RMS Ambient Temperature	42°C			
RMS Controller Status	1	0.5	1.5	RMS electronics failure
RMS Digital Input 2	(Fail)			
RMS Digital Input 3	(Fail)			
RMS H1 Current	245.0 mA	50.0 mA		Heater failure
RMS Sample Tube Current	350.0 mA	50.0 mA		Heater failure
RMS Flow Sensor	(OK)		(OK)	Low sample flow
RMS Position	31	0.5		Optical sensor / detector needs cleaning
Faraday Deflection	559.0 volts			
Multiplier Deflection	0.0 volts			
Multiplier Voltage	828.0 volts			

Device	Typical Reading	Set Point Low	Set Point High	Typical Cause
Negative Quad Lens Voltage	-5.38 volts			
Positive Quad Lens Voltage	5.33 volts			
Resolution	0.0 volts			
Amplifier +24V	14.9 volts	14.0 volts	16.0 volts	PDU or switch mode ps
Analyser Supplies 24V	23.9 volts	22.5 volts	25.5 volts	PDU (fuse) or switch mode ps
Fans 24V	24.0 volts	22.5 volts	25.5 volts	PDU (fuse) or switch mode ps
Instrument CPU 5V	4.8 volts	4.8 volts	5.3 volts	PDU (fuse) or switch mode ps
Magnet AC	38.8 volts	12.0 volts	50.0 volts	PDU or switch mode ps
Magnet Current	2.84 amps	0.10 amps	10.00 amps	Magnet drive or magnet controller
Power Controller State	1	0.5	1.5	PDU or switch mode ps
Turbo 24V	24.2 volts	22.5 volts	25.5 volts	PDU (fuse) or switch mode ps
User IO 24V	24.2 volts	22.5 volts	25.5 volts	PDU or switch mode ps
Vent Valve 24V	24.1 volts	22.5 volts	25.5 volts	PDU or switch mode ps
Host Comms Quiet	0 seconds			
User I/O 0 Comms Quiet	0 seconds			
User I/O 1 Comms Quiet	0 seconds			
VGiNet / Opto22 Comms Quiet	0 seconds		20 seconds	Cable fault

Leak Detection

Warning! High Voltages! To avoid flash over, exercise extreme caution when probing with helium in the vicinity of the ion source feed through. Helium is extremely susceptible to “break down” when exposed to high electric fields.

Abnormally high atmospheric peaks and a slight rise in the background pressure, as displayed by the Penning gauge reading, may indicate a small leak in the instrument. In general, the most convenient method for leak detection is to use the extreme sensitivity of the instrument itself. Tune the mass spectrometer to a suitable gas (helium, mass 4, is particularly effective). Then probe around the vacuum chamber using a fine collimated stream of the same gas. A sharp increase in the ion current of the monitored peak indicates that the search gas has entered the vacuum system. Finer gas flows may be required to help pinpoint the exact location of the leak.

Leaks may sometimes occur at elastomer seals, particularly after they have been disturbed. First check that the seal fixings are not loose. If they are, tightening may cure the problem; however, care should be taken not to over tighten as damage to the seal or fixings may result.

In most instances this is due to contamination of the sealing surface by dust, particulates or fibres. These can normally be removed by use of clean, filtered compressed gas or wiping the surface with a lint free cloth. A small amount of solvent, such as IPA, applied to the cloth may assist in the process.

A larger leak which results in a pressure rise above 3×10^{-5} mbar will cause the emission to trip or prevent the system from attaining the trip pressure on pumping down from atmosphere. This would prevent switch on of the mass spectrometer. In this case, the leak can potentially be located by probing around the vacuum system with helium and looking for slight changes in the Penning gauge reading.

Mass Spectrometer Faults

The complete loss of mass peaks or a loss of resolution / peak shape or sensitivity indicates the majority of faults. The following sections are a list of possible causes of the above conditions.

Warning! High Voltages! Voltages of up to 1 kV are present! Exercise extreme caution at all times.

No Peaks

If there are no mass peaks, complete the following:

- Check that the electronics are powered and that all leads are connected and that all fuses are intact.
- Check the Instrument Status light is green.
- Check for satisfactory electron emission, filament current, source current, trap current, and electron energy on the GasWorks Status Window. If filament is short circuit the filament current will be at filament limit with no trap current.
- In Control Centre, set the Ion Energy and Focus Voltages to **zero**. Disconnect the socket from the source flange plug.
- Check each pin on the source flange plug to all others and earth (see Figure 9–1). Continuity should only be found between:
  - 7 and 12 Filament 1
  - 4 and 8 Filament 2
  - 9 and 16 PRT
  - 10 and 11 Source Heater
- A plate in the source or collector assemblies may be disconnected and / or charging up. This can only be confirmed by examining the individual assemblies, which requires a full instrument shutdown and removal of the assemblies from the vacuum chamber.
- Access the GasWorks Instrument Status window and check that the magnet current is approximately 2.3 amps with mass 28 selected in Control Centre.
- If the noise displayed on the computer screen does not increase as the gain is increased, it is likely that the detector system is inoperative.

Poor Sensitivity

For low or unstable sensitivity and / or poor peak shape:

- Ensure that the source and lens tuning is optimized.
- Check for contamination of the source or lens. If necessary, remove and clean.
- Ensure that an electrode in the source or collector assembly is not disconnected and / or charging up. Normally, in this condition, no peaks are observed.
- Try the second filament. Sensitivity can be degraded after a filament has been in use for a long period, often due to contamination or physical movement of the filament wire.
- Check that the Penning gauge pressure reading is normal.
  An extremely low pressure may indicate blockage of the inlet – blocked RMS and / or screen filter and / or capillary and / or glass leak.

Poor Stability on Startup

There are a number of factors that influence the initial stabilization (settling) of the mass spectrometer following startup or restart. Filament emission characteristics are altered by reactions with the sample gas and depend on the specific chemistry of the gas. Different gas mixtures will have differing effects on the filament. Gas mixtures that have a net oxidizing effect tend to be the reverse of mixtures that are net reducing. Where an instrument has either been idle or analyzing a radically different sample stream, there will be an initial stabilization in the filament characteristics as they adjust to the average composition of the samples. This will normally result in an overall general drift in absolute peak heights. From cold startup, there will be other factors contributing to this effect such as out-gassing from the enclosure and various system elements as they come up to working temperature.

If there is a very large change in the average gas composition with time, for example from one day to the next, then it may be necessary to recalibrate on a daily basis to maintain system accuracy within an acceptable bandwidth. Most processes, however, involve a stable average composition of process gas samples.

Reactions between the sample gases and filament result in changes in the electron work function of the filament emissive material. The electron flux through the ionization region is regulated by control of the trap current, which will adjust filament current to affect any necessary compensation. This in turn changes the filament temperature and could affect the temperature distribution of the ion source, even though the overall source temperature may be regulated to ± 0.1°C by a source temperature controller.

There can also be changes to the spatial distribution of electron trajectories and in the electron energies following impact with the sample gas molecules. The nature and efficiency of ionization, as well as the energies and trajectories of ions exiting the source, can all be affected. As a result of all these effects, there can be changes in the absolute sensitivities, relative sensitivities, and cracking patterns. The magnitude of the effects depends very much on the types of gases being analyzed –the more overlaps or the greater the variation in ion types (e.g. mass and multiplicity of charges) the greater the potential impact.

Contamination can result in sample reacting with or being absorbed on the walls of the ionization chamber, which can modify the surface potentials due to localized charging effects.

Poor Results Following Calibration

Check the calibration method has been correctly configured. The general “rules” for configuring calibration gases are as follows:

Fragmentation: Can only be selected when there are no overlaps in the spectra of the gas component, at the selected masses, with those of any other gas in the cylinder.

Background: Enable for background measurement only when there are no contributions at any of the selected peaks in the cracking pattern for the gas component in the calibration cylinder.

Sensitivity: Can be calibrated only when there are no overlaps at the principle peak.

Linearity: Can always be calibrated provided the component is present at a significant and accurately known level in a gas mixture.

Check that there is flow when the calibration gases are selected. If a non-hazardous calibration gas is selected and no hazardous sample gases are flowing through the RMS, removal of the RMS cover allows verification that the RMS is aligned correctly on the calibration stream.

Check that the Penning pressure reading is consistent with that obtained during original installation. Depending on the molecular leak type used, the Penning pressures for nitrogen or air are typically:

70 micron pinhole leak	ca. 5×10^{-6} mbar
50 micron pinhole leak	ca. 3×10^{-6} mbar
30 micron pinhole leak	ca. 1×10^{-6} mbar

Note these pressure readings are gas dependent.

A good test for system integrity is to introduce helium, which will typically give the following pressure readings:

70 micron pinhole leak	ca. 2 x 10^{-6} mbar
50 micron pinhole leak	ca. 1 x 10^{-6} mbar
30 micron pinhole leak	ca. 5 x 10^{-7} mbar

The above is only a rough guide but can be used to establish if the micro capillary is blocked, significantly lower readings, or there is a gross leak, significantly higher readings.

If the capillary is blocked or the ion source is contaminated, the linearity factors obtained during calibration will depart greatly from unity; typically they are close to 1 (e.g. 0.7 to 1.3) but in the event of a blockage or contamination, could be as low as 0.3 or as high as 3.0.

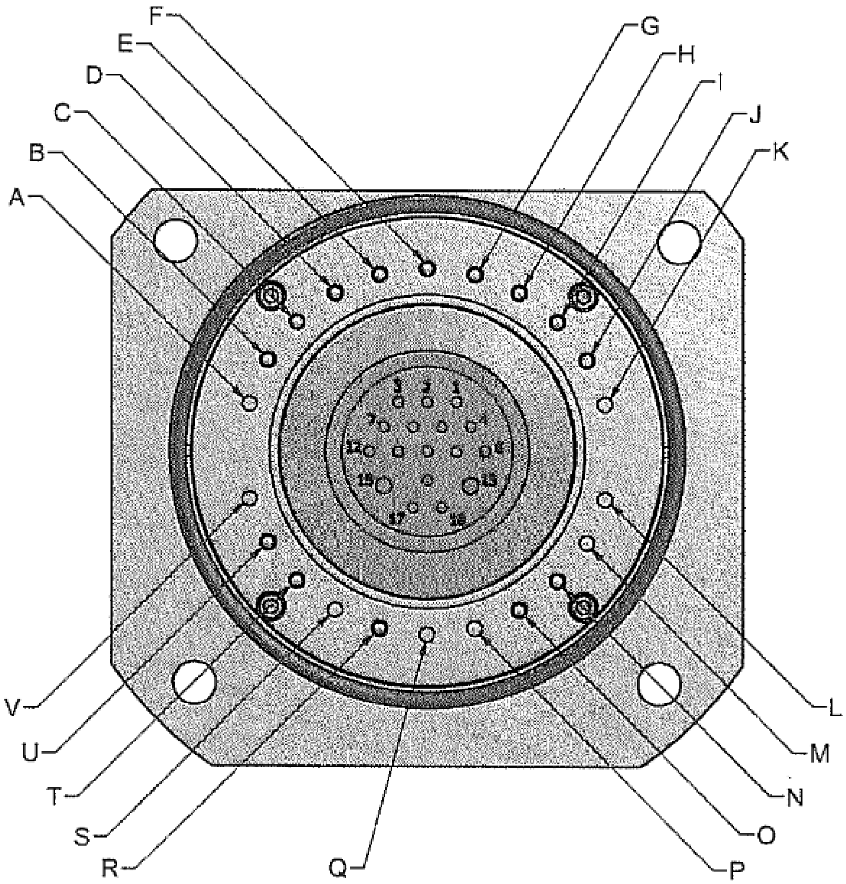
Ion Source and Magnet Pole Pieces Assembly

After prolonged operation of the unit (typically between six months and two years) a build-up of contamination may occur on the ion source electrodes, the magnet pole pieces, and Z-restrictor. This can result in the build-up of insulating films on metal surfaces, which in turn can charge up and disturb the local field potentials. This usually results in a reduction in sensitivity and / or a distorted peak shape. In this event, it will be necessary to clean the source, pole pieces, and the Z-restrictor. Refer to Chapter 10: Maintenance for cleaning procedures.

Filament Failure

Typical filament lifetime is six months to two years. The instrument status flags indicate filament failure. Filament current, trap current, and source current will all be zero. Filament failure can be checked (after setting the ion energy and focus voltages to zero), by testing the source flange plug for continuity between pins 7 and 9 for the case of filament 1 or pins 16 and 17 for the case of filament 2 on the source plug. However, since the source is the dual filament type then, provided one filament is still intact, one can simply switch operation to the other filament. Filament replacement can then be carried out at any convenient time, such as during a routine preventive maintenance service.

Note: The filament current will be switched off if the Penning pressure rises above its set point. This should be checked before dismantling the system to change the filament.



WIRING TABLE		
PEEK RING CONNECTION	FEEDTHROUGH PIN No.	FUNCTION
A	N/C	N/C
B	12	FIL 1 +
C	7	FIL 1 -
D	3	FIL TRAP 1
E	6	HALF PLATE 2
F	2	SOURCE
G	5	REPELLER
H	1	FIL 2 TRAP
I	4	FIL 2 -
J	8	FIL 2 +
K	N/C	N/C
L	N/C	N/C
M	N/C	SOURCE SLIT (GND)
N	9	PRT LOW
O	16	PRT HIGH
P	N/C	N/C
Q	N/C	N/C
R	14	HALF PLATE 1
S	N/C	N/C
T	10	HEATER LOW
U	11	HEATER HIGH
V	N/C	N/C

Figure 9–1. Source plug pinout (external feed-through)

Chapter 10
Maintenance

Caution! Before commencing any maintenance work on this instrument, refer to instructions in Chapter 2: Safety Information.

Maintenance Schedule

The maintenance schedule is summarized in the table below.

Table 10–1. Maintenance schedule

Frequency*	Item	Task
Weekly	Rotary pump	Check oil level, and replenish if necessary.
Monthly	Fan / Filter	Check external air circulation, and clean as required.
Every 6 months	Rotary pumps	Change oil.
Every 3 years	Turbo molecular pump	Service pump.

* May vary depending on application and environment.

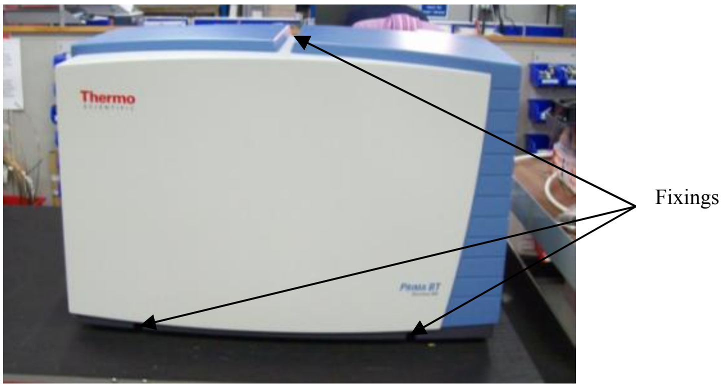
Procedures

Instrument Access

Caution! Access to the analyser and electronics should only be undertaken by suitably trained personnel. There are no user serviceable parts or assemblies which require access under normal operation of the analyser.
The instrument should be isolated from the mains power supply before outer covers are removed.

Removal of outer cover

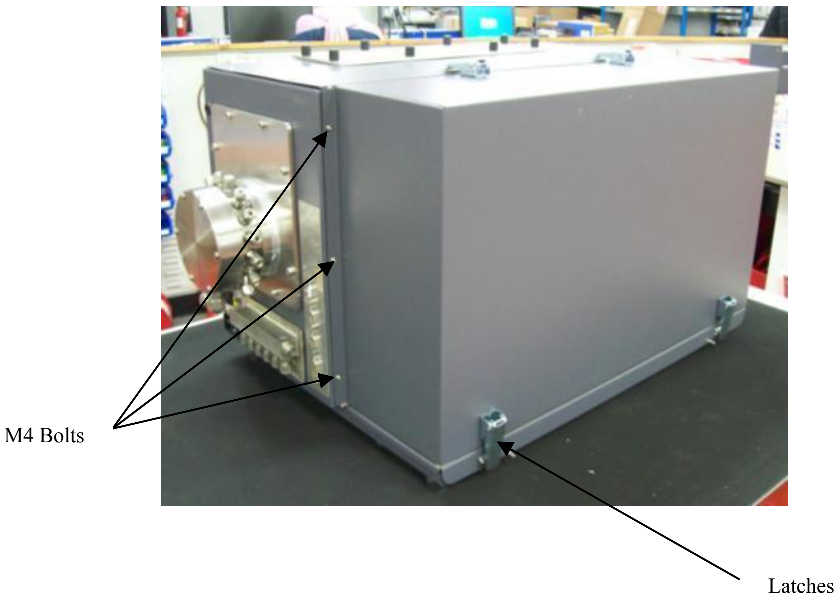
The outer cover is secured in place at three points. Two M5 nuts in the front lower plinth and one M5 bolt top centre back. Undo the two plinth nuts, use a suitable nut driver, then remove the M5 bolt, hex driver. The outer cover is now free. Pull forward to clear the front studs and lift clear of the instrument.



Caution! The outer cover weighs 11Kg and care should be exercised when removing from the instrument. Ensure there is a suitable surface or cleared area to set the cover down.

Internal Covers

The two internal covers form the instruments EMC shield. Removal of the front cover provides access to the analyser, Turbomolecular pump, Amplifier, PDU and ASU electronics.
First remove the three M4 hex bolts securing the left hand face to the chassis plate. Undo the six retaining latches, two top, two front and two right hand face, remove the cover and set aside.
To gain access to the fan, DIN rail and inlet assemblies as well as the CPU, Inlet and magnet drive PCA assemblies remove the rear EMC shield. This is secured in place by 8 M4 hex bolts and the two rear centre thumb nuts on the fan filter assembly. It is NOT necessary to remove the remaining 6 thumb nuts of the fan / filter assembly as these simply secure the filter to the removable cover.

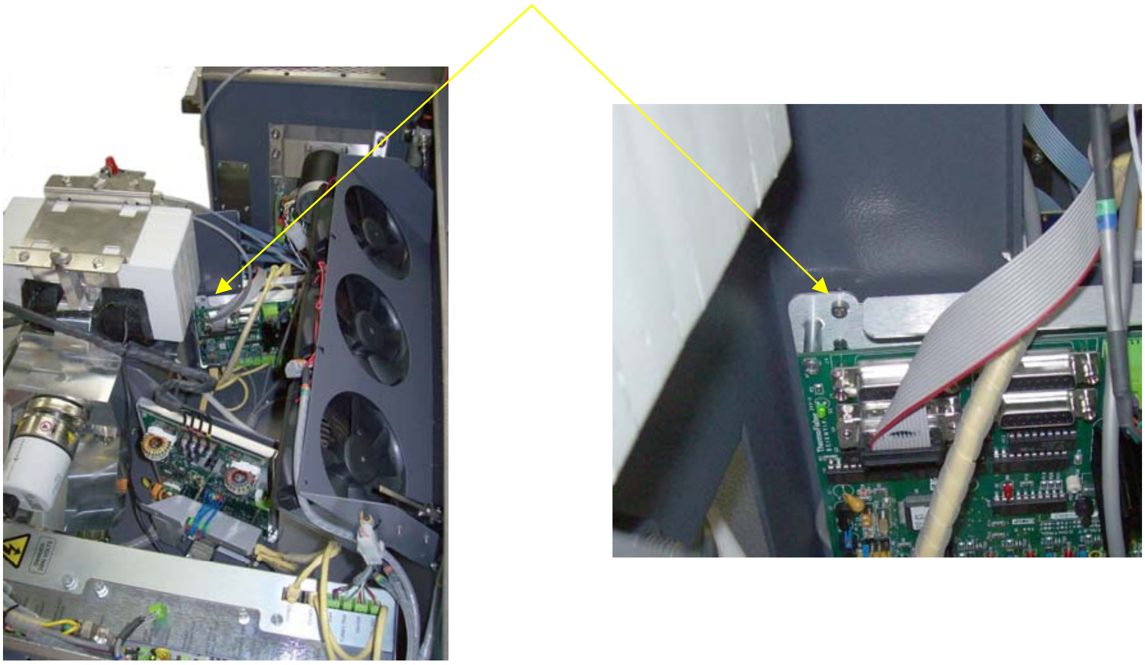


Removal of the fan / filter assembly or the rear EMC cover will expose the cooling fan assembly. The fans are controlled by a micro switch which will break the supply when either of these covers is removed.

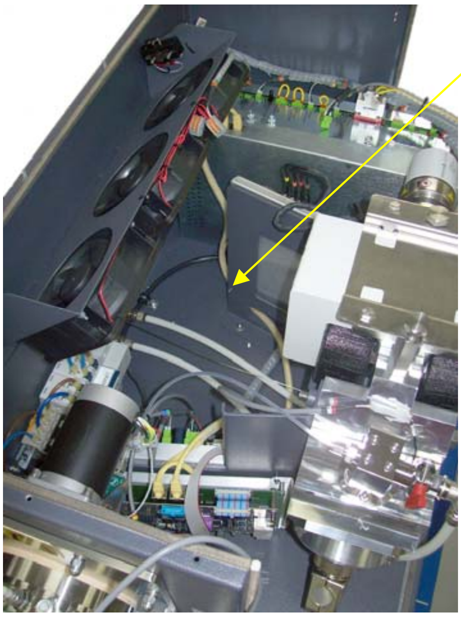
Caution! If the instrument is powered when these covers are removed the fans may still be rotating. Care should be taken to keep tools or fingers away from the assembly until the fans have stopped.

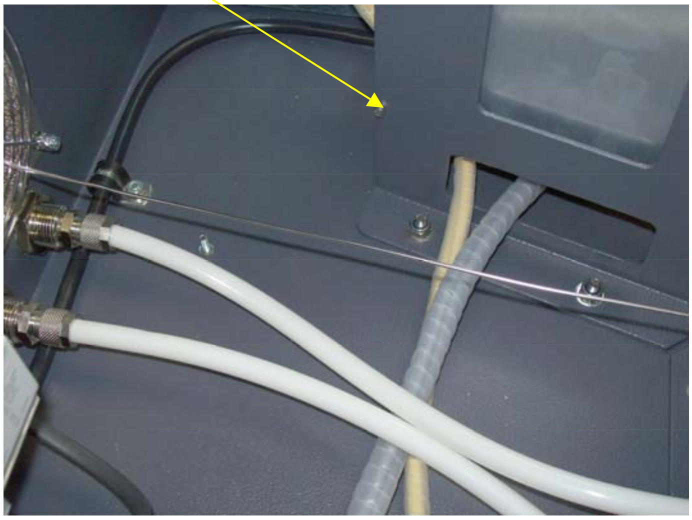
Electronics Assemblies

Shipment locking bolts are fitted to the following PCA assemblies; CPU_I/O and Inlet Controller; Magnet power & control. Should it prove necessary to remove any of these assemblies these locking bolts must be removed before the carrier plates can be un-clipped. The CPU_I/O and Inlet Controller PCAs are clamped by a single bolt which is located through the top left corner of the Inlet Controller



The Magnet power & control assembly is fixed with a single bolt through the lower left side of the support frame as shown below





Rotary Pump Oil Level

Warning! Hazardous Substance!

- Pump oil must not be ingested.
- Protective clothing must be worn when replenishing the oil in the rotary pumps.
- Keep away from skin, especially open wounds. Pump oil is a possible irritant.
- Waste oil is to be disposed of in an environmentally responsible manner.

Warning! High Temperatures!

- Rotary pumps run hot during operation.
- Allow pumps to cool before draining oil.

The oil level in the rotary pump should be checked in accordance with the maintenance schedule and replenished if necessary. If a significant loss of oil has occurred, this should be investigated according to instructions provided by the pump manufacturer (manual supplied).

Routine oil changes are not required for rotary pumps filled with Fomblin oil.

Figure 10–1. Rotary pump drain plug

Cleaning the Cooling Fans

To maintain adequate cooling of the instrument, ensure that the fan filter covers are not obstructed and are free of dust. Clean as required.

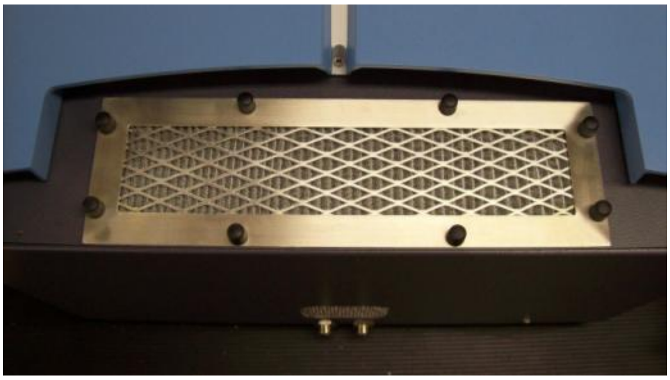


Figure 10–2. Fan cover guard

Lubricating the Turbo Molecular Pump

Warning! Lethal Voltages! Lethal voltages are present in the vicinity of the turbo molecular pump. Isolate the mains electrical supply before attempting any maintenance activity on the pump.

Maintenance procedures for the turbo molecular pump are given in the instructions provided by the pump manufacturer (manuals supplied). The procedures are quite involved and, in general, it is preferable to return the pump for maintenance.

Dismantling and Cleaning Procedures

Warning! Lethal Voltages! Lethal voltages are present within the Prima BT enclosure. Isolate the mains electrical supply before attempting any maintenance activity.

Filament

Warning! Hazardous Substance!

- The filament contains hazardous substances and must not be ingested.
- Protective clothing (appropriate gloves) must be worn when handling the filament.

Note: All parts are clean assemblies and gloves should be worn during the following procedure.

To **remove** the filament, proceed as follows.

1. Using Control Centre, switch off the filament.
2. Switch off the electrical power. Wait approximately five minutes for the turbo molecular pump to slow down and vent the system to atmospheric pressure.
3. Remove the source connector and 4 x M6 nuts shown in Figure 10–3.

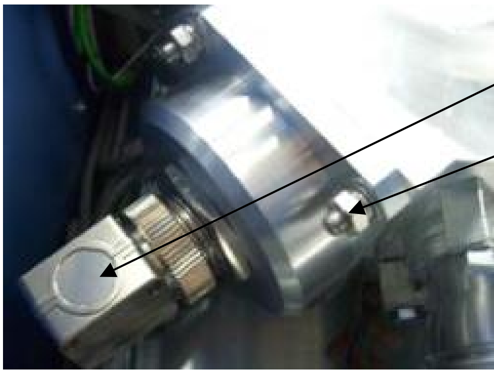


Figure 10–3. Removing the source connector and 4 x M6 nuts

4. Carefully lift the source flange complete from the analyser chamber.

5. Remove the 3 x M3 cap heads that secure the source connection ring in the analyser housing and carefully withdraw the source.

Note: The filaments are positioned against a "stop" on one side and are "hooked" over the anodized aluminium insulating plates. They are secured by slotted knurled nuts. The filament supply and trap leads are connected to the filament assembly pins by gold socket connectors. Note the position of the three connectors. ♦

6. Remove the socket connectors from the failed filament assembly and the knurled nut. See Figure 10–4.

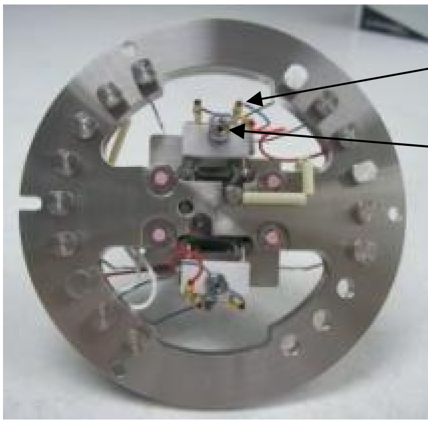


Figure 10–4. Removing the failed filament

7. Using tweezers, withdraw the filament.

Note: The anodized aluminium insulating plate is left in place. This must be reused or replaced when replacing the filament.

To **replace** the filament, proceed as follows.

1. Position the filament as shown in Figure 10–4 and fix in place with the knurled nut.

   Note: The alignment of the filament in the plane of the ion beam is preset.

2. Reconnect the supply leads (socket connectors shown in Figure 10–4).

3. Check the filament connections for continuity.

4. Check the filament to ion source and trap to ion source and ensure these are open circuit (> 50 MΩ).

5. Refit the source connection ring in the analyser housing and secure with 3 x M3 cap head screws. A dowel pin in the analyser housing ensures the correct orientation.

6. Refit the source flange and tighten the M6 nuts evenly.

Ion Source

Caution!

- This operation should only be undertaken by trained personnel with knowledge of the requirements for the cleanliness of vacuum components.
- This operation should ONLY be performed in a dust free, clean environment. Failure to observe this requirement could result in contamination of the assembly with particulates, and / or materials, which could degrade or prevent subsequent operation of the source.
- Magnetic particulates will easily attach to the collimating magnets and will be very difficult to remove.

To **dismantle** the ion source, follow the procedure below.

1. Remove the ion source as previously described in the previous section.

2. Disconnect the source wiring leads from the feed-through.

3. Remove the filaments as described in an earlier section titled "Filament".

4. The source assembly is built on four ceramic rods secured by a clamp.

Caution!

- The electron beam collimating magnets are fixed to the bracket at the top of the assembly nearest the connection ring. These magnets should NOT be removed.
- Care must be taken to prevent contact between magnetic materials (e.g. a screwdriver) and the magnets.
- These magnets have a high field strength but are very brittle and easily damaged if subject to impact.

5. Disconnect all leads from the source assembly, not the connection ring.

6. Remove the insulating rod clamp furthest from the connection ring. See Figure 10–5.

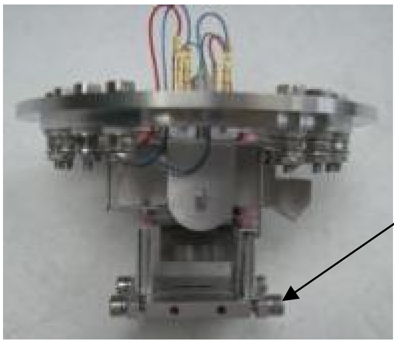


Figure 10–5. Removing the insulating rod

7. Take careful note of the sequence and orientation of the components. Detail changes to the exploded view below (Figure 10–6) may occur in special applications.

8. Remove the source component items in sequence.

9. Metal electrode surfaces can be cleaned with a fine abrasive such as diamond paste. Aim to remove discolourations and achieve a “shiny finish.” Ceramics can be cleaned with suitable detergent or acid.

10. Final clean as follows. If possible, use of an ultrasonic bath is recommended.
    a. Wash off all traces of abrasive pastes according to type. Use deionised water for water soluble pastes or recommended solvents as per manufacturer’s instructions.
    b. Final rinse with clean isopropyl alcohol or a similar solvent.
    c. Dry thoroughly, preferably in an oven at 100°C for one hour.

11. Reassemble the source assembly and refit to the analyser flange in reverse order.

12. Once reassembled, ensure that all plates are parallel. Check that the two focus half plates are aligned with each other.

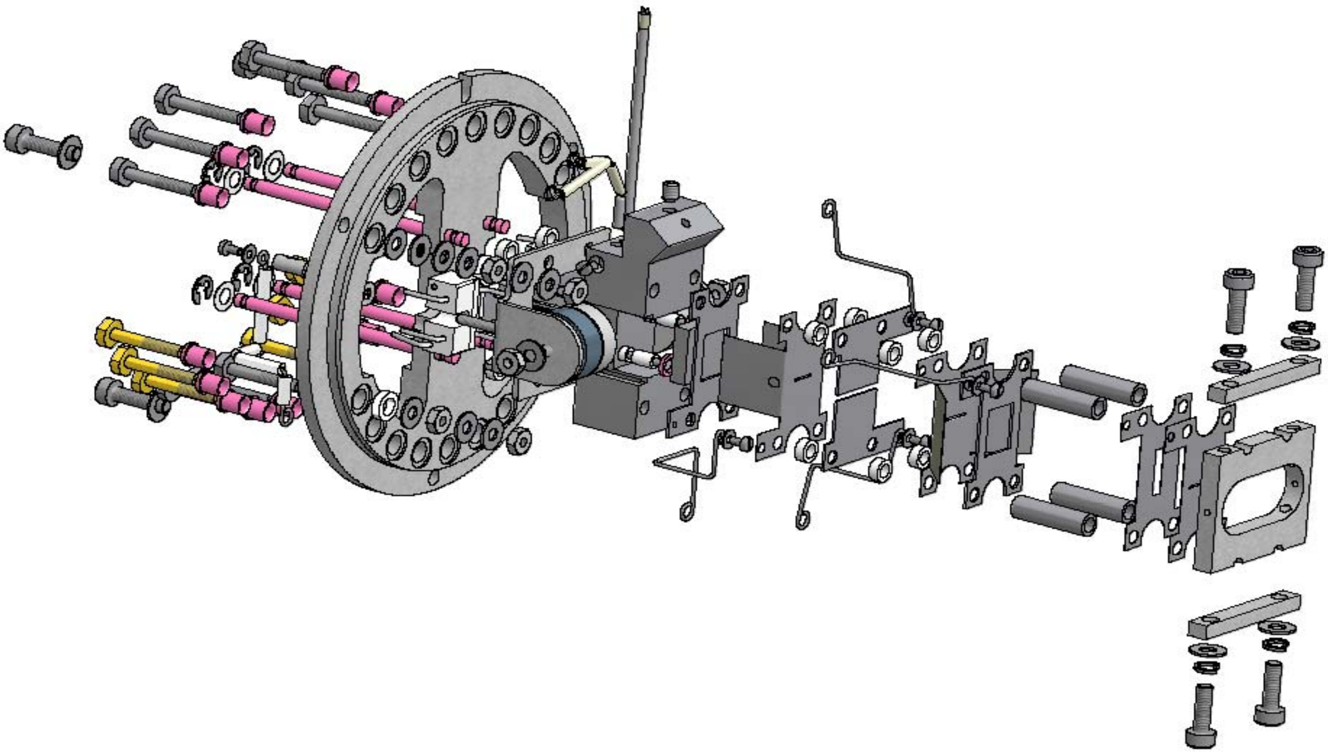


Figure 10–6. Source assembly

Magnet Pole Piece Assembly

To clean the magnet pole pieces, proceed as follows.

1. Disconnect the wiring for the electromagnet coils from the magnet power and control PCAs, top RH side of the enclosure (Figure 10–7). Note the wiring sequence of the main coils to the power PCA. The electromagnet will fail to operate if these are not refitted in the correct sequence.

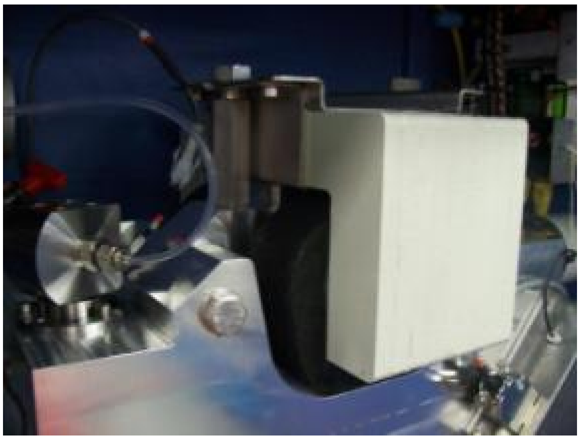


Figure 10–7. Disconnecting the wiring

2. Using two 19 mm ratchet ring spanners, jack up the coils and yoke assembly until it clears the pole pieces (Figure 10–8). Ensure that the bolts are unscrewed evenly to avoid unnecessary strain to the threads. The bolts will now be free from the threaded hex pillars.

Figure 10–8. Jacking up the coils and yoke assembly

3. Lift out the coil assembly (Figure 10–9).

Caution! Heavy!

- The electromagnet assembly weighs approximately 16 kg.
- Protective footwear should be worn when performing this action.

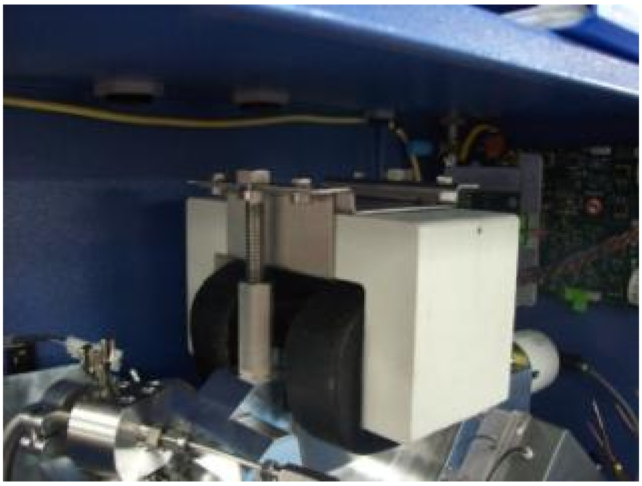


Figure 10–9. Lift out coil assembly

4. Unscrew the four pole piece locating screws and remove the front magnet pole piece (Figure 10–10).

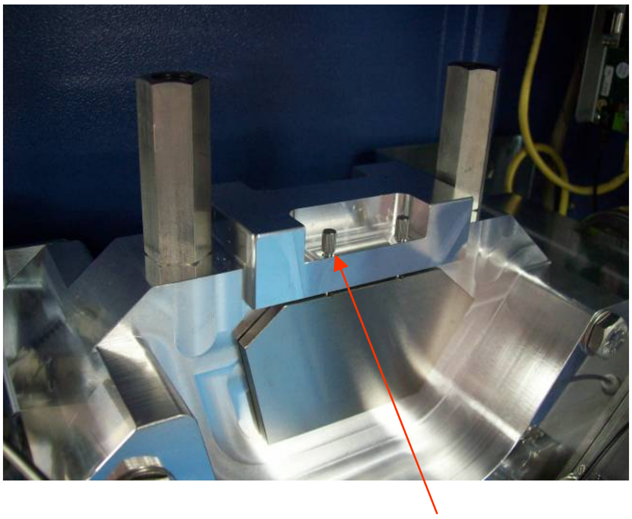


Figure 10–10. Removing the front magnet pole piece

5. The Z-restrictor is a thin aperture plate positioned at the beam entrance to the electromagnet, left hand, source side, of the magnet assembly and held in place by the pole pieces. When the front pole piece is removed, slide out the Z-restrictor plate before removing the rear pole piece.

6. Use diamond or other suitable abrasive paste to polish the surfaces of the magnet pole pieces and the Z-restrictor. Pay particular attention to the rectangular slot in the Z-restrictor.

7. Final clean as follows. If possible, use of an ultrasonic bath is recommended.
   a. Wash off all traces of abrasive pastes according to type.
   b. Use deionised water for water soluble pastes or recommended solvents as per manufacturer's instructions.
   c. Final rinse with clean isopropyl alcohol or a similar solvent.
   d. Dry thoroughly, preferably in an oven at 100°C for one hour.

8. Visually inspect the magnet pole pieces for any remaining contamination, and repeat steps 6 and 7 if necessary.

9. Refit the rear magnet pole piece and Z-restrictor.

10. Refit the front magnet pole piece, taking care not to dislodge the Z-restrictor.

11. Lift the magnet / coil yoke assembly back into position. Engage the threaded bolts into the threaded hex pillars, and use the ratchet spanners to evenly jack down the assembly.

12. Reconnect the coil and search coil wires to the magnet power and control PCAs.

Fuses

Maintenance Hazard! Qualified Personnel Only! Before changing a fuse, isolate the instrument from the mains power supply and read the safety instructions provided in Chapter 2.

Fuses fitted are one of two body types. See Figure 10–11 and Figure 10–12.

Figure 10–11. ATO (low voltage DC supplies only)

Figure 10–12. 5 x 20 mm ceramic (mains supplies)

Table 10–2. Power distribution unit fuses

Fuse Number	Supplied Unit	Voltage	Rating	Type	Manufacture
F1	Magnet	30 V	10 A	ATO	Littelfuse 257 010
F2	Probe heater	24 V	2 A	ATO	Littelfuse 257 002
F3	Auxiliary heater	24 V	2 A	ATO	Littelfuse 257 002
F4	Inlet power	24 V	4 A	ATO	Littelfuse 257 004
F5	Amplifier	24 V	1 A	ATO	Littelfuse 257 001
F6	Turbo pump	24 V	4 A	ATO	Littelfuse 257 004
F7	Vent valve	24 V	1 A	ATO	Littelfuse 257 001
F8		24 V	2 A	ATO	Littelfuse 257 002
F9	CPU carrier	24 V	1 A	ATO	Littelfuse 257 001
F10	User I/O	24 V	4 A	ATO	Littelfuse 257 004
F11	Analyser	24 V	2 A	ATO	Littelfuse 257 002

Table 10–3. CPU and user I/O carrier fuses

Fuse Number	Supplied Unit(s)	Voltage	Rating	Type	Manufacture
F1 to 3	Digital O/Ps, Individual	24 V	1 A	ATO	Littelfuse 257 001

Table 10–4. Inlet control fuses

Fuse Number	Supplied Unit(s)	Voltage	Rating	Type
F1	Mains heater	115/230 V	5 A	5 x 20 mm FF
F2	Solenoid valves	24 V	1 A	ATO
F3	Inlet control and RMS drive	24 V	4 A	ATO

Appendix A

Technical Description: System

Introduction to the Prima BT Hardware

The Prima BT instrument is an industrial laboratory gas analyser based on a scanning magnetic-sector mass spectrometer with a 6 cm radius. The variable magnetic field is produced by an electromagnet with a laminated core that enables rapid and extremely stable analysis of multiple user defined gases.

The instrument is designed for continuous operation in a process environment and is configured for ease of maintenance and simplicity of operation. With the exception of the host PC, all components are mounted either within or on the instrument enclosure and are modular for ease of replacement.

Main Components

The main components are provided in the following table.

Table A–1. Prima BT hardware - Main components

Component	Description
Data system	Control of all the Prima BT functions is via an integral embedded processor in the instrument enclosure. This processor also handles communication with the host PC, DCS, etc.
Electronics	Provides the required power supplies, etc. for the mass spectrometer.
Analyser	Comprises ion source housing, magnet assembly, and detector housing (collector area).
Rotary pump	Provide backing pressure for evacuating the analyser and also provides inlet pumping.
High vacuum pump	Turbo molecular pump and controller.

Figure A–1. Hardware components

Principle of Operation

Sample gas is ionised by electron impact in the source. Ions injected into a magnetic field will describe a circular orbit where the radius is dependent on the initial velocity of the ion, its mass, charge and the strength of the magnetic field.

This relationship is described in Equation 1:

$$R^2 = \frac{MVx}{H^2 z} \quad \text{constant}$$

Equation 1.

Where:

R = radius of orbit

M = mass of the particle

V = voltage applied to the particle to accelerate it to velocity ‘v’

H = magnetic field

z = charge on the ion

For a fixed radius, the equation reduces to Equation 2:

$$\text{constant} = \frac{MV}{H^2}$$

Equation 2.

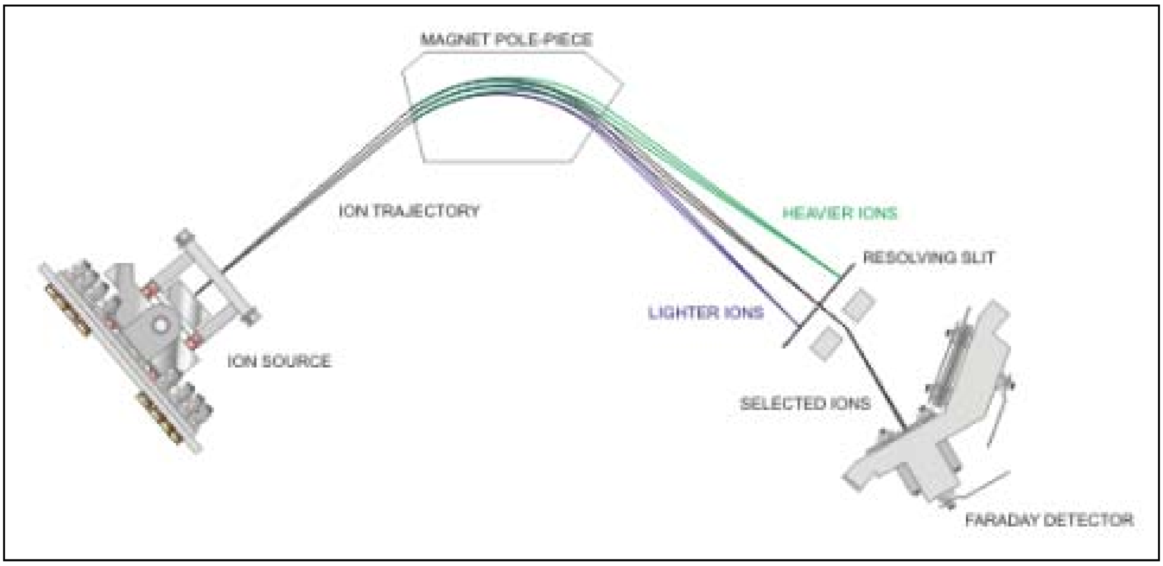


Figure A–2. Separation of ions through the magnetic sector

For an electromagnet with a fixed accelerating voltage the equation can be reduced to Equation 3:

$$M \propto H^2$$

Equation 3.

This is the basic mass spectrometer equation used by Prima BT instrument.

If a fixed position ion detector is placed at the exit from the electromagnet, the mass of the ion arriving at the detector can be selected by varying the magnetic field. Refer to

Figure A–**2**. The ion signal measured at the detector is then proportional to the concentration of the corresponding component in the sample gas. This enables the mass spectrometer to be used as a compositional gas analyser. The constant of proportionality is determined by calibration against known gas standards.

Analysis and calibration may be complicated by overlapping mass spectra as a result of the sample gas mixture. Methods to make such overlaps less convoluted are discussed later.

Vacuum System

The Prima BT vacuum system consists of the listed in the following table.

Table A–2. Prima BT vacuum system components

Component	Description
Analyser assembly	The chamber that contains the ion source, the magnetic sector flight-tube, and collector assembly.
Turbo molecular pump	A mechanical pump capable of producing and sustaining ultra high vacuum (hydrocarbon-free) in a relatively short period. The pump is mounted vertically below the main analyser vacuum housing. All turbo molecular pumps must be used in conjunction with a mechanical rotary pump that maintains the backing pressure in the molecular flow regime required for the turbo molecular pump to operate.
Rotary pump	An $8 m^3 h^{-1}$ two-stage rotary pump is fitted below the analyser enclosure. The turbo molecular pump and inlet bypass are backed by both stages of the rotary pump.
Penning gauge	A glow discharge device that operates within the pressure range 5×10^{-5} mbar to 1×10^{-7} mbar and monitors the pressure in the analyser vacuum chamber. The gauge is controlled by the ASU and the output relayed to the control PC for display on the user interface status page. The primary function is to interlock the ion source filament and analyser high voltages supplies in the event of accidental vacuum loss or high pressure.
Vent valve	To prevent migration and loss of bearing lubricant, the turbo molecular pump should always be held at atmospheric pressure when switched off. To ensure this happens, the pump is fitted with a valve that automatically vents to atmosphere when the pump is switched off or power is lost to the system. The vent line is protected by a filter which protects the valve from contamination and throttles venting rate. The vent line opens into the compression stages of the turbo molecular pump so that the gas is distributed evenly to the high vacuum and backing pressure vacuum sides. A dry gas vent connection, 1/4” / 6 mm Swagelok fitting, is provided on the upper left hand side of the cabinet.

Ion Source

Atoms or molecules of gas must carry an electrical charge before they can be analysed, i.e. they must be ionised. Ionisation takes place in the ion source where a beam of electrons is passed through the sample gas. Electrons that either collide with or pass in close proximity to the sample gas molecules cause electrons to become detached from or adhere to the molecule. In the former case, the ion formed will be positively charged; in the latter, negatively charged. Positive ions are used exclusively for the analysis.

The electron beam is derived from a hot wire filament and is focused magnetically through the ion source. The effect of the magnetic field on the electron beam is similar to that experienced by ions in the magnetic sector and they will describe a circular path. Due to the smaller mass of the electron, however, the paths have a much smaller radius and the beam tends to spiral along the lines of magnetic field.

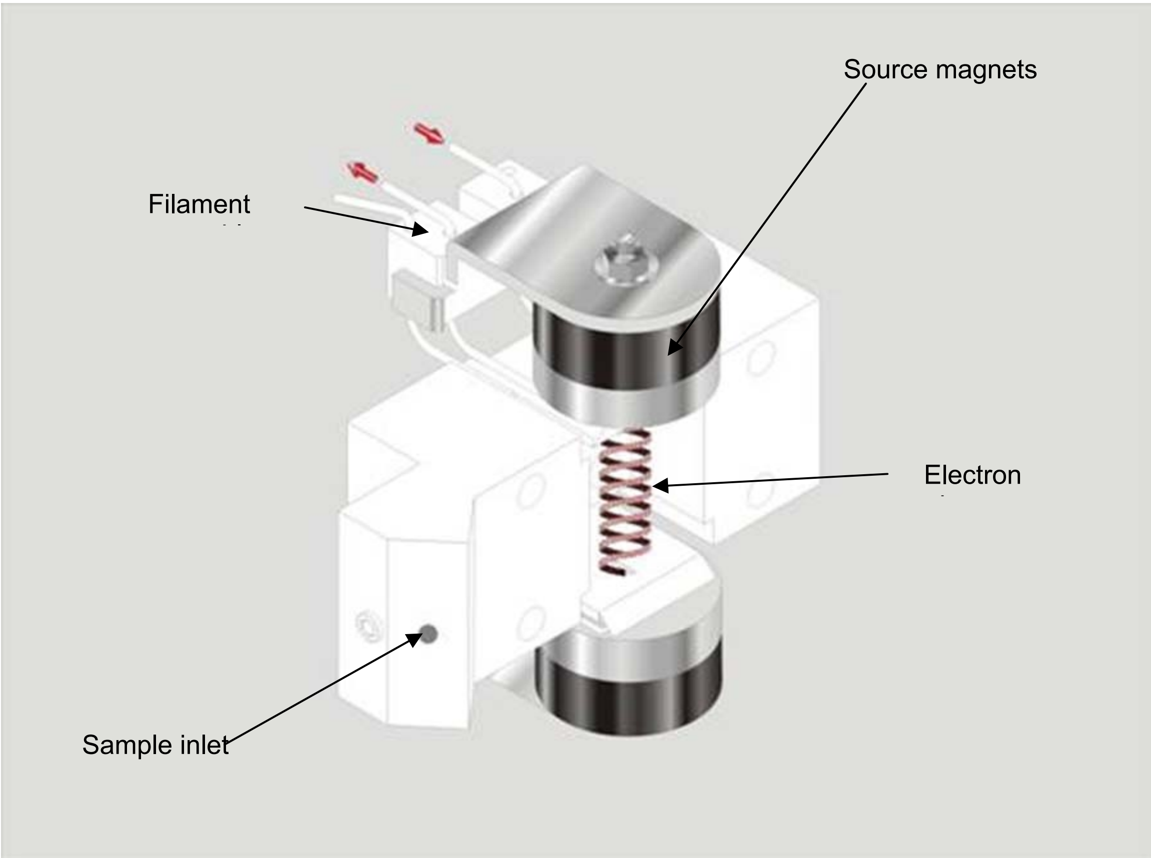


Figure A–3. Ion source

Figure A–3 illustrates the ion source. The electrons are drawn to the source by a positive potential and then pass through the source via apertures on either side. They are collected on an electrode, referred to as the electron trap, on the opposite side of the source. The ribbon of electrons interacts with some of the molecules of the sample stream and forms a cloud of ions in the source cavity.

A single exit slit from the source is positioned close to the path of the electron beam, and therefore the ionisation region. By creating a negative potential difference between the source block and an additional electrode, the source slit, positive ions are drawn out of the ion source. The ion source is held at a potential above ground termed the ion energy: for the Prima BT this is 1 kV.

Ions leaving the source through the exit slit pass between two half plates which focus and steer the beam. The potential difference between the two plates is varied to steer the beam on to the source slit. The voltage difference between the mean half plate voltage and the ion source block is varied to focus the ion beam. The ion beam leaves the source assembly via the alpha slit, which trims the beam height prior to injection into the sector.

Inside the ion source an additional electrode, referred to as the repeller, is positioned directly opposite the ion exit slit. Operating at a positive potential relative to the ion source helps move positive ions out of the source and improves the ion extraction efficiency.

The number of ions formed depends upon both the quantity of molecules in the ion source and the electron current. The electron current is monitored on the trap electrode and a feedback circuit in the analyser supplies unit (ASU) adjusts the filament current so that electron flux is maintained at a constant level.

The electron beam and ion exit apertures are the only openings in the ion source which is otherwise fully enclosed. Introduction of a sample gas to this enclosure results in a higher pressure environment inside the source enclosure compared with that of the surrounding vacuum chamber where the filament assemblies are positioned and operate. This pressure differential helps suppress instrument backgrounds, inherent in the vacuum chamber, and reduce effects due to sample / (hot) filament interactions.

Figure A–4 illustrates the arrangement of the Prima BT ion source assembly. This is of the Nier type, in which ionisation is achieved by collision of the sample molecules with a collimated electron beam; the source materials are primarily ceramic (alumina) and stainless steel. All the source components are aligned by four precision-ground ceramic rods fixed to the source mounting flange and the assembly is held in place by the source clamp.

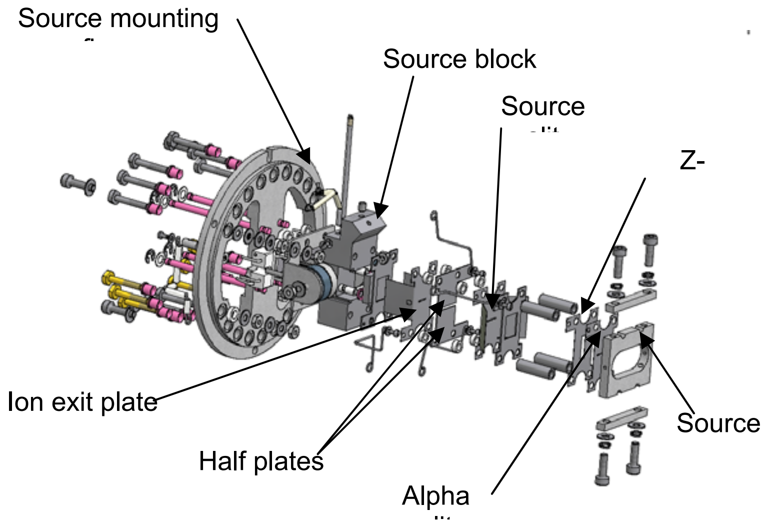


Figure A–4. Prima BT ion source assembly

Filaments

Selection of a filament

There are two commonly used filament materials on the Prima BT (others may occasionally be used for special applications).

Thoriated Iridium

The emissive material is a thoria (thorium oxide) coating on an iridium wire support. This has a lower work function than other common filament materials such as tungsten and rhenium. Thoriated iridium filaments are recommended for operation at low pressures and low pumping speeds since less out gassing occurs due to the lower working temperature of the filament. This also makes the filament more tolerant of operation under conditions of overpressure and is less likely to “burn out” and fail. Thoriated iridium filaments are used for the majority of applications on the Prima BT.

Tungsten Thoria Alloy

This material is a 1% ThO / 99% W alloy and is formed into a coiled wire, which gives prolonged life for hydrocarbon applications, particularly where there is greater than 20% of hydrocarbon compounds of molecular weight higher than 40.

Filament Current Limit

The filament current limits for each of these materials are given in the table below.

Table A–3. Filament current limits

Filament	Current Limit (Amps)
Thoriated iridium	3.1
Tungsten / ThO alloy	3.0

Figure A–5 shows how the filament is mounted to the source block.

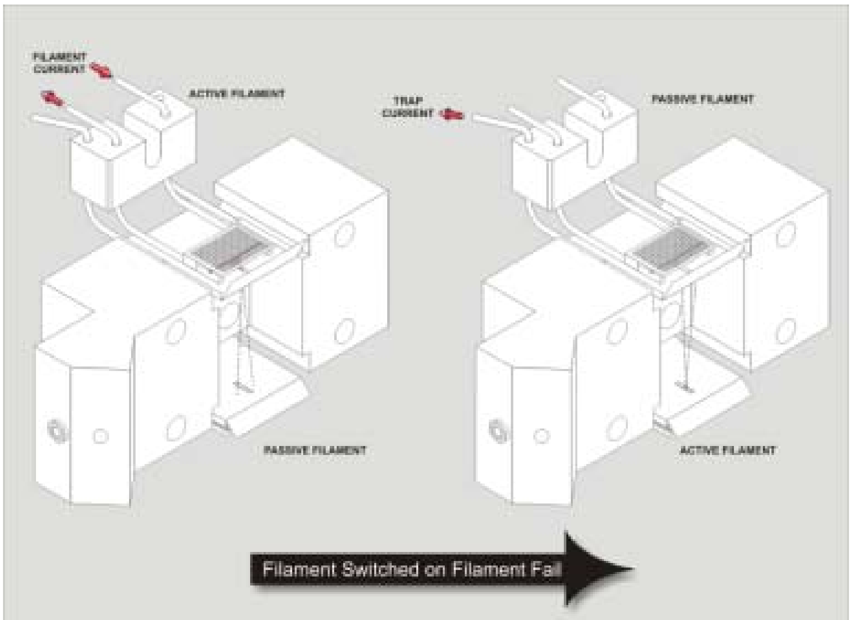


Figure A–5. Mounting of the filament onto source block

Mass Analyser

Refer to Figure A–6.

For a given magnetic field (i.e. a given magnet current), only ions of a specific mass to charge ratio will reach the mass spectrometers collector. Masses that are lighter than that for which the instrument has been tuned will be deflected more and masses that are heavier deflected less. Therefore, the higher the mass of ions to be detected the higher the magnetic field and consequently the higher the magnet current.

$$H \propto I_m$$

Equation 5.

Where:

I_m = magnet current

H = field created by the electromagnet

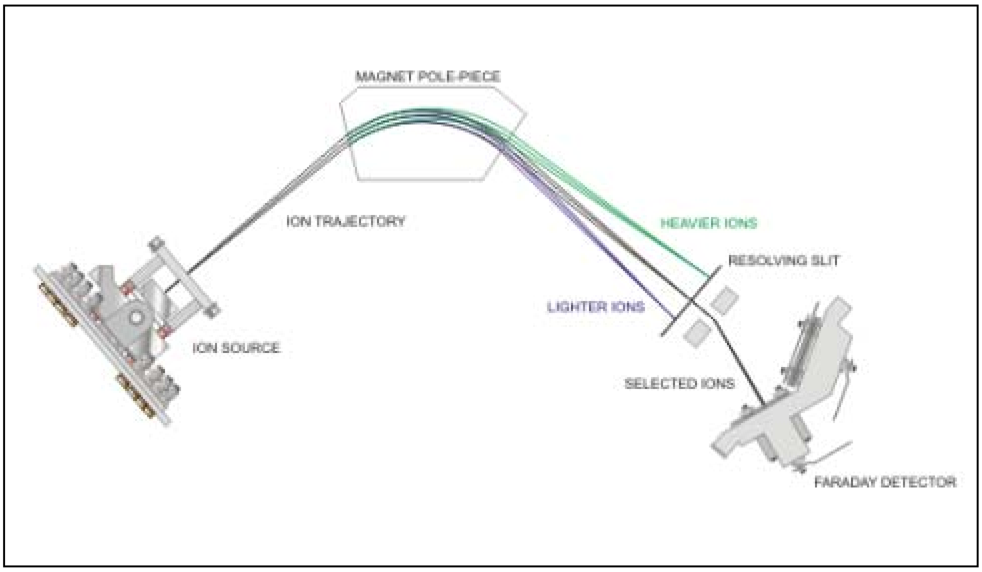


Figure A–6. Separation of ions through the magnetic sector

Collector

Two detectors are available in the collector assembly (the Faraday detector and the optional MCP, the operation of which are described below.

In conventional sector mass spectrometers, compensation for errors in analyser geometry, magnet fringing field effects and beam aberrations, resulting from localized, contamination induced charge effects is achieved by adjusting the magnet position. In the Prima BT instrument, this is addressed by fixing the position of the magnet and using a quad lens positioned before the collector to compensate for and adjust the position of the refocused beam image.

This lens is capable of producing either a converging or diverging beam profile as required to adjust the effective focal length of the instrument and thereby trim minor changes electronically. The voltages applied to the two pairs of quadrupole rods are equal and opposite.

The quad lens DAC voltage can be varied from -25 V to +25 V but, in practice, a variation of a few volts positive or negative is usually sufficient to achieve a symmetrical, flat-topped, peak that provides good, long-term precision from the analyser.

The deflector lens is used to divert the ion beam through one of two resolving slits. The size of the slit is generally chosen so the instrument can resolve adjacent peaks at the high mass end of the scale. The higher the mass, the wider the peaks become. A disadvantage of using a single slot is that low mass peaks are, in comparison, very narrow making these peaks more susceptible to drift in mass scale. On the Prima BT instrument, two resolving slits are fitted – one for low mass peaks e.g. typically below mass 5, and one for high mass peaks. The normal resolving powers used are 20 and 60 for the Prima BT.

The Faraday / multiplier deflectors divert the ion beam onto either the Faraday or optional MCP detector. With a positive voltage applied to the Faraday deflector (F-def) and the multiplier deflector (M-def) grounded, the beam is diverted into the Faraday detector. When a positive voltage is applied to (M-def) and (F-def) is grounded, the beam diverts on to the MCP detector.

The resolving stack and detector assemblies are illustrated in Figure A–7.

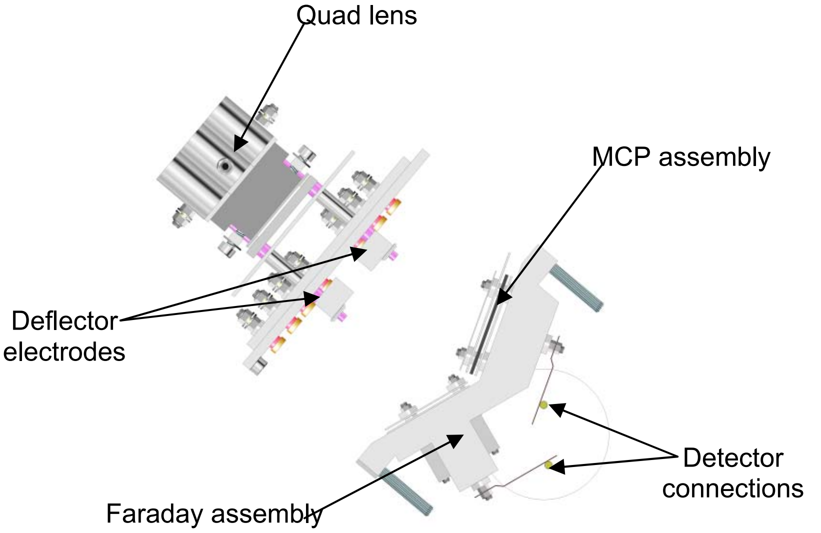


Figure A–7. Resolving stack and detector assemblies

Ion Detection

MCP Detector Assembly

The output of the MCP detector assembly is an electron current with a magnitude 103 or 104 times that of the ion current striking the plate and depends on the voltage applied. Operation of the MCP is described below. Electron current exiting the MCP is picked up on the detector electrode which connects directly to the amplifier mounted on the front right face of the vacuum housing.

Figure A–9 illustrates an MCP that consists of an array of millions of glass capillaries, each of which acts as secondary electron multipliers to form a large secondary electron multiplier array.

Each channel has an internal diameter of 25 µm and are fused together to form a thin disk 0.8 mm thick. The inside wall of each channel is coated with an electron emissive material the ends of each channel is covered with a thin metal film to form an electrode. Each channel is therefore an independent secondary electron multiplier.

When a voltage is applied across the ends of the capillaries, an electric field is generated along the axis of each channel. When an ion strikes the entrance wall of the channel, secondary electrons are produced. The secondary electrons are accelerated by the electric field and travel along the parabolic trajectories determined by their initial velocity. When these electrons strike the opposite wall of the channel, additional secondary electrons are produced. This process is repeated many times along the channel and as a result, the electron current increases exponentially towards the output of the channel.

The capillary bundles are sliced into thin disks with a slice angle chosen to ensure that primary electrons cannot pass through the channel without subsequently colliding with the channel wall.

MCP Operation

The MCP should not normally be operated at voltages exceeding 1200 V. Optimum results are normally obtained at a gain setting of 1000. At this value detector noise is negligible and the life of the MCP assembly is prolonged.

Caution! Damage to the MCP!

Ion currents in excess of 1 x 10^{-12} A should never be applied to the MCP as irreversible damage, requiring replacement of the MCP, will occur. For normal analysis the MCP should only be used for peaks less than 100 ppm.

Calibration of the MCP requires that the voltage needed to produce a certain gain is established. This is achieved by measuring the ratio of peak intensity on the Faraday and MCP array as follows.

1. From the Instrument Control Panel, select the Detectors tab and Multiplier Calibration from the drop down function box (Figure A–8).

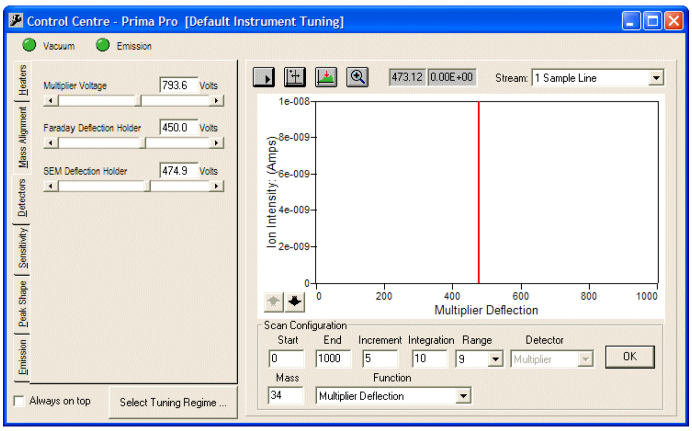


Figure A–8. Multiplier Calibration

2. Choose a suitable mass with a peak height in the range of 1 to 8 x 10^{-13} amps on the Faraday detector. Start the scan (by clicking on the **Start Scan** button) and observe the multiplier gain as a function of the multiplier voltage.

3. Click on the **Add Cursor** button and set the cursor to the required gain.

4. Click on the **Set DAC** button to set the voltage to give the required gain.

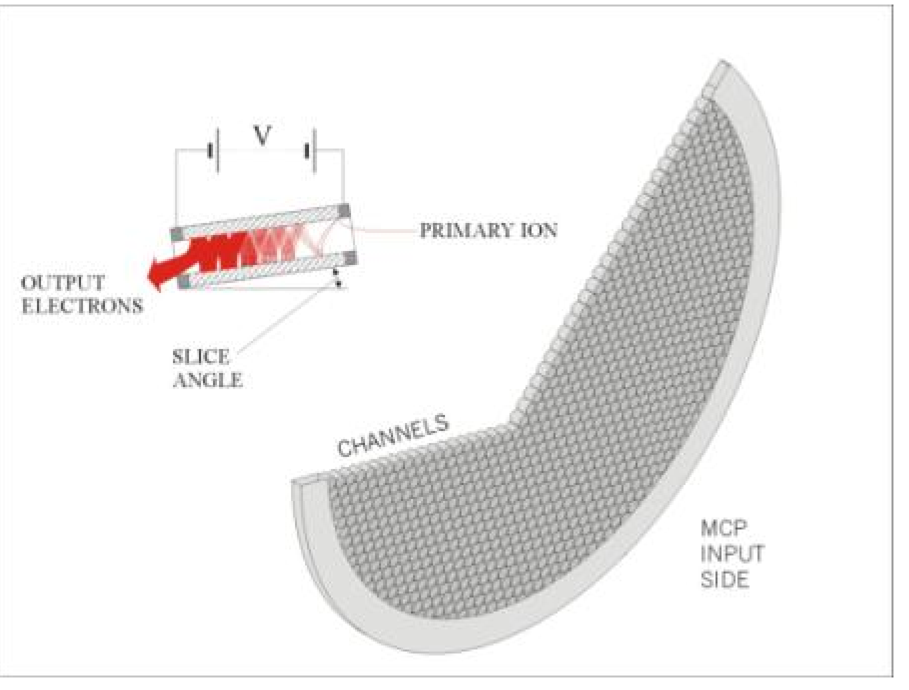


Figure A–9. Micro channel plate

Faraday Detector

Ions collected in the Faraday bucket result in a current that is sensed and then amplified to a suitable level in the instrument amplifier. The Faraday bucket is light tight and coated with colloidal graphite to reduce emission of secondary electrons that would otherwise cause a false indication of ion current. Further suppression is provided at the entrance to the detector by a suppressor plate, biased at approximately -40 volts and magnets that create a field around the bucket to ensure any electrons emitted remain confined.

Introduction of a Gas Sample

Gas samples are introduced into the ion source through the inlet probe assembly. For a Prima BT, gas is sampled through a micro capillary in the stream selector assembly (RMS, solenoid, or single point), passes along a transfer line, into the inlet probe, and finally down a bypass line to the rotary vacuum pump. In the inlet probe, a small portion of the gas flowing through the assembly passes through a further leak element into the ion source.

The flow of gas through the capillary and inlet probe is viscous in nature. It is desirable that the flow through the leak element should be molecular, i.e. of the same characteristics as the flow out of the ion source. Failure to achieve this can result in a distortion to the composition of the gas as presented to the source, specifically a depletion of light gases with respect to heavier gases. The leak elements used are typically a pinhole type (commonly 70 µm diameter, but other diameters may be used depending on application). For leaks of this diameter, flow is predominately molecular if the pressure of gas at the leak is around 1 mbar. The length and diameter of the bypass line are chosen such that this pressure is achieved. The leak element therefore forms the transition point between viscous and molecular flow.

The flow through the capillary is 7 to 10 ml/min, while the flow rate through the leak element is around 5×10^{-4} mbar l/s, i.e. only a small fraction (<1%) of the gas entering the system through the capillary passes to the ion source, the bulk of the sample passes to exhaust through the rotary pump. The flow path through the capillary, transfer line, inlet probe, and bypass line to the pump can be considered as a fast loop, so that changes in sample gas composition are rapidly transmitted to the inlet probe.

A typical arrangement (in the case of an RMS inlet system) is shown in the following figure.

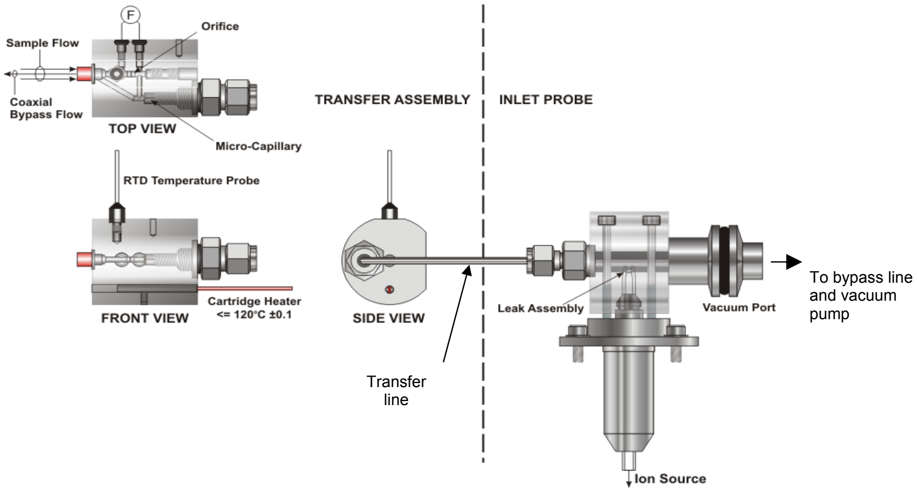


Figure A–10. Typical arrangement of RMS inlet system

Mass Spectrometer Parameters

Sensitivity

Whenever the instrument completes a calibration, a calibration report is produced that details the sensitivity and relative sensitivity of each analyte. The sensitivity indicates the ion current as measured at the collector and is proportional to the percentage of a particular analyte in the gas used for calibration. Equal pressures of different gases in the ion source do not necessarily produce an equal ion current. This is due partly to differences in molecular ionisation cross-sectional area, those with a larger area being more easily ionized, and in part due to the different transmission efficiency of the analyser for ions of different mass. Consequently a sensitivity factor, relative to a reference analyte, is calculated. This factor is known as the relative sensitivity.

For the calibration report, relative sensitivity is calculated as being the sensitivity of the analyte relative to the base gas defined in the gas database. For example, if the relative sensitivity of nitrogen measured by the instrument is 1.0 and helium measured by the instrument is 0.2, for equal quantities of nitrogen and helium in the ion source the ion current measured for nitrogen will be five times that of helium.

Resolution

The resolving power or resolution indicates the highest adjacent mass numbers that a particular instrument can separate. The definition used is known as the 10% Valley definition. This states that the resolving power of a mass spectrometer is the highest mass number at which peaks of adjacent molecular weight and equal heights have a valley between them of 10% of the peak height. Figure A–11 shows the appearance of two such mass peaks. The intensity of the two peaks is additive, and if each contribute 5% at the midpoint between them, then the net effect is a 10% valley.

For example: If an instrument has a resolving power of 100 by this definition then, if peaks of equal height could be formed at mass 100 and mass 101, the valley between them would be 10% of the height of either peak. At lower masses, the valley between adjacent masses is reduced, and at higher mass numbers, the valley between adjacent masses increases.

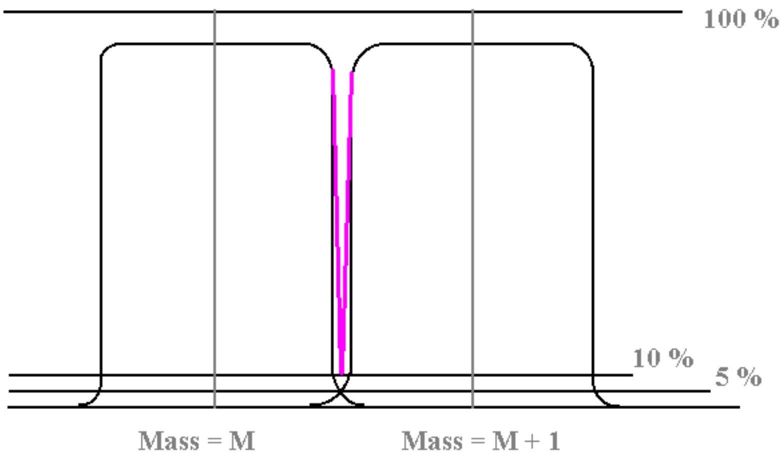


Figure A–11. Two mass peaks of adjacent molecular weight and equal heights

A 5% Peak Height definition is exactly the same as the 10% Valley definition and is illustrated in Figure A–12. This is an easier definition to apply in practice since the peak width at 5% level of the peak height can be easily measured. For example, in air when examining the nitrogen 28 peak, the peak width at the 5% height can be measured and related to the difference between mass 28 and mass 29. If the 5% width proved to be equal to 0.7 of the distance between mass 28 and mass 29 (assuming the peak is centred exactly on 28), then the resolving power is given by 28/0.7 = 40. This indicates that the 10% Valley definition would apply to peaks of equal height at masses 40 and 41.

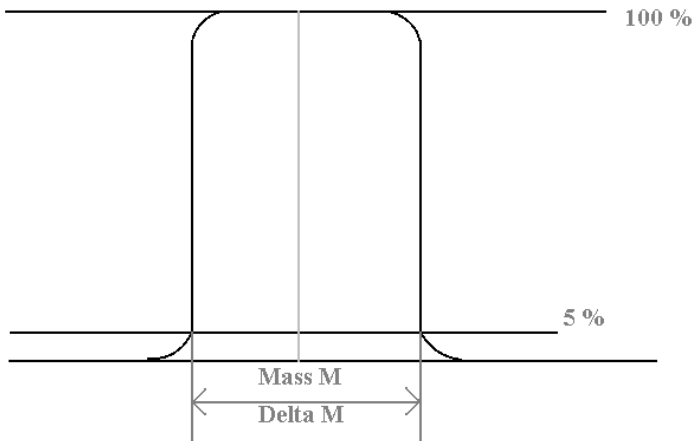


Figure A–12. Comparison of 5% Peak Height definition and 10% Valley definition

Cracking Patterns

The sequence of peaks representing a particular gas species is a direct result of how that species reacts under electron impact in the source. Depending on the species, the molecules can undergo a variety of reactions within the ion source, and the resultant set of peaks is referred to as the "cracking pattern."

A single gas species may produce a variety of different ions giving rise to several peaks of different mass-to-charge ratio and intensity. The relative intensities of these various peaks are characteristic of the gas species in question, although the peak intensities may be affected by changes in source conditions (such as electron energy). The relative intensities of the different peaks can be used to de-convolute the composition of a gas stream containing components of the same molecular weight.

Ion Types

Some of the different types of ions that can be produced within the ion source are described below.

Molecular Ion

This is the simplest reaction to understand and results from the gas molecule acquiring a single positive charge as a result of electron impact in the source. An example of this type of ion would be the $^{14}N_2^+$ ion for nitrogen at m/e 28 amu.

The molecular ion peak may not always be the peak with the highest intensity in the mass spectrum for a given component. This is especially true for hydrocarbons, where the fragment peaks are usually statistically more likely to occur than the molecular ion.

Isotope Ions

Isotope ions are produced when one or more of the atoms that comprise a molecule are of a different isotope from that most commonly found. The relative intensities of these ions are defined by the relative abundance of the different isotopic species for the component atoms in question. The relative intensities of isotope peaks are not normally affected by changes in source conditions.

An example of an isotope peak is the m/e 29 amu peak observed in the nitrogen mass spectrum. This peak results from a single charge on a nitrogen molecule where one of the two nitrogen atoms is the ^{15}N isotope. The relative intensity of this peak to the molecular ion at m/e 28 amu is twice the natural abundance of the ^{15}N atom (since the ^{15}N atom could be either of the two atoms).

Fragment Ions

These ions result from fragmentation of the molecule under electron impact in the ion source. In hydrocarbon species these ions are normally the most abundant since longer chain molecules are statistically more likely to be broken. The relative intensities of fragment peaks are dependent on the flux of electrons within the ion source (trap current) and the energy of the electrons from the filament (electron energy).

An example of a fragmentation ion would be the m/e 27 amu peak in ethylene. The molecular ion for ethylene has a mass of 28 amu. The 27 amu ion arises from the loss of a hydrogen atom from the molecule.

Multiple-Charged Ions

These ions result from the sample gas molecule losing more than one electron during the ionization process which gives rise to multiple positive charges. The relative intensities of multiple-charged ions depend upon the energy of the electrons from the filament. It should be noted that multiple-charged isotope and fragment peaks can also occur further complicating the spectrum. An example of a multiple-charged ion peak is the $^{40}Ar^{++}$ ion at m/e 20 amu.

In some species it is possible to produce ion peaks that are a combination of multiple-charged ions and fragment ions. An example of this would be the m/e 14 amu peak for nitrogen, where the peak is produced from a combination of $^{14}N_2^{++}$ ions and $^{14}N^+$ ions. Since the multiple-charged ions have a different energy from the singly charged ions, the observed peak shape resulting from combinations of the two types may be slightly degraded.

Rearrangement and Recombination Peaks

These peaks arise from a variety of causes, such as ion-molecule interactions within the ion source. The product ions can appear to have strange (and perhaps seemingly impossible) chemical structures. The relative intensities of these ions are usually very low since they normally result from collisions within the ion source between molecules in the gas stream. Relative intensities are very strongly dependent on the partial pressure of the different gas species present, and on the total gas pressure within the ion source. The intensities of these peaks are usually unstable in time and should be avoided for quantitative analytical work.

Examples of these types of ions are the m/e 30 amu peak observed in air due to the interaction of nitrogen and oxygen within the ion source to produce NO and the m/e 29 peak observed in methane produced by the interaction of a methane ion with another methane molecule to give a $C_2H_5^+$ ion at m/e 29 amu.

Definitions

The following definitions are commonly used when defining cracking patterns for a gas species.

Base or parent peak: The ion that exhibits the most intense peak in the mass spectrum for that gas species.

Cracking fragment: The cracking fragment is given by:

$$C = \frac{\text{Intensity of the peak for ions of the required m/e}}{\text{Intensity of the base peak for the gas component}} \times 100\%$$

These are normally expressed in percentages, though some references normalize to 1000. Although this is the most common definition used, the intensities are normalized to the molecular ion rather than to the base peak in some text books and library reference sources.

On dual detector systems it is possible that different cracking patterns are observed on the two detectors (and consequently two different cracking patterns stored in the software gas database). These different patterns result from the mass and energy-selective characteristics of the secondary electron multiplier, i.e. the detector has a different gain for ions of different masses and charges.

In cases where more than one ion peak is required to identify a component in a mixture with others that have overlapping peaks in their spectra, there is no real alternative to measuring the individual cracking patterns.

If a single peak can be used to uniquely identify a species in a mixture then there is little point in measuring a cracking pattern.

The system configuration can be set to measure the cracking pattern for a gas species in a given calibration gas bottle by ticking the

Fragmentation box for that gas in that bottle (see the Calibration Parameters menu in the software). The cracking pattern for a gas should only be measured in a calibration mixture for which the required ion peaks are unique for the gas species of interest. This is known as a non-overlapping mixture.

Table A–4. Example of a cracking pattern for ethylene (C_2H_4)

Mass (m/e)	Relative Intensity (%)
2	0.06
12	0.51
13	0.87
14	2.09
15	0.25
24	2.27
25	7.77
26	52.85
27	62.65
28	100.00
29	2.31

It is neither necessary nor recommended to include all of the peaks for a particular gas component. Users should employ the minimum possible number of peaks within the gas database that results in the best conditioning of the sensitivity matrix discussed in the software manual.

Examples of application specific calibration gas mixtures and a recommended cracking pattern matrix are normally provided with quotation.

Appendix B
Technical Description: Inlet

Rapid Multi-Stream Sampler

Introduction

The Rapid Multi-Stream Sampler (RMS) inlet system has been used successfully with a range of Prima and Sentinel gas analysers. The unit selects a single gas stream from a number of incoming sample or calibration gas streams, and some of the selected gas stream is then delivered to the mass spectrometer for analysis.

Safety

Warning! The notes below must be considered in addition to safety notes elsewhere in this manual covering the system as a whole.

Depending on the nature of the gases in use (sample and calibration gases), precautions may need to be taken to avoid inhalation of, contact with, or ignition of these gases. In general, these precautions are as follows.

- Always use the correct tubing size for the fittings supplied and the correct nut / ferrule assembly.
- Do not exceed the pressure and flow specifications detailed below.
- Leak check all external pipe work, and recheck periodically.
- Leak check inlet system periodically, as defined in this manual.
- Before opening up any gas handling part of the system, it may be necessary to purge the system with a suitable inert or non-hazardous gas. Gas lines and the exhaust line should be isolated. Consideration should be given to the possibility and implications of failure of the isolation mechanism used.

Warning! It is the responsibility of the instrument user to assess the potential hazards involved in working on any gas handling parts of the system.

Warning! Before working on the inlet assembly the following points should be noted.

- The inlet is heated at temperatures up to 80°C. Care must be taken to avoid burns.
- The inlet contains moving mechanical parts. Movement could start unexpectedly.
- Mains voltages exist in the supply to the main heater.
- The remainder of the inlet (including the sample tube heater and stepper motor) is driven from a 24 Vdc supply.
- In general, it is recommended that the inlet is switched off, the mains and 24 Vdc cables disconnected, and the unit be allowed to cool before carrying out any service work.

Inlet Specification

The following table summarizes the specifications for this inlet system. Detailed discussion can be found in following sections.

Table B–1. Inlet system specification

Number of inlets	16
Gas inlet connection (compression type tube fittings)	1/4”, 1/8”, or 6 mm
Sample / calibration gas filtration requirement	≤ 2 µm[(1)]
Sample gas inlet flow rate (regulated externally)	Typical: 0.5 Nl/min Range: 0.1 to 5.0 Nl/min[(2)]
Sample gas inlet pressure. Maximum, exhaust blocked to give above typical flow.	0.2 bar(g)[(3)] <0.1 bar[(4)]
Maximum sample gas / line temperature	80°C[(5)]
Maximum operating temperature	80°C
Exhaust requirements	Pressure range: 0.8 - 1.2 bar(abs)[(3)] Stability: ±3 mbar/min
Exhaust connection	1" O.D. tube stub[(6)]
Materials exposed to sample gas	Stainless steel, Viton, PTFE / graphite composites
Number of calibration inlets drives	6
Flushing time required ($t_{99.9\%}$ decay time, non-polar gases)	5 seconds

Notes:

[1]Gas must be free from liquids or materials that will condense at the operating temperature of the inlet.

[2]The total sample flow (i.e. total over all streams) should be considered with respect to the exhaust line, such that back pressure is within the exhaust pressure limits specified above.

[3]Limit is to prevent over-pressurisation of and possible degradation in performance of the mass spectrometer.

[4]Pressure relative to exhaust pressure.

[5]Higher temperatures may be possible. Consult factory.

[6]Suitable for hose or compression fitting, depending on application.

Inlet Operation

General Description

Gas samples connected to the inlet ports are allowed to flow into the common exhaust chamber of the RMS. Process gas samples are typically allowed to flow continuously to minimize the gas response time between the process sample point and the RMS. Calibration gases normally only flow when required (see below).

The sample gas streams pass through the inlet ports and enter the RMS common exhaust chamber via a series of holes (connected to the inlet ports) in the RMS main body, or stator (see Figure B–4). A rotary sample arm diverts the flow of one of these sample (or calibration) gases through the sample arm and along an axial sample tube (outer part) to the sample tube main body, which is on the inside of the instrument enclosure. A 2µm screen filter is built into the sample arm so that the selected gas stream is filtered before passing on to the analyser. It should be noted, however, that this is not intended as an active filter, and is only a backup to the main gas conditioning filters installed elsewhere (upstream). It is useful for trapping any debris that may be left over from initial fitting of sample lines.

The inlet assembly is attached to a mounting plate, which in turn forms part of the instrument enclosure wall. All of the gas connections are on the outside of the cabinet. All electrical, electronic, and moving mechanical parts are internal.

Connection to the Mass Spectrometer

The selected gas flows around a loop contained entirely within the sample tube body and returns to the main RMS exhaust chamber via the inner tube of the sample tube assembly. A small fraction of the gas in the loop is sampled by the system capillary, which is built in to the sample tube body, and flows on to the mass spectrometer inlet probe via the transfer line. Flow through the capillary is in the range 7 to 10 atm cm^3/min, and the pressure in the transfer line is approximately 1 mbar. Compared to the quantity of gas in the sample tube, the flow rate is very high, resulting in a rapid response to sample gas composition changes. Temperature control of the sample tube assembly (see “Heating” later in this chapter) means that flow through the capillary is very stable with minimal dependence on ambient temperature.

Seals

The seal between the stator and the sample arm (at the point where the selected sample gas is collected) consists of a spring loaded seal. The spring loading of the seal means that wear is not a significant issue, and no degradation of performance is expected over a period of several years. Furthermore, the flow of the selected gas through the filter, sample arm, sample tube, and pressure sensing orifice generates a slight overpressure within the sample arm. As a result, any leakage at this seal will be out from the sample arm and into the common exhaust. Cross contamination of the selected gas from other streams is thus avoided.

The main drive shaft for the sample arm is sealed through the stator by standard rotary shaft seals. These seals have integral spring loading for wear compensation and will give many years of service. The sample tube is sealed within the main shaft by similar (smaller) seals.

Inlet Selection

The sample arm is driven by a stepper motor connected via toothed pulleys and drive belt. The pulleys are positively mechanically attached to their respective shafts so that slippage is negligible. Inlet position indexing is achieved by optical encoding. An encoder disc is attached to the main drive pulley on the RMS. The encoder is a stainless steel disc with a series of holes close to its circumference, one for each inlet. An additional hole (located on a slightly greater circumference) identifies inlet position #1. A pair of optical transmitter / receiver sets is used to sense the individual inlet positions and the inlet #1 position respectively.

This type of encoder does not give absolute position indexing. On startup, the main shaft is rotated slowly so that position #1 can be located. To move to other inlets, the controller steps the drive motor to the calculated correct position. At the same time, the second sensor counts the number of inlet positions moved. If there is a mismatch, an error is flagged and the position #1 alignment procedure is repeated. The motion is always clockwise, and position #1 is checked each time the inlet rotates through this point. This procedure has been found to give an extremely high reliability in terms of the accuracy of inlet selection.

Calibration Gas Control

Calibration gases are expensive and required on a relatively infrequent basis. It would clearly be wasteful to have these gases flowing continuously so additional gas control is required in the calibration gas lines.

The main control element is a solenoid valve (24 Vdc, normally closed), located in the calibration gas line before the RMS. The valve is energized (and the gas allowed to flow) by the inlet controller only when the gas is required.

The solenoid valves are a manifold of six valves, the common port connecting to a port 16 on the RMS valve. This means that up to six calibration gases can be handled through one port on the RMS. The manifold is fitted to the LH side of the instrument below the RMS valve. This minimizes the line length between the panel and the RMS, which in turn minimizes the flushing time requirement when sampling a calibration gas. The tubing material used for this connection is FEP. This has excellent chemical resistance and mechanical properties, is non-porous, and has very favourable adsorption and permeation properties.

A restriction element (an orifice) is built into the six-way assembly such that a line pressure of 1.0 bar(g) will give a flow around 150 cm^3/min. Fine adjustment of flow is achieved by adjustment of line pressure at the calibration bottle regulator.

The incoming calibration gases connect to compression fittings on the bottom of the calibration manifolds(s). By default, the fittings will be the same size / type as fitted to the main RMS body.

Gas Connection

Sample gases are connected directly to the RMS ports. The connections are compression type. The standard types are 1/4”, 1/8”, and 6 mm Swagelok, but other options may be available. The lines may need to be heated, depending on the gas mixtures to be used.

Sample gas flow / pressure should be regulated externally to values within the range indicated in “Inlet Gas Connections” in Chapter 3 (the total flow should also be considered when selecting the exhaust line size). Since the pressure required is quite low, it is preferable to regulate the flow rather than the pressure. In the event of a flow failure, the gas mixture in the common exhaust will diffuse back into that port, and this will be the measured gas if that port is sampled. In order to avoid this (and the possible reporting of false results), the built-in flow sensor is used to verify that flow is within acceptable limits (see “Connection to the Mass Spectrometer”), but note that this can only be used to check the flow of the selected sample stream, i.e. the flow can only be checked when that stream is sampled. External flow monitoring (as part of a sample conditioning system) can also be used and has the advantage that all streams can be continuously monitored, but does require a flow monitor for each sample line.

Gases must also be conditioned to remove particulate material down to
2 µm. Liquids or materials that could condense at the operating temperature of the RMS should also be removed. Consideration should be given to fault conditions that may occur in the process or sample lines and maximum and minimum ambient temperatures to which the sample lines and RMS may be exposed.

Exhaust Connection

The exhaust connection is a 0.5" or 12mm compression fitting and it may be required to heat the exhaust line if condensable vapours are being monitored. This can be just as critical as the sample lines but is often neglected. Alternatively, the exhaust line can be configured with a continuous downward slope so that any condensate will drain away from the valve. If this is not possible, a trap should be installed to prevent condensate draining back to the RMS.

The exhaust line size should be selected, giving consideration to the maximum total sample gas flow, and the length of the line. The requirement is to generate a negligible back pressure, as any such back pressure will be flow dependant and may result in variable conditions at the analyser.

The pressure at the exhaust should be within the values specified in "Inlet Specification". Any pressure variation will result in a variation at the analyser, so the pressure stability should also be within specified limits. Longer term variations outside this range may be acceptable but will require more frequent instrument calibration. The easiest way to achieve the required stability is to use a direct atmospheric vent. However, this is not always possible depending on the nature of the gas mixture, local environmental regulations, etc. An alternative, such as a line to a flare, may be used if the above conditions can be met.

If the exhaust line is shared with other equipment, consideration should be given to pressure fluctuations that might be caused by that equipment. In conditions where all sample streams to the RMS are off, the possibility of back diffusion of gases from the exhaust must also be considered. For this reason, extreme care must be taken if the analyser rotary pump is vented to the same exhaust system. The pump exhaust will contain oil vapour that could contaminate the RMS and subsequently the mass spectrometer. Where possible, it is preferable to avoid such a connection.

In some cases, it may be necessary to use a pump to produce a negative pressure at the RMS exhaust to induce flow in the sample lines particularly when sample points are at low pressures.

It is important to remember that all sample gases are mixed in the common exhaust. There may be situations where combustible or reactive mixtures are generated. To avoid problems of this nature, a diluting gas (commonly nitrogen) can be added to the exhaust line or via spare RMS ports (the latter may be more convenient if spare ports are available).

Heating

The inlet assembly can be heated up to 80°C to avoid condensation, polymerization, etc. of sample gas mixtures. Two heaters are required to heat the main body and peripheral parts of the inlet as follows. Control of the heaters is discussed later in this chapter.

Main Body Heater

The bulk of the inlet body is maintained at the required temperature by a set of 2 cartridge heaters located in bores in the stator itself. The two heaters are wired as effectively a twin element assembly, with a three-wire connection.

The three-wire connection allows for series or parallel connection for 230 V or 115 V applications respectively, selected by a switch on the inlet controller PCB. Insulation is provided to minimize heat loss into the enclosure.

A K-type thermocouple is used to measure and thereby control the temperature of the valve body via the control electronics.

Sample Tube Heater

The sample tube body carries the selected gas stream flow and so also requires heating.

A 30 W, 24 V cartridge heater heats the sample tube body, and a PRT is provided for measurement and control. Typically, this assembly is set to the same temperature as the main body. Additional temperature limiting is not required as the heater power is insufficient to take the assembly to a dangerous temperature. The actual sample tube between the sample tube body and the valve body is clad in copper, which greatly improves the thermal conductivity and helps maintain a uniform temperature along its length.

Gas Flow versus Temperature

The above heaters were specified assuming that the incoming sample gases are heated to a temperature similar to that set for the RMS. However, there may be situations where this is not the case. For instance, all streams may be cold or there may be a combination of hot and cold streams. Consideration must be given to the possible general cooling of the RMS body or to localized cooling – for example, the cooling effect of a cold gas flow on an adjacent port or the cooling of the common flow regions (sample tube, etc.). The latter could be important if a cold stream is sampled immediately prior to sampling a temperature critical hot stream. The following examples illustrate the limits of cold gas flows (the cold gas streams are assumed to be air or nitrogen at a temperature of 20°C).

a. To maintain a body temperature of 80°C, the maximum cold gas flow is 500 Nl/min.

b. A cold gas flow of 10 Nl/min on a single port (the maximum allowed) will cool an adjacent port by approximately 4°C.

c. A cold gas flow of 10 Nl/min on a single port (the maximum allowed) will cool the sample tube by approximately 10°C (over a period of several minutes). Similarly, the temperature takes several minutes to recover. This situation can be avoided by reducing the cold gas flow rate or bypassing approximately 80% of the cold stream to a spare (non-sampled) RMS port.

Potential problems can be minimized by careful selection of which ports hot and cold gas streams are connected to and the sequence of sampling of these ports.

Inlet Controller

General

The inlet controller handles all the control functions required for the RMS and its associated calibration panels. The controller PCA is mounted on the left hand side of the enclosure below the inlet assembly. Most connections are via pluggable screw terminals.

The RMS sampling head drive is unidirectional and indexed at port 1 every rotation (see “Inlet Operation”). The controller drives the sample arm to port 1, which is indicated by both the red and the green LED being illuminated on the sensor PCB. If the reference aperture cannot be found, the controller will stop searching after 60 seconds. The sample arm completes any port to port transit in less than two seconds. Positioning failure (due to mechanical or electronic failure) is communicated to the inlet controller and the unit will re-index on the next commanded move.

The controller is powered from the main enclosure power supply (24 Vdc). Mains power is required for the main body heater (see below).

Fuses (all on PCB):

F1: 5 A (mains)

F2: 1 A (24 Vdc, calibration gas valves). Supply OK indicated by green LED D2.

F3: 4 A (24 Vdc, all other functions). Supply OK indicated by green LED D19.

A watchdog function is used to show that the processor is active (red LED D6).

Calibration Valve Drives

There are 24 outputs, individually addressable with up to 4 switched on at any one time (limited in software). Each is designed to deliver up to
300 mA at 24 Vdc and is intended to drive a single calibration gas control solenoid valve.

Main Body Heater

Mains power is connected to J18 along with the heater(s). This allows 115 Vac or 230 Vac operation to be selected by means of a switch on the PCB. Unplugging J18 or switching off the Inlet circuit breaker in the main enclosure will remove mains power from the PCB.

Warning! Removal of F1 will only break the L/L1 line. Power might still be on the assembly from the N/L2 line.

The main body thermocouple connects to J13. Power to the heater is switched via a solid state relay on the PCB, giving a control precision of ±1°C, accuracy ±3°C. A red LED (D25, adjacent to the solid state relay) flashes when the SSR is driven on.

Sample Tube Heater

The sample tube PRT connects to J15 and the heater output (24 Vdc) to J16. Control precision is ±1°C, accuracy ±3°C. A red LED (D24) flashes when the sample tube heater is driven on.

Flow Sensor

Connect the flow sensor to J17.

Digital Inputs

Four opto-isolated digital inputs are available (J5, J6, J9, J10). Generally, these are used for internal monitoring of inlet functions They can be configured as requiring 24 Vdc or voltage free contact input by means of jumpers on the PCB. A red LED adjacent to each connector indicates the status of the input.

VGiNet

The red LED adjacent to the VGiNET sockets (D3) indicates data received by the unit (not necessarily for it or processed by it) and the green LED (D4) indicates data transmitted from it. When the unit detects a message intended for it, it will respond with a reply message and so a full message cycle will appear as a red flash followed quickly by a green flash as the reply is transmitted.

Inlet Testing

Testing of the inlet system may be required in order to confirm correct performance, in the event of unexpected analytical results, suspected contamination, or following a major service. The areas to be checked are as follows. It is advisable to switch off the inlet heaters before starting any work.

Caution!

- Before starting any work, switch off all gas lines.
- If required, the system should be purged with a suitable inert, safe gas and the system exhaust isolated.
- It is the responsibility of the user to assess the hazard involved in breaking into any gas handling part of the system.

Functional Testing

Inlet Alignment

Remove the stator cover from the RMS body.

Warning!

- Possible gas hazard. See Chapter 2: Safety Information.
- Moving mechanical parts exposed. Movement could start unexpectedly.
- Hot surfaces. Check RMS temperature settings before starting work. Adjust as required.

Select a port using GasWorks (**Control Centre** > **Analogue Scan**, etc.). Check that the sample arm moves to the correct physical port and check that the sample arm is aligned over that port. It should be (visually) equally spaced between the two adjacent ports. A slight misalignment can be corrected by loosening the four screws retaining the sample arm to the main shaft and moving the arm to the correct position before re-tightening. Check several ports, including 1, in this way.

The probability of the inlet operating with no reported errors but giving significant mechanical misalignment is extremely low as all parts are mechanically keyed together. Check that the main pulley is secure on the drive shaft and that the sample arm is aligned with the position 1 cutout on the encoder disc (incorrect reassembly after servicing could result in 90°, 180°, or 270° misalignment).

If the controller reports positional errors, check if position 1 can be located correctly. If it can, but subsequent inlets are showing errors, there may be a motor or driver fault (or, very unlikely, a mechanical fault making the shaft difficult to rotate).

If position 1 cannot be located (the motor drives slowly as it searches for position 1) then the most likely cause is a problem with the optical encoder. There are two LEDs on the encoder PCB. The green one is lit whenever the RMS is “on” an inlet. The red LED indicates position 1 (i.e. both LEDs will be lit at position 1). For ease of checking the functionality of the optical sensors, the RMS can be rotated by hand: switch off the controller, unplug the motor connector, and switch back on. If there appears to be a problem, check that there is no dust around the holes in the disc and around the optical transmitters / receivers. If there is no activity, remove the screws fixing the encoder disc to the pulley (note the orientation). Both LEDs now should be lit. Slide a piece of paper between the optical transmitters and receivers, and both LEDs should be off.

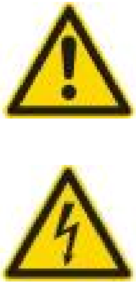
If it appears that the LEDs (and hence the optical sensors) are working correctly, then the bracket holding these sensors may need adjustment. Refit the encoder disc. Loosen the nuts holding the bracket in place and adjust until both LEDs are on at position 1 (Adjustment will affect the alignment of the sample arm with the port. Remember that the arm position can be adjusted slightly, as noted above). Switch off, reconnect the motor, and switch on. Check that alignment is still OK when the motor is driven. Check a selection of other ports.

Heating

If there is a reported fault, the following checks can be carried out.

Main Heater

Warning!



- Mains voltages present.
- Switch off at the instrument before removing the inlet controller PCA or disconnecting any heaters.

Verify that the temperature readout is correct. Use an independent temperature measurement if required. Check F1 on the controller and then check that the controller is delivering power to the heater connections on the controller (J18) (This may not be a continuous supply if the controller is attempting to control. See also “Main Body Heater”).

Check the status of the heater and thermal trip by disconnecting at J18. The wires (1-5) from the heaters and trip are:

- Wire 1: Heater 1
- Wire 2: Heater common
- Wire 3: Heater 2
- Wire 4 & 5: Thermal trip

The electrical resistance between wires 4 and 5 should be approximately 0 Ω, unless the RMS body is over temperature or the thermal trip is faulty. The resistance between wires 1 and 2 and between 2 and 3 should be the same – around 140 Ω for a system with cartridge heaters, either 70 Ω or 35 Ω (depending on type) for a ring heater system.

Sample Tube Heater

Verify that the temperature readout is correct. Use an independent temperature measurement if required. Unplug the heater and check the heater resistance. The standard heater should be approximately 19 Ω. Reconnect the heater, and switch on the voltage across the heater. It should be 24 Vdc. Check fuses if required voltage is not present.

Cross Port Leakage

This test should be carried out with the stator cover fitted.

Set up a continuous flow of air on one or more ports (flow within the specified limits) and a helium flow on another port. (The helium flow does not need to be a continuous flow and can be a calibration port. Helium should flow at typical calibration gas flow rates.) Sample the air inlet and measure the mass 32 peak height. Next, sample the helium inlet and check that the mass 32 peak height drops by the required factor. For a Prima BT the factor is system pressure (mbar) × 108.

If it is not possible to introduce air into the system, the test should be carried out with process gas flowing. Measure the peak heights of masses in the analysis and check against historic values (e.g. from a calibration report from when the instrument was initially installed). Backgrounds should be similar to the original values.

If the inlet fails to meet specification, follow the procedures for inspection and / or replacement of seals detailed later in "Maintenance".

External Leakage

A limited amount of leak checking can be carried out with process gasses connected and flowing e.g. leak checking of external connections using a proprietary bubble type leak detector, such as Snoop. It should be noted that the pressure in the RMS body may be low or below ambient and it is very difficult to test rotary seals by this method.

The best method involves a positive pressure test. Isolate or disconnect / plug all sample lines and the exhaust line (<see the note at the beginning of this appendix). Disconnect the transfer line from the RMS, and plug the RMS port(s). The RMS body should now be pressurised to 0.5 bar(g) with a clean inert gas (e.g. nitrogen) via one of the inlet ports. The RMS body pressure should be monitored via one of the unused inlet ports.

Once pressurised, isolate from the gas supply, and monitor the RMS pressure as a function of time. The test is passed if the pressure falls by no more than 0.05 bar in a 1 hour period.

If the test fails, leak check the accessible parts of the system while still under pressure, including any external connections which may also be pressurised. If the leak cannot be found, follow the procedures for inspection and / or replacement of seals detailed in "Maintenance".

Maintenance

Little maintenance is required on the system providing that the incoming gases are correctly conditioned. The system will require attention if there is a failure of the gas conditioning system resulting in excessive liquid or particulate contamination.

The seals in hazardous area systems should be inspected on a six monthly basis.

Caution!

- Before starting any work, switch off all gas lines.
- If required, the system should be purged with a suitable inert, safe gas and the system exhaust isolated.
- Liquid or solid contaminants might be encountered within the RMS assembly during maintenance operations. The likely composition of any such contaminants should be determined so that appropriate safety measures can be put in place before further work is carried out.
- It is the responsibility of the user to assess the hazard involved in breaking into any gas handling part of the system.
- It is also recommended that the inlet is fully switched off and allowed to cool.

Routine Maintenance

Capillary Replacement

To replace the capillary assembly, the vacuum system will need to be shut down so that the transfer line can be disconnected from the sample tube.

Unscrew the transfer line fitting from the sample tube to expose the capillary assembly. The capillary can be removed using a 3mm hexagon key.

Figure B–1. Capillary assembly

Note that the capillary assembly has a small o-ring seal fitted to the inner face. This may remain behind in the sample tube when the capillary is removed. Carefully remove the o-ring prior to fitting a new capillary assembly taking care not to scratch the sealing surface in the bottom of the bore.

When replacing the capillary, take care to locate the o-ring on the end of the assembly, and keep it in place during the refitting process. Reinstall the transfer line fitting and the transfer line.

Filter Replacement

Remove the stator cover to reveal the sample arm. Note the position of the sample arm (check which port it is positioned over, and that it is central over that port). Remove the two screws holding the sample return seal retainer in place and then carefully remove the seal. Undo the four screws fixing the sample arm to the main shaft and carefully pull off the sample arm.

Note:

- The seal in the end of the sample arm is spring loaded and could drop out when the sample arm is removed.
- Loss or damage to this seal and or spring will render the sampler unusable.
- Exercise care when removing the sample arm and ensure the spring and seal are set aside for refitting later. ♦ ♦

The above procedure exposes the double rotary seals, two back-to-back seals on the main axis, between the sample arm and the sample tube / main shaft. Typically, one of the seals is retained in the sample arm assembly and the other remains on the sample tube / main shaft assembly. In some instances both seals may remain on the sample tube / main shaft and can be left in place as long as they are firmly seated.

Removal of the spring-loaded seal from the sample arm exposes the
2 µm stainless steel mesh screen which acts as the filter. The filter can be removed with a pointed tool but will be destroyed or irreparably damaged in this process therefore ensure a spare is available before removal. Care should be taken to avoid damaging or scratching of the sample arm.

Press the new filter into place, a flat ended piece of 1/4" / 6 mm diameter rod serves as a suitable tool, so that it is well seated in the bottom of the hole. Replace the spring / seal assembly and position the sample arm over the main body assembly in approximately the correct orientation. Push into place. Ensure the seal remains in place during this operation. Refit the four retaining screws ensuring the head is correctly positioned over the inlet port it was on when removed. Refit the sample return seal and cover and finally, refit the stator cover.

General Inspection

Remove the stator cover. A good impression of the cleanliness of the system can be made by inspection of the common exhaust chamber and the inside of the stator cover. If there is evidence of dust, liquid, or other residues then it may be advisable to inspect the RMS further as detailed in the following sections. It is often possible to determine which inlet ports are the source of any contamination, particularly in the case of liquid residues. Check the gas conditioning for any suspect streams.

Seal Inspection and Replacement

The rotary seals are difficult to remove, and could be damaged in the process. It is advised that spares are available before starting work.

Sample Arm Seal

Follow the instructions in “Filter Replacement” above to the point where the sample arm is removed. Inspect the sealing edge of the sample arm seal and if there is serious scoring or other damage the seal should be replaced. Note that the design of the inlet means that it is tolerant of leakage at this point and any leakage would be in the cross port sense. There is no danger of an ambient leak.

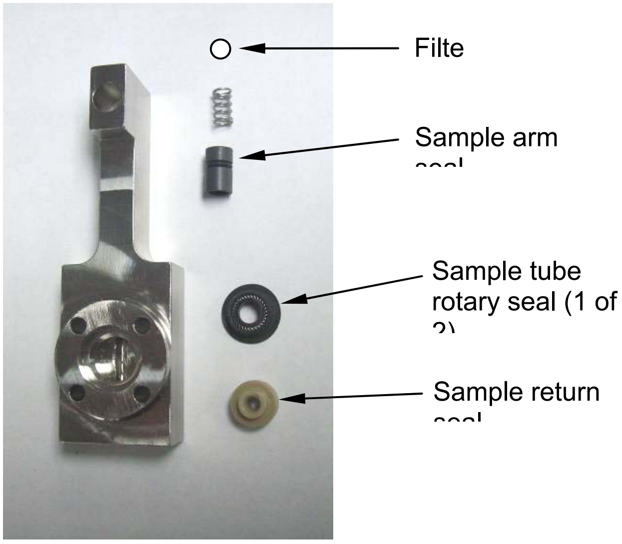


Figure B–2. Sample arm

To reassemble, follow instructions in “Filter Replacement” above.

Sample Tube Seals

As noted in above, removal of the sample arm exposes the sample tube rotary seals. If the sample tube is to be removed for any reason, this should be done as it is then easier to remove these seals for inspection. Remove the seals being careful not to distort them in the process and check their condition.

A polished region should be visible in the central bore, where the sample tube has been running. No other signs of wear should be evident. Also check the surface of the sample tube which protrudes out of the end of the main shaft. There may be some slight deposit on the sealing surface due to material worn from the seals. This can be wiped off with a tissue moistened with isopropanol. The surface should appear highly polished. Replace any worn or damaged parts. Reassembly is the reverse of disassembly.

Main Shaft Seals

Follow the procedure above to point of removal of the sample arm.

Remove the six screws holding the main shaft seal retaining ring in place (the six screws can be seen when looking into the RMS exhaust chamber), and withdraw the retaining ring. The seal is now visible.

Note:

- There are two possible orientations for the shaft seal either facing in or out depending on application.
- If a plain black surface is revealed then the seal is configured for negative pressure applications
- If the lips of the seal and its internal spring can be seen, the seal is configured for positive pressure applications, most Prima BT applications.
- If seals are removed and or replaced ensure the seals are refitted in the correct orientation for the application
- If a change of application is anticipated, adding a sample pump for example, consult the factory before changing the seal orientation.

Prize out and inspect the seal and the sealing surface of the main shaft as for the sample tube seal.

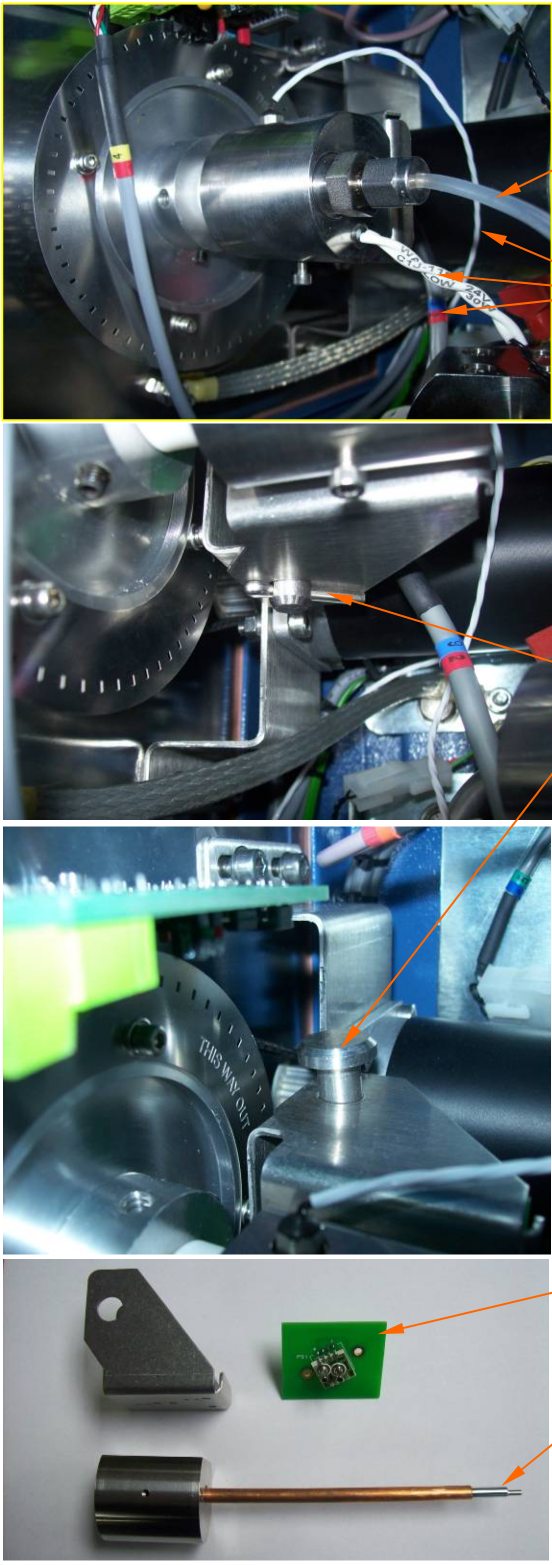
Replace any worn or damaged parts.

Reassembly is the reverse of disassembly.

Sample Tube Removal

The sample tube connects to the rotating sample arm and conducts the selected sample down the main shaft axis out of the sampler assembly and on to a connection point in the instrument enclosure. If any contamination was noted in other parts of the inlet while performing the above procedures, it is recommended that the sample tube is also removed for inspection and / or cleaning.

In the event that the differential pressure sensor has to be replaced, the sample tube will need to be removed.



Disconnect the transfer line / sample lines from the sample tube inside the cabinet.

Disconnect the heater, temperature sensor, and flow sensor cables.

Remove the bracket split pin and hinge pin holding the sample tube in place. The sample tube can now be withdrawn. The white PTFE bush is a fairly loose fit and care should be taken to avoid mislaying it.

Remove the flow sensor PCB assembly.

The sealing surface at the tip of the tube can be easily cleaned / inspected now. Further cleaning can be carried out as per the instructions below. It is advisable to remove the capillary before any such cleaning.

Figure B–3. Removing the flow sensor PCB assembly

To reassemble the flow sensor, first locate the o-rings in the recess in the sample tube body. Then refit the bracket but leave the screws quite loose at this point. Fit the flow sensor PCB assembly to the bracket. Tighten down the PCB retaining screws and finally the bracket retaining screws (via the holes in the PCB). This procedure ensures the o-rings are correctly centred on the pressure transducer tubes.

To refit the sample tube into the RMS, first remove the sample arm so that the various seals can be assembled on to the end of the sample tube without damage. Otherwise, reassembly is the reverse of disassembly

Inlet Ports

The gas inlet fittings are screwed into the main body. The thread type is 1/8" BSPP (alternatively G1/8, or ISO 228/1). This is a parallel thread, so a gasket type seal is used between the fitting and the RMS body, and not formed on the thread itself.

Caution! Alternative threads such as 1/8” NPT or 1/8” BSPT are similar but not identical. Damage WILL result if these are used. ♦ ♦

A fitting may need to be removed in the event of damage to the compression faces of the fitting, or in the event of contamination (a pocket of contaminants may reside behind the fitting even after flushing through).

Unscrew the fitting and replace / clean as required. The seal is generally reusable but replace if in doubt.

Cleaning Procedure

The above procedures will have exposed all areas that can be wetted by sample gas, and no further disassembly is required for complete cleaning.

The RMS stator should be cleaned in situ.

Particulate contamination should be removed initially by blowing with a clean inert gas, such as nitrogen. More stubborn contamination can be removed by wiping with a tissue moistened with clean isopropanol (specifically, free from non-volatile impurities) or another suitable solvent. Difficult to access regions may require cleaning in an ultrasonic bath. The ports in the main body should be flushed through, working from the top downwards. Solvents in pressurised (aerosol) containers are convenient for this providing they meet the above requirements.

Do not use excessive solvent for cleaning elastomeric seals as it can cause the seal to harden and subsequently fail or leak. All metallic parts can be cleaned with solvents.

To remove contamination from process fluids, some knowledge of the contaminant is required, both from a safety point of view and for choice of a suitable solvent or cleaning agent. Accessible areas can be mechanically cleaned, providing sealing surfaces are not damaged.

Final cleaning should always involve the use of a clean solvent (such as isopropanol as above) to remove any cleaning residues, grease, etc. Excess solvent should be blown from the inlet components using an inert gas. If possible, components should be dried in an oven at temperatures up to 80°C.

Clean, or replace if necessary, external pipe work and gas conditioning equipment to avoid recontamination. In cases of severe contamination, it is possible for lines not directly responsible as sources of contamination to have become contaminated. This can include calibration lines.

The cause of the contamination should be investigated and rectified.

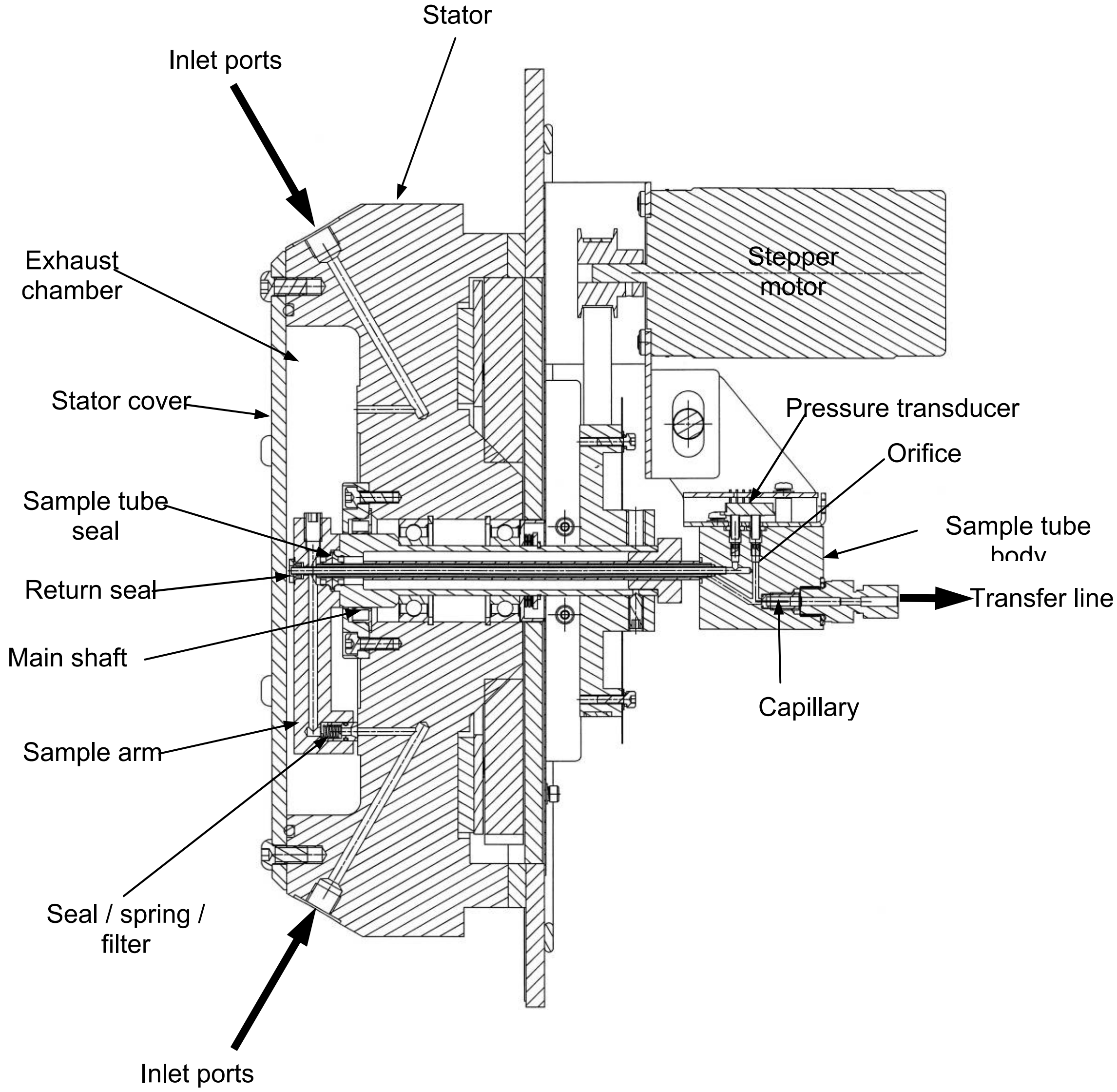


Figure B–4. RMS assembly

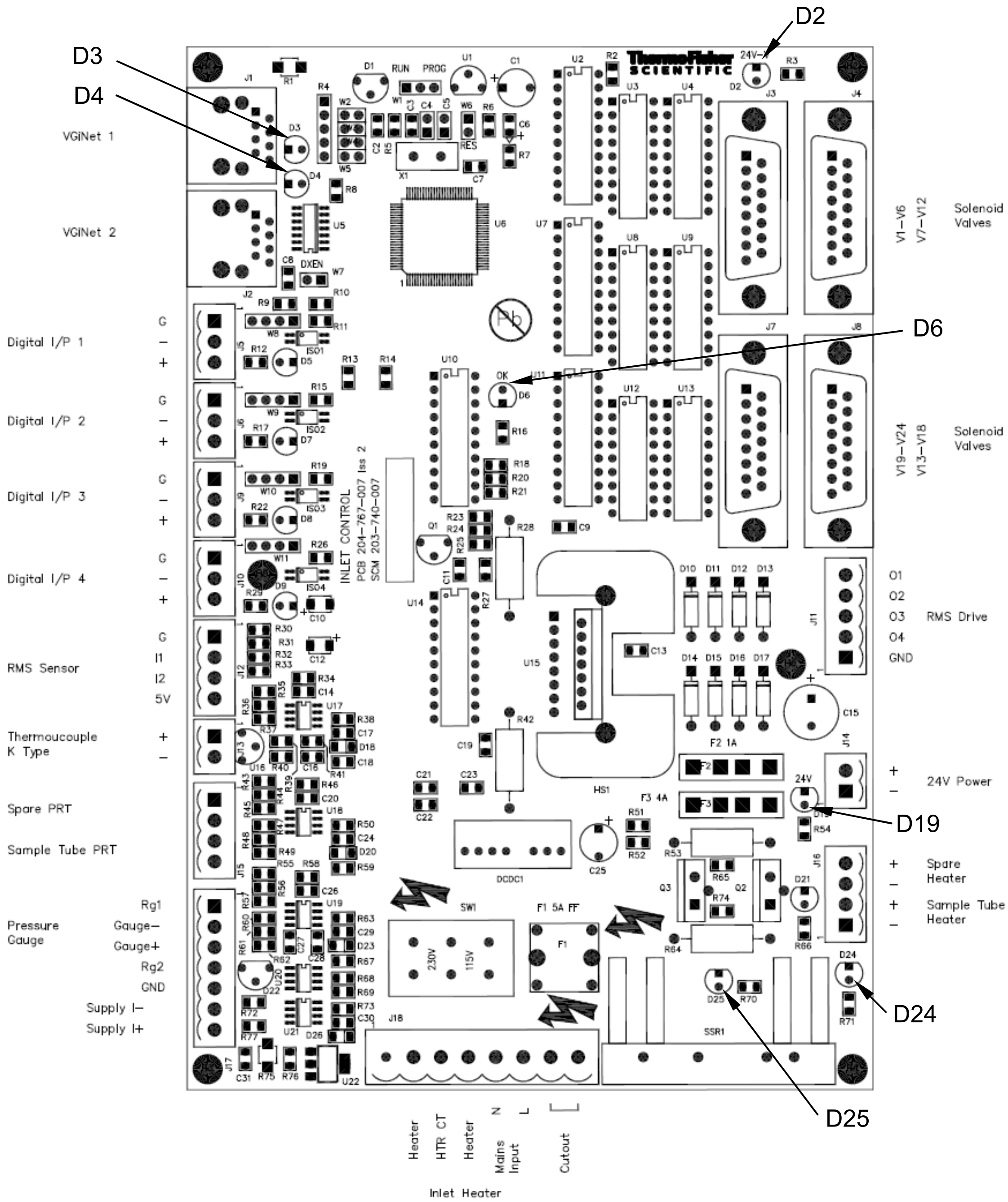


Figure B–5. Controller PCB

Table B–1. List of service replacement parts

Quantity	Description	Part number
1	Sample arm seal	203-217-013
1	Sample arm seal o-ring	201-120-197
1	Sample arm seal spring	201-135-085
1	Filter (2µm)	201-110-292
1	Shaft seal	201-110-200
2	Shaft bearings	201-106-019
1	Sample tube bearing (PTFE)	203-217-020
2	Sample tube seal	201-110-199
1	Return seal	201-110-489
1	Sample tube	204-764-002
1	Sample tube heater	202-310-075
32/64	1/4” compression fitting	201-110-219
32/64	1/8" compression fitting	201-110-245
32/64	6 mm compression fitting	201-110-725
32/64	Seal ring PTFE	201-120-138
32/64	Seal ring stainless steel	201-120-137
1	Thermal switch (140°C)	202-190-013
1	Thermocouple	203-217-120
1	Opto encoder PCB	204-766-009
1	Controller PCB	204-766-007
1	Ring heater	202-320-017
4	Cartridge heater	202-310-067

Solenoid Inlet

Introduction

The solenoid inlet is a sample gas selector that can be used in place of the RMS valve assembly on applications with a limited number of sample inlets.

There are several bulk gas handling components and connections inside the instrument enclosure, and therefore this inlet cannot be used with hazardous area systems.

The stream selector assembly is available in two sample line configurations with either a 6-port manifold or fast response single stream inlet. Each version comes with 6 calibration gas ports. The inlet uses the same micro capillary assembly and transfer line technology as fitted to other inlet systems. Gas flow (sample and calibration) is monitored with a differential pressure flow sensor, similar to the RMS.

Safety

Warning! The following must be considered in addition to all safety notes elsewhere in this manual covering the system as a whole.

Depending on the nature of the gases in use (sample and calibration gases), precautions may need to be taken to avoid inhalation of, contact with, or ignition of these gases. In general, these precautions are as follows.

- Always use the correct tubing size for the fittings supplied and the correct nut / ferrule assembly.
- Do not exceed the pressure and flow specifications detailed below.
- Leak check all external pipe work and recheck periodically.
- Leak check inlet system periodically, as defined in this manual.
- Before opening up any gas handling part of the system, it may be necessary to purge the system with a suitable inert or non-hazardous gas. Gas lines and the exhaust line should be isolated. Consideration should be given to the possibility and implications of failure of the isolation mechanism used.

Warning! It is the responsibility of the instrument user to assess the potential hazards involved in working on any gas handling parts of the system.

Table B–2. Solenoid inlet system specification

Number of inlets (sample)	1 or 6 (options)
Number of inlet (calibration)	6 (standard)
Sample / calibration gas filtration requirement	≤ 2 µm[(1)]
Sample gas inlet flow rate (regulated externally)	0.1 to 2.0 l/min
Sample gas inlet pressure. Maximum, exhaust blocked to give above typical flow.	0.2 bar(g)[(2)] < 0.1 bar[(3)]
Maximum sample gas / line temperature	60°C
Calibration gas flow (regulated externally)	0.1 to 0.25 l/min
Calibration gas pressure	0.5 to 2.0 bar(g)
Exhaust requirements	Pressure range: 0.8 - 1.2 bar(g) Stability: ±3 mbar/min
Gas connection	1/4” or 6 mm
Solenoid valve drive	24 Vdc, 7 W
Flushing time required ($t_{99.9\%}$ decay time)	Sample inlet (1 inlet option): 5 seconds Sample inlet (6 inlet option): 20 seconds Calibration inlet: 45 seconds

Notes:

[1]Gas must be free from liquids or materials that will condense at the operating temperature of the inlet.

[2]Limit is to prevent over-pressurisation of the mass spectrometer.

[3]Pressure relative to exhaust pressure.

Inlet Operation

Refer to Figure B–6.

The principle element of the inlet is the micro capillary block assembly that incorporates a 3/2 selector valve, micro capillary, and flow sensor. The 3/2 valve selects gas from either the sample inlet section or the calibration inlet section, to flow across the micro capillary and through the flow sensor.

In the six inlet stream configuration a six-way manifold of 3/2 solenoid valves provides sample stream selection, sample gases being connected directly to ports on this manifold. The normally open ports of all six valves are connected to a common exhaust that provides for continuous flow of sample even when not selected.

When a valve is actuated, sample gas from that port is directed through the manifold to the above 3/2 selector valve and hence to the capillary assembly for sampling.

In the single inlet stream version, the six-way manifold is omitted and the sample stream connects directly to the selector valve.

Warning!

- All unused inlets must be fitted with blanks and sealed.
- Gases from the manifold exhaust can enter the local environment from an unsealed inlet port and will pose a safety risk if the sample is toxic, corrosive, or an asphyxiant.

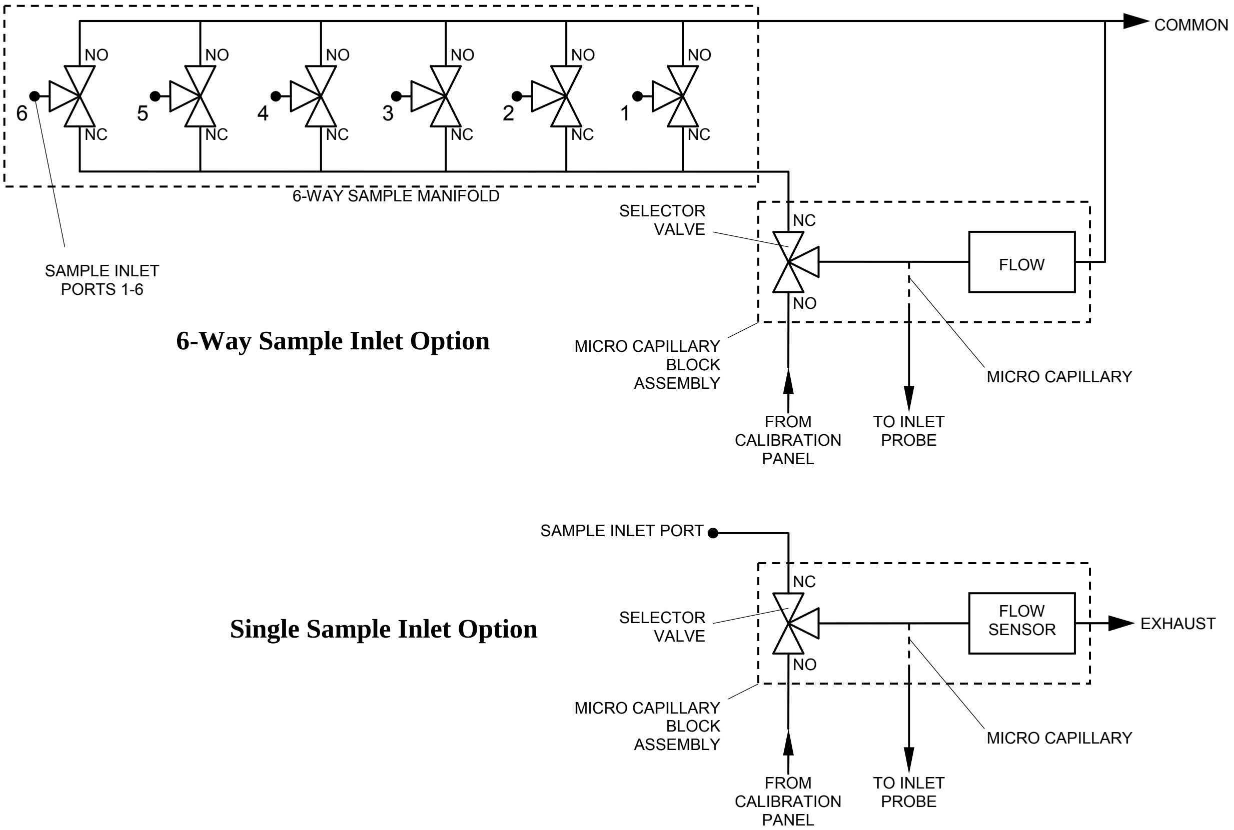


Figure B–6. Solenoid inlet schematics

Calibration gases are required on a relatively infrequent basis and can be costly. To avoid unnecessary usage, the calibration valve assemblies are fitted with 2/2 valves such that calibration gas will only flow when a valve is energized. The calibration assembly is a manifold of six solenoid valves with the common outlet port connected to the normally open port of the solenoid inlet selector valve. The tubing material used for this connection is FEP which has excellent chemical resistance and mechanical properties, is non-porous and has very favourable adsorption and permeation properties. The calibration manifold is fitted to the left-hand side of the instrument enclosure. A single calibration manifold is supplied with either configuration of the solenoid inlet although additional calibration lines can be added. (Consult factory for details.)

A restriction element (orifice) is built into the six-way manifold assembly, such that a line pressure of 1.0 bar(g) will give a flow around 150 cm^3/min. Fine adjustment of flow is achieved by adjustment of line pressure at the calibration bottle regulator.

The incoming calibration gases connect to compression fittings on the bottom of the calibration panel(s). By default, the fittings will be the same size / type as specified for the solenoid inlet assembly.

Note: The calibration panels are identical to those used with the RMS inlet system.

In order to provide the best possible dynamic response to sample gases, a selector valve with a very low internal volume (< 0.1ml) is used. Sample streams connect to the normally closed port, calibration streams to the normally open port. To further optimize the response time of the system, both the calibration and sample manifolds are fitted with an internal flow splitter to ensure that the entire length of the manifold is dynamically flushed regardless of which inlet is selected.

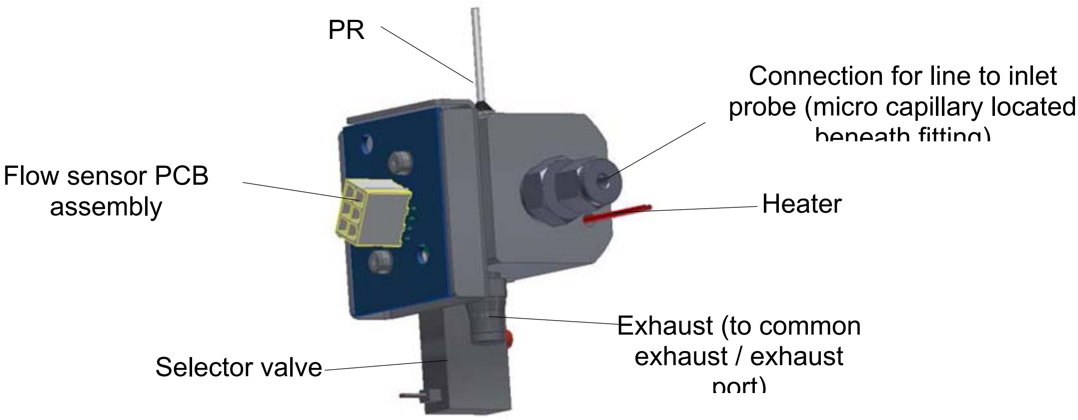


Figure B–7. Micro capillary block assembly

Valve Operation

All valves are 24 Vdc solenoid actuated and are driven directly from the inlet control PCA mounted below the inlet assembly in the top left of the instrument enclosure.

Capillary Connection

A small fraction (7 to 10 $cm^3 min^{-1}$) of the selected gas stream is sampled by the micro capillary (and subsequently by the mass spectrometer). Excess gas flows to the common exhaust.

For a detailed description of the gas transfer system into the mass spectrometer refer to "

Introduction of a Gas Sample" in Appendix A.

Heating

Heating of the system is split into two parts, the micro capillary block assembly and the six-way sample manifold. In the case of a single sample inlet system, the six-way manifold is not used and an adaptor block of the same dimensions is fitted in its place, such that this can be heated in exactly the same way. The capillary block and manifold (or adaptor block) can be heated up to 60°C.

The micro capillary assembly is heated by means of a 24 V 30 W (19 Ω) cartridge heater. Temperature is monitored using a PRT. See Figure B–7.

The valve manifold (or adaptor block) is heated by means of a heater block fixed along one side. The heater block has two cartridge heaters embedded in it, operating at mains voltage. The heaters are connected in parallel or series to allow 115 Vac or 230 Vac operation respectively (see "Inlet Controller" below). The total power dissipated is 50 W. Individual heaters are 576 Ω.

The heater block contains a thermocouple to allow temperature monitoring. A thermal switch is also attached to the heater block (fixed set point of 70°C) to provide a heater cut-out in the event of a controller fault (runaway condition).

Inlet Controller

General

The inlet controller handles all the control functions required for the solenoid inlet and its associated calibration panels. The controller PCA is mounted on the left side of the enclosure below the inlet assembly. Most connections are via pluggable screw terminals. The controller PCA is the same as that used for other inlet systems, such as the RMS.

The controller is powered from the main enclosure power supply (24 Vdc). Mains power is required for the main body heater (see below).

Fuses (all on PCB):

F1: 5 A (mains)

F2: 1 A (24 Vdc, calibration gas valves). Supply OK indicated by green LED D2.

F3: 4 A (24 Vdc, all other functions). Supply OK indicated by green LED D19.

A watchdog function is used to show that the processor is active (red LED D6).

Valve Drives

There are 24 outputs, individually addressable with up to 4 switched on at any one time (limited in software). Each is designed to deliver up to 300 mA at 24 Vdc and is intended to drive a single gas control solenoid valve. For the six-way inlet version, when a sample inlet is selected the controller will energize the specific inlet valve as well as the selector valve.

Manifold Heater

Mains power is connected to J18 along with the heaters and over-temperature thermal switch. This allows 115 Vac or 230 Vac operation to be selected by means of a switch on the PCB. Unplugging J18 or switching off the Inlet circuit breaker in the main enclosure will remove mains power from the PCB.

Warning! Removal of F1 will only break the L/L1 line. Power might still be on the assembly from the N/L2 line.

The heater block thermocouple connects to J13. Power to the heater is switched via a solid state relay on the PCB, giving a control precision of ±1°C, accuracy ±3°C. A red LED (D25, adjacent to the solid state relay) flashes when the SSR is driven on.

Micro Capillary Block Heater

The micro capillary block PRT connects to J15 and the heater output (24 Vdc) to J16. Control precision is ±1°C, accuracy ±3°C. A red LED (D24) flashes when the sample tube heater is driven on.

Flow Sensor

Connect the flow sensor to J17.

Digital Inputs

Four opto-isolated digital inputs are available (J5, J6, J9, J10). They can be configured as requiring 24 Vdc or voltage free contact input by means of jumpers on the PCB. A red LED adjacent to each connector indicates the status of the input.

VGiNet

The red LED adjacent to the VGiNET sockets (D3) indicates data received by the unit (not necessarily for it or processed by it) and the green LED (D4) indicates data transmitted from it. When the unit detects a message intended for it, it will respond with a reply message and so a full message cycle will appear as a red flash followed quickly by a green flash as the reply is transmitted.

Testing

Testing of the inlet system may be required in order to confirm correct performance, in the event of unexpected analytical results, suspected contamination or following a major service. The areas to be checked are as follows (it is advisable to switch off the inlet heater before starting work). If any valves have been stripped down, or had parts replaced, the valve should be operated a number of times before carrying out any test work.

Warning!

- It is advisable to switch off the manifold heater before performing work on the assembly.
- The manifold heaters are Mains voltages rated and should be isolated from the inlet control PCA prior to exist in the supply to the main heater.
- The remainder of the inlet (including the capillary assembly heater) is driven from a 24 Vdc supply. ♦ ♦

Capillary Testing

To test the micro capillary assembly, set the inlet to calibration mode or power down the inlet by removing 24 V from the inlet controller PCA. As noted above, either disconnect the mains heaters from the PCA or switch off the inlet breaker on the mains distribution DIN rail. Disconnect the transfer tube from the calibration manifold at the CALIBRATION INLET fitting where it enters the solenoid assembly. The micro capillary will now effectively be open to room air. Select a suitable display mode in the system software (e.g. Control Centre or MIM) and display the mass 32 peak or a measurement of the mass 32 peak height. Make a note of the mass 32 peak height value (air).

If possible, helium should be used for the next step. Adjust the cylinder regulator to give a gentle flow of helium at the open end of the line (approximately 0.5 to 1.0 lmin^{-1}) and connect the cylinder to the CALIBRATION INLET fitting on the front panel of the solenoid assembly. The mass 32 peak height reading should now drop as helium passes over the micro capillary. The reading should change by a factor of 10^{-8}/pressure (air) within 7seconds, i.e. if the system pressure reading in air was 10^{-5}mbar, the mass 32 peak height should drop by a factor of 1000. If this is not achieved, it may be necessary to leak check the analyser itself.

When complete reconnect the calibration manifold line to the CALIBRATION INLET on the front panel of the solenoid assembly.

Valve Function

The following checks are to ascertain that each solenoid valve is being driven correctly and that the correct gas is being sampled. It should be noted that if any particular solenoid valve does not operate correctly the system will sample the gas mixture in the common exhaust. This condition may not be easy to detect if the composition of the sample gases on the manifold are similar.

For the calibration inlets, select calibration gas inlet #1 from the system software. A flow should be indicated on the flow sensor. Verify that if the gas supply is switched off externally the indicated flow drops to zero (allowing for the line to depressurise). Repeat for all (used) calibration inlets.

For a sample inlet, connect a ‘distinctive’ gas (e.g. helium), set to a typical flow rate, to the inlet and monitor the system in Histogram Scan mode 0-50 amu. When the helium inlet is switched in the spectrum of helium (mass 4 only, plus a few background peaks) should be visible. As with the calibration inlets the flow sensor should also register a reading. Purge with air or nitrogen and then repeat the process for the remaining inlets.

If there appears to be a problem, check that there are no external blockages. Remember that, for sample inlets, the selector valve must be driven in addition to the individual inlet valve. Possible failure of either the selector valve or inlet manifold valve must be considered.

If any given valve fails to operate, first check the 24 V drive signal to that valve. If there is a problem trace this back, as far as the inlet controller PCA if required. If the drive signal is present, check the solenoid coil. The resistance should be approximately 115 Ω.

If there is still a problem, there may have been some contamination causing a valve to ‘stick’. Follow the maintenance procedures in Chapter 10 for cleaning / replacement.

Cross Seat Leakage

Cross seat leakage can result from contamination on the valve seat (as a result of inadequate gas filtering) or simply following prolonged use resulting in minor deformation of the valve seat. As a result a valve may not shut off fully, resulting in cross contamination of samples or loss of valuable calibration gas if this occurs on a calibration manifold valve.

Switch off all calibration and sample gases, and depressurise, the lines.

Ensure the inlet is powered off, i.e. no valves selected. Connect helium to each port on the calibration manifold in turn and pressurise the line to normal operating pressure. Set the mass spectrometer to monitor mass 4 and look for traces of helium. If the valve seat is good there should be no increase in the signal level at mass 4.

To check the sample valves the inlet needs to be powered and any external sample lines removed. Select an inlet and set the mass spectrometer to mass 4 as above. Blank the selected inlet and then connect the helium supply to the remaining, unselected, inlets in turn and pressurise each to the normal operating pressure. A change in mass 4 would indicate a cross seat leakage. Blank off one of the tested inlets and select it through the system software. Connect the helium to the previously blanked inlet to check it for cross seat leakage.

To check the selector valve couple the helium to the CALIBRATION INLET port on the solenoid inlet front panel while the blanked sample inlet is still selected. Pressurise to normal operating pressure and monitor for changes in mass 4 signal. Any change would indicate a cross seat leakage of the selector valve into the sample stream.

If there is a change in signal in any of the above tests, check that it is reversible and repeatable, and check that the inlet is not being pressurised beyond normal operating conditions. If a change is confirmed, the valve may be faulty and require cleaning or replacement. See Chapter 10.

External Leakage

A leak of this type involves gas leaking into the atmosphere, and is only likely to occur following disassembly of the system. The significance of such a leak will depend on the nature of the gas involved. External leakage may give rise to contamination of gas samples with air, but is unlikely as the system normally runs at a slight positive pressure.

Warning!

- It is advisable to switch off the manifold heater before performing work on the assembly.
- The manifold heaters are Mains voltages rated and should be isolated from the inlet control PCA. Switch off the instrument and allow to cool before performing work on the assembly.

- The remainder of the inlet (including the capillary assembly heater) is driven from a 24 Vdc supply.
- In general, it is recommended that the inlet is switched off, the mains and 24 Vdc cables disconnected, and the unit be allowed to cool before carrying out any service work.

Maintenance

No routine maintenance is required on the system providing that the incoming gases are correctly conditioned. The system will require attention if there is a failure of the gas conditioning system resulting in excessive liquid or particulate contamination. The system should be tested periodically as indicated in the previous section to identify issues that could compromise analytical performance.

In general it will be necessary to strip down the inlet system to clean or replace affected part or parts. There are different procedures to follow depending on whether a calibration or sample inlet is involved. See below.

Caution!

- Before starting any work, switch off all gas lines.
- If required, the system should be purged with a suitable inert gas, and the system exhaust isolated.
- Liquid or solid contaminants might be encountered within the inlet assembly during maintenance operations. The likely composition of any such contaminants should be determined so that appropriate safety measures can be put in place before further work is carried out.
- It is the responsibility of the user to assess the hazards involved in breaking into any gas handling part of the system based on knowledge of the sample and calibration gases used.

Calibration Inlet

On calibration manifolds, the solenoid actuator is assembled through the mounting plate so that operationally the electrical coil is inside the instrument enclosure and gas handling parts of the manifold are outside. A consequence of this is that it is not possible to remove and inspect the solenoid valve assemblies of the calibration manifold individually. When the mounting plate is removed all six actuator assemblies will be exposed. The actuators are spring loaded and once the manifold and mounting plate are separated they can detach from the manifold body. Take care to ensure that parts are not lost or contaminated. Work on a clear, clean work surface.

1. Disconnect the cable from the inlet controller PCA (ribbon cable).

2. Disconnect the common line from the calibration manifold that runs to the solenoid inlet assembly Calibration Inlet port.

3. Disconnect the external calibration gas lines.

4. Unscrew the mounting plate from the enclosure and remove the assembly from the system. Ensure the sealing gasket is not damaged or discarded. Replace if necessary.

5. Unscrew the knurled nuts from the six actuator coils. The coils and the connection PCB to which they are attached can be removed as one piece.

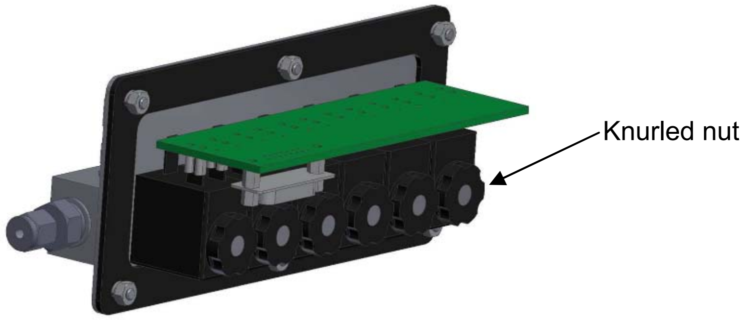


Figure B–8. Knurled nut

6. To remove the manifold from the plate, unscrew the 12 M3 countersunk screws that are now exposed. Note that when the plate is removed the actuator bodies are free to detach from the manifold body. Care must be exercised to ensure these do not spring loose and parts are lost, damaged, or contaminated.

7. Inspect the sealing gasket positioned between the manifold block and the mounting plate. If this is damaged in any way it must be replaced.

8. Carefully pull out the actuator body from the manifold and inspect the valve parts, particularly the sealing surfaces. Clean or replace as required before reassembly.

9. Reverse the above procedure to reassemble the system. Always leak check the system after reassembly.

Sample Inlet

The following description only applies to the six-way manifold version of the solenoid inlet assembly. As this operation will require the transfer line from the micro capillary to the inlet probe to be disconnected, it will be necessary to shut down and vent the instrument vacuum system.

1. Switch off the instrument.

2. Disconnect the heater, sensor, and valve cables from the inlet controller PCA. Take carful note of cable / plug positions to aid in reassembly.

3. Disconnect the transfer tube from the micro capillary assembly to the mass spectrometer inlet probe assembly.

4. Disconnect the external sample gas line, calibration gas inlet, and the exhaust line.

5. The assembly can now be removed from the system. Unscrew the eight M6 nuts holding the mounting plate in place and carefully remove from the enclosure.

6. To remove the exhaust manifold disconnect the 4 mm FEP tube from the push in connector in the exhaust manifold block.

7. Unscrew the Exhaust fixing on the front panel. Retain the sealing ring.

8. Unscrew the six screws below the manifold to release the manifold from the valve assemblies. Take great care not to lose the o-rings that are on both sides of the manifold.

9. Unscrew the knurled rings on the end of the solenoid coil assemblies to release the coils and connection PCB. Remove to expose the actuator assemblies.

10. To inspect individual valves unscrew the valve actuator from the manifold using a suitable spanner. Take care to avoid dropping the valve plunger / spring assembly as the actuator is withdrawn.

11. Inspect the valve parts, particularly the sealing surfaces, clean as required, replacing any damaged seals.

12. Reverse the above procedure to reassemble the system. Always leak check the system after reassembly.

13. If there is any sign of contamination in the valve manifold it is advisable to remove all valves from the manifold to enable full cleaning. Keep the valve assemblies separate and in sequence, so that the valves can be replaced into the same manifold ports.

Cleaning Procedure

Particulate contamination should be removed initially by blowing with a clean inert gas such as nitrogen. More stubborn contamination can be removed by wiping with a tissue moistened with clean isopropanol (specifically, free from non-volatile impurities) or another suitable solvent. Difficult to access regions may require cleaning in an ultrasonic bath. Do not use this procedure for cleaning elastomer seals as it can cause the seal to harden. All metallic parts can be solvent cleaned.

To remove contamination by process fluids, some knowledge of the contaminant is required, from both a safety perspective and in order to choose a suitable solvent or cleaning fluid. Accessible areas can be mechanically cleaned, providing sealing surfaces are not damaged.

Final cleaning should always involve the use of a clean solvent (isopropanol etc., as above) to remove any cleaning residues, grease, etc. Excess solvent should be blown from the inlet components using an inert gas. If possible, components should be dried in an oven at temperatures up to 60°C.

Remember to clean external pipe work and gas conditioning equipment to avoid recontamination. The cause of the contamination should be investigated and rectified before putting back into service.

Appendix C
Local I/O

Introduction

The Prima BT has a number of channels of both serial I/O and discrete I/O built in to the instrument to allow connection to external equipment.

Refer to the instrument specific installation drawing for location and connection details of all local I/O.

Serial I/O

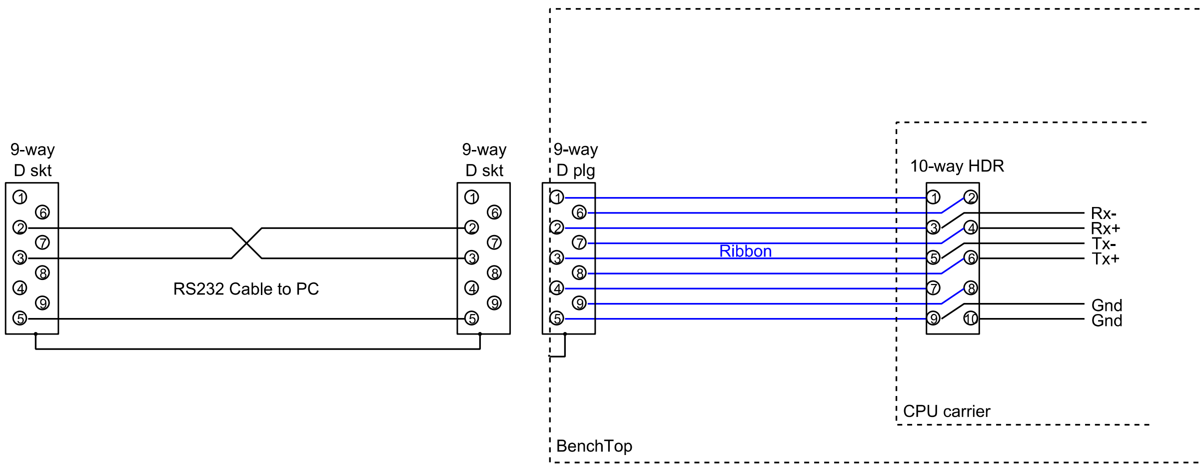
Three channels of serial I/O are available.

One channel marked HOST is specifically for connection to a PC running GasWorks software.

The other two, channels SER 0 , SER 1, are available for communication to external expansion I/O systems, a plant DCS, or other external data gathering or control equipment.

A number of protocols are available for communication to the latter items, including Modbus and simple ASCII data logging. Other (optional) protocols (e.g. Profibus) are implemented using third-party converters, typically converting from Modbus RTU to the required protocol.

HOST connection shown below (as RS232)
Serial 0 & 1 have the same pin assignments.



Serial Link Type Selection

Individual serial channels can be configured as RS232, RS422 (i.e. full duplex) or RS485 (i.e. half duplex, not available for the HOST link) by jumpers on the I/O CPU board. See table below.

Table C–1. Serial communication jumpers

Jumper Function	I/O Port Jumper ID			Jumper Fitted (Y/N)		
	HOST	I/O 0	I/O 1	RS232	RS422[1]	RS485[1, 2]
RS422/485 select	W18	W6	W12	N	Y	Y
Tx+ pull up	W17	W5	W11	N	N[1]	N[3]
Tx- pull down	W16	W4	W10	N	N[1]	N[3]
Rx- pull up	W15	W3	W9	N	N[1]	N[1]
Rx+ pull down	W14	W2	W8	N	N[1]	N[1]
Rx termination	W13	W1	W7	N	N[1]	N[1]

[1] Typical for point to point connection of up to a few hundred metres. Longer lengths, or line drivers, may require other settings; consult factory for advice.

[2] Not available for the HOST connection.

[3] RS485 is Half Duplex. Rx+ must be linked to Tx+, and Rx- must be linked to Tx- (via the external serial cable). Consequently only one set of bias jumpers is fitted (if required – see note 1)

Indicators

On the CPU carrier board, each serial port has two bi-colour LEDs to indicate activity: one on the transmit line and one on the receive line.

Digital I/O

This provides digital and (as an option) analogue connectivity for a wide variety of functions.

Table C–2. Discrete I/O

I/O Type	Function	Channels	Assigned as
PP/SP In-built User IO: Digital IN	Digital Input	1	0
PP/SP In-built User IO: Digital OUT	Instrument fail	1	1
PP/SP In-built, DIP	Control, etc	3	0,1,2
PP/SP In-built, DOP	Alarm, status	2	0,1

*Instrument fail will change to the alarm state if any hardware alarm is activated or if the main CPU 'watchdog' fails.

All I/O is isolated to at least 1kV.

Digital inputs can be jumper-configured to activate via an external volt-free contact, or by an applied 24V dc (+/- 4V) signal (~ 5mA per channel). The default setting is for volt-free contact.

Digital outputs are via a volt-free relay contact, each fused at 1A, and can be software-configured as 'pulsed' or 'static'.

D I/O Indicators

On the CPU carrier board, all digital channels have an associated LED showing the state of the input or output. When an input is energised (by external contact closure or applied 24V dc, dependent on configuration) the LED is ON. When an output channel is driven, the LED is ON and the output contact is closed. Note that the logic of each I/O channel (i.e. 'positive' or 'negative') can be changed in the software's hardware configuration, but the LED will always indicate the state of its associated input or output circuit.

Appendix D
Regulatory

FCC

Federal Communications Commission Radio Frequency Interference Statement.

Information to the user:

Warning! This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions:

(1) This device may not cause harmful interference, and

(2) This device must accept any interference received, including interference that might cause undesired operation. ♦

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at their expense.

WEEE

European Union Waste Electronic & Electronic Equipment Directive.

This product is required to comply with the European Union's Waste Electrical & Electronic Equipment (WEEE) Directive 2002/96/EC and is marked with the following symbol:

This product contains electronic parts and assemblies and should be disposed of at end of useful life in an appropriate and responsible manner.

Thermo Fisher Scientific has contracted with one or more recycling / disposal agencies in each EU Member State, and this product should be disposed of or recycled through them. For further information, contact your distributer or reseller in the country of use. In the UK, contact Thermo Fisher Scientific, Ion Path, Road Three, Winsford, CW7 3GA.